FISON
91349

Isoniazid (INH)

Specimen Requirements:

Specimen Type: Serum

Container/Tube: Red-top

Specimen Volume: 2 mL

Collection Instructions: Draw blood in a plain red-top tube(s), serum gel tube is not acceptable. Spin down and send 2 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	14 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80299

ISPG
82768

Ispaghula, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FITZS
75882

Itraconazole Susceptibility Testing

Specimen Requirements:

Container/Tube: Agar slant or other appropriate media

Specimen Volume: Organism in pure culture

Complete and submit with specimen:

1. Fungus Testing Lab request form

Collection Instructions:

1. Organism must be in pure culture, actively growing.

2. Place specimen in a large infectious container (T146) and label as etiologic agent/infectious substance.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient	

CPT Code Information:

87999-Unlisted Microbiology Procedure (Refer to patient report to apply the appropriate CPT code in place of this unlisted CPT code based on specimen organism.)

87188 - mould - MIC microdilution or agar dilution (if appropriate)

87186 - yeast - MIC microdilution or agar dilution (if appropriate)

ITCON
81247

Itraconazole, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top (serum gel/SST are **not** acceptable)

Acceptable: None

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.18 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	29 days
	Ambient	29 days
	Frozen	29 days

CPT Code Information:

80189

JCHAI 7048 J-Chain Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

JMACK
82819

Jack Mackerel, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

JAKXB
89189

JAK2 Exon 12 and Other Non-V617F Mutation Detection, Blood

Specimen Requirements:

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 10 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 8 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole blood	Refrigerated (preferred)	5 days	PURPLE OR PINK TOP/EDTA
	Ambient	5 days	PURPLE OR PINK TOP/EDTA

CPT Code Information:

0027U-*JAK2* (Janus kinase 2) (eg, myeloproliferative disorder), exon 12 sequence and exon 13 sequence, if performed

JAKXM 60025 JAK2 Exon 12 and Other Non-V617F Mutation Detection, Bone Marrow

Specimen Requirements:

Container/Tube:

Preferred: EDTA (lavender top)

Acceptable: ACD (yellow top)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Bone Marrow	Refrigerated (preferred)	5 days	PURPLE OR PINK TOP/EDTA
	Ambient	5 days	PURPLE OR PINK TOP/EDTA

CPT Code Information:

0027U-*JAK2* (Janus kinase 2) (eg, myeloproliferative disorder), exon 12 sequence and exon 13 sequence, if performed

JAKXR 44178 JAK2 Exon 12-15 Sequencing, Polycythemia Vera Reflex, Varies (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information, see PVJAK / Polycythemia Vera, *JAK2* V617F with Reflex to *JAK2* Exon 12-15, Sequencing for Erythrocytosis, Varies.

Submit only 1 of the following specimens:

Specimen Type: Blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD-B)

Specimen Volume: 10 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Label specimen as blood.

Specimen Type: Bone marrow aspirate

Container/Tube: Lavender top (EDTA) or yellow top (ACD-B)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Minimum Volume: Whole blood: 8 mL; Bone marrow: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	5 days
	Ambient	5 days

CPT Code Information:

0027U-*JAK2* (Janus kinase 2) (eg, myeloproliferative disorder), exon 12 sequence and exon 13 sequence, if performed

JAK2B
88715

JAK2 V617F Mutation Detection, Blood

Specimen Requirements:

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole Blood EDTA	Ambient (preferred)	7 days	PURPLE OR PINK TOP/EDTA
	Refrigerated	7 days	PURPLE OR PINK TOP/EDTA

CPT Code Information:

81270-*JAK2* (Janus kinase 2) (eg, myeloproliferative disorder) gene analysis, p.Val617Phe (V617F) variant

JAK2M 31155 JAK2 V617F Mutation Detection, Bone Marrow

Specimen Requirements:

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Bone Marrow	Ambient (preferred)	7 days	PURPLE OR PINK TOP/EDTA
	Refrigerated	7 days	PURPLE OR PINK TOP/EDTA

CPT Code Information:

81270-*JAK2* (Janus kinase 2) (eg, myeloproliferative disorder) gene analysis, p.Val617Phe (V617F) variant

JAKFM 618647 JAK2 V617F Mutation Detection, Bone Marrow

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot**.

3. Label specimen as bone marrow.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Bone Marrow	Ambient (preferred)	7 days
	Refrigerated	7 days

CPT Code Information:

81270

JAK2V
31156

JAK2 V617F Mutation Detection, Varies

Specimen Requirements:

Specimen Type: Extracted DNA from blood or bone marrow

Container/Tube: 1.5- to 2-mL tube with indication of volume and concentration of the DNA

Specimen Volume: Entire specimen

Collection Instructions: Label specimen as extracted DNA from blood or bone marrow and indicate volume and concentration of the DNA.

Specimen Stability Information: Refrigerated/Ambient

Specimen Minimum Volume: 50 microliter at a concentration of 20 ng/microliter

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81270-*JAK2* (Janus kinase 2) (eg, myeloproliferative disorder) gene analysis, p.Val617Phe (V617F) variant

FJPE
57921

Jalapeno/Chipotle (Capsicum annuum) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

JCEDR
82865

Japanese Cedar, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL

For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FJCQP 75758 JC Polyoma Virus DNA, Quantitative Real-Time PCR, Serum/Plasma

Specimen Requirements:

Collection Container/Tube: Lavender top (EDTA), yellow top (ACD-A), white top (PPT) tube

Submission Container/Tube: Plastic vial

Specimen Volume: 0.7 mL Plasma

Collection Instructions:

1. Centrifuge and aliquot plasma into a plastic vial.

2. Send frozen.

Specimen Minimum Volume: Plasma: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen (preferred)	30 days
	Refrigerated	7 days
	Ambient	48 hours

CPT Code Information:

87799

JCV 70475 JC Virus Detection by In Situ Hybridization

Specimen Requirements:

Specimen Type: Tissue

Supplies: Pathology Packaging Kit (T554)

Submit:

Formalin-fixed, paraffin-embedded tissue block (preferred)

OR

4 Unstained glass, positively-charged slides with 5 (+ or -1)-micron formalin-fixed, paraffin-embedded tissue

Additional Information: Broken or cracked unstained slides will be rejected.

Transport Temperature:

Specimen Type	Temperature	Time
Special	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88365-Primary

88364-If additional ISH

LCJC 800170 JC Virus, Molecular Detection, PCR, Spinal Fluid

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5mL (T914)

Container/Tube:

Preferred: Aliquot tube

Acceptable: Sterile screw cap vial

Specimen Volume: 0.5 mL

Collection Instructions: Do not centrifuge.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798

JCPCR
618305

JC Virus, Molecular Detection, PCR, Spinal Fluid

Specimen Requirements:

Collection Container/Tube:

Preferred: Vial number 2

Acceptable: Any vial number

Submission Container/Tube: Sterile screw cap vial

Specimen Volume: 0.5 mL

Collection Instructions: Do not centrifuge.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798

JO1
80179

Jo 1 Antibodies, IgG, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.35 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86235

JOHN
82900

Johnson Grass, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

JIP
621699

Joint Infection Panel, PCR, Synovial Fluid

Specimen Requirements:

Specimen Type: Synovial fluid

Container/Tube: Sterile vial

Specimen Volume: 1 mL

Collection Instructions:

1. Do not process or treat sample in any way.

2. Label specimen as synovial fluid.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Synovial Fluid	Refrigerated (preferred)	7 days
	Ambient	4 hours

CPT Code Information:

87627

JUNE
82893

June Grass, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FWJR 57953 Juniper Western (Juniperus occidentalis) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

KAIHC 70482 Kappa Light Chain Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

KAIPC
113330

KappaLambda IHC (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342

KBAL
623286

Karius Focus, Bronchoalveolar Lavage

Specimen Requirements:

Container/Tube: Sterile, leak-proof freezable container

Specimen Volume: 1.5 mL

Collection Instructions:

1. Collect bronchoalveolar lavage fluid (BALF) specimen according to your institutional guidelines.

2. Within 12 hours of specimen collection, transfer at least 1.5 mL of BALF specimen to a sterile container and freeze.

3. See Karius Focus: BAL Specimen Collection and Preparation Process for complete instructions.

Additional Information:

1. **Specimen cannot be shared with other tests**.

2. **Specimen trap collection containers (with suction catheters attached) will be rejected** due to the high risk of leakage and contamination upon opening. Avoid use of these for BALF specimens.

3. If specimen transfer into an acceptable sterile container is necessary, perform specimen transfer in a biosafety cabinet. Place the container in a separate sealed plastic bag.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Bronchoalveolar Lavage	Frozen	180 days

CPT Code Information:

87999

KDBAL
623288

Karius Focus, Direct Ship, Bronchoalveolar Lavage

Specimen Requirements:

Supplies: Karius KDBAL Kit (T1015)

Container/Tube: Streck CP tube

Specimen Volume: 1 mL

Collection Instructions:

1. Collect bronchoalveolar lavage fluid (BALF) specimen according to your institutional guidelines.

2. Within 12 hours of specimen collection, transfer at least 1 mL of BALF specimen to the Streck CP tube using disposable sterile pipette, taking care to avoid spilling storage reagent in Streck CP tube.

3. Screw the cap tightly and gently invert the tube 8 to 10 times.

4. **Do not freeze the Streck CP tube.**

5. See Shipping Instructions for how to package and ship the specimen to Karius Labs.

Additional Information:

1. **Specimen cannot be shared with other tests**.

2. **Specimen trap collection containers (with suction catheters attached) will be rejected** due to the high risk of leakage and contamination upon opening. Avoid use of these for BALF specimens.

Alternate Collection Instructions for when storage is needed before shipment

1. Collect BALF specimen according to your institutional guidelines.

2. Freeze BALF in the sterile collection tube. **Do not use the Streck CP tube**, as it must not be frozen.

3. When ready to prepare for shipment, thaw the frozen BALF specimen.

4. Label the Streck CP tube with the patient's first and last name, a unique patient identifier (eg, date of birth or medical record number), the collection date and time, **and the thawing date**.

5. Use a disposable pipette to carefully transfer 1 mL of BALF into the labeled Streck CP tube containing storage reagent. Avoid spilling the reagent.

6. Securely tighten the cap and gently invert the tube 8 to 10 times to mix.

7. **Do not freeze the Streck CP tube**.

8. See Shipping Instructions for how to package and ship the specimen to Karius Labs.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Bronchoalveolar Lavage	Ambient	7 days

CPT Code Information:

87999

KDPLA
623289

Karius Spectrum, Direct Ship, Plasma

Specimen Requirements:

Supplies: Karius KDPLA Kit (T1014)

Collection Container/Tube:

Preferred: Plasma Preparation Tube (PPT)

Acceptable: Lavender top (K2 EDTA)

Submission Container/Tube:

Preferred: PPT

Acceptable: Sterile polypropylene tube

Specimen Volume: 1 mL Plasma

Collection Instructions:

1. Gently invert tube 8 to 10 times to mix whole blood.

2. Centrifuge specimen as follows:

a. For PPT: Within 6 hours of collection, centrifuge at 1100 x *g* for 10 minutes.

b. For K2 EDTA:

i. Within 24 hours of collection, centrifuge at 1600 x *g* for 10 minutes. For tubes less than 4 mL, refer to tube manufacturer's instructions for centrifugation speed and time.

ii. Aliquot 1 mL of plasma into a sterile polypropylene tube, taking care not to disturb the buffy coat.

3. See Shipping Instructions for how to package and ship the specimen to Karius Labs.

Specimen Minimum Volume: Plasma: 0.7 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Ambient	4 days

CPT Code Information:

0152U

KPLA
623287

Karius Spectrum, Plasma

Specimen Requirements:

Collection Container/Tube:

Preferred: Plasma preparation tube (PPT)

Acceptable: Lavender top (K2 EDTA)

Submission Container/Tube:

Preferred: PPT

Acceptable: Sterile polypropylene tube

Specimen Volume: 1 mL Plasma

Collection Instructions :

1. Gently invert tube 8 to 10 times to mix whole blood.

2. Centrifuge specimen as follows:

a. For PPT: Within 6 hours of collection, centrifuge at 1100 x *g* for 10 minutes.

b. For K2 EDTA:

i. Within 24 hours of collection, centrifuge at 1600 x *g* for 10 minutes. For tubes less than 4 mL, refer to tube manufacturer's instructions for centrifugation speed and time.

ii. Aliquot 1 mL of plasma into a sterile polypropylene tube, taking care not to disturb the buffy coat.

3. For complete instructions see Karius Spectrum Specimen Collection and Preparation Process.

Specimen Minimum Volume: Plasma: 0.7 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Frozen (preferred)	180 days
	Ambient	4 days

CPT Code Information:

0152U

KLHCS 622117 Kelch-Like Protein 11 Antibody, Cell Binding Assay, Serum

Specimen Requirements:

Patient Preparation: For optimal antibody detection, specimen collection is recommended before starting immunosuppressant medication or intravenous immunoglobulin (IVIg) treatment.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

0432U

KLHCC 622118 Kelch-Like Protein 11 Antibody, Cell Binding Assay, Spinal Fluid

Specimen Requirements:

Patient Preparation: For optimal antibody detection, specimen collection is recommended before initiation of immunosuppressant medication, or corticosteroid or intravenous immunoglobulin (IVIg) treatment.

Container/Tube: Sterile vial

Preferred: Collection vial number 1

Acceptable: Any collection vial

Specimen Volume: 1 mL

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

0432U

KRT34 Keratin (34BE12) Immunostain, Technical Component Only

70492

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

KRTAE 70493 Keratin (AE1/AE3) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

KRTOS 70495 Keratin (OSCAR) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

KRT19 70490 Keratin 19 (KRT19) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

KRT20
70491

Keratin 20 (KRT20) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time

TECHONLY	Ambient (preferred)
	Refrigerated

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

KRT5 606999 Keratin 5 (KRT5) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

KRT7 70488 Keratin 7 (KRT7) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

K818 622524 Keratin 8 and 18 (KRT8/18), Technical Component only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

FKEMS
75730

Ketamine and Metabolite Screen, Plasma

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Specimen Type: Plasma

Collection Container/Tube: Lavender-top or pink top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL

Collection Instructions:

1. Draw blood in an EDTA (lavender top or pink top) tube(s). **Plasma gel tube is not acceptable.**

2. Centrifuge and send 3 mL of plasma refrigerated in a preservative-free plastic vial.

Specimen Minimum Volume: 0.9 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Refrigerated (preferred)	14 days

	Frozen	270 days
	Ambient	14 days

CPT Code Information:

80307-Screen

80357-Confirmation, if appropriate

KETGP 608024 Ketone Disorders Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a hematopoietic stem cell transplant.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight..

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Prenatal cultured amniocytes

This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 Flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81479

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

SKETC 606546 Ketones, Urine

Specimen Requirements:

Container/Tube: Plastic urine container

Specimen Volume: 20 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated	72 hours

CPT Code Information:

81003

KI67
70481

Ki-67 (MIB-1) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-Primary

88341-If additional IHC

KI67B
70588

Ki-67(MIB-1), Breast, Quantitative Immunohistochemistry, Automated

Specimen Requirements:

Supplies: Pathology Packaging Kit (T554)

Specimen Type: Preferred: Formalin-fixed, paraffin-embedded tissue block containing

invasive or metastatic breast carcinoma

Acceptable: 2 Unstained sections, containing invasive or metastatic breast carcinoma, on charged slides cut at 4 microns less than 1 month ago. Tissue on the slides should have been fixed in 10% neutral buffered formalin. Also send one hematoxylin and eosin-stained slide if possible.

Submission Container/Tube: Pathology Packaging Kit

Collection Instructions: Submit paraffin-embedded invasive or metastatic breast carcinoma tissue.

Additional Information: Paraffin block will be returned with the final report.

Transport Temperature:

Specimen Type	Temperature	Time
Special	Ambient (preferred)	
	Refrigered	

CPT Code Information:

88361

KIBM 71668 Ki-67(MIB-1), Breast, Semi-Quantitative Immunohistochemistry, Manual

Specimen Requirements:

This is not an orderable test. Order PATHC / Pathology Consultation. The consultant will determine the need for special stains.

Supplies: Pathology Packaging Kit (T554)

Specimen Type:

Preferred: Formalin-fixed, paraffin-embedded tissue block containing invasive or metastatic breast carcinoma

Acceptable: 2 unstained sections, containing invasive or metastatic breast carcinoma, on charged slides cut at 4 microns <1 month ago. Tissue on the slides should have been fixed in 10% neutral buffered formalin.

Submission Container/Tube: Pathology Packaging Kit (T554)

Collection Instructions: Submit paraffin-embedded invasive or metastatic breast carcinoma tissue.

Additional Information: Paraffin block will be returned with the final report.

Transport Temperature:

Specimen Type	Temperature	Time
Special	Ambient (preferred)	
	Refrigered	

CPT Code Information:

88360

KINET 71503 Ki-67(MIB-1), Gastrointestinal/Pancreatic Neuroendocrine Tumors, Quantitative Immunohistochemistry, Automated

Specimen Requirements:

Supplies: Pathology Packaging Kit (T554)

Specimen Type:

Preferred: Formalin-fixed, paraffin-embedded tissue block containing neuroendocrine tumor of the pancreas or gastrointestinal (GI) tract including metastases.

Acceptable: 2 unstained sections, containing neuroendocrine tumor of the pancreas or GI tract including metastases, on charged slides cut at 4 microns <1 month ago. Tissue on the slides should have been fixed in 10% neutral buffered formalin.

Submission Container/Tube: Pathology Packaging Kit

Collection Instructions: Submit formalin-fixed, paraffin-embedded tissue block.

Additional Information: Paraffin block will be returned with the final report.

Transport Temperature:

Specimen Type	Temperature	Time
Special	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88361

KINM 71667 Ki-67(MIB-1), Gastrointestinal/Pancreatic Neuroendocrine Tumors, Quantitative Immunohistochemistry, Manual

Specimen Requirements:

This is not an orderable test. Order PATHC / Pathology Consultation. The consultant will determine the need for special stains.

Supplies: Pathology Packaging Kit (T554)

Specimen Type:

Preferred: Formalin-fixed, paraffin-embedded tissue block containing neuroendocrine tumor of the pancreas or gastrointestinal (GI) tract including metastases.

Acceptable: 2 unstained sections, containing neuroendocrine tumor of the pancreas or GI tract including metastases, on charged slides cut at 4 microns <1 month ago. Tissue on the slides should have been fixed in 10% neutral buffered formalin.

Submission Container/Tube: Pathology Packaging Kit

Collection Instructions: Submit formalin-fixed, paraffin-embedded tissue block

Additional Information: Paraffin block will be returned with the final report.

Transport Temperature:

Specimen Type	Temperature	Time
Special	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88360

KI67P 72130 Ki-67(MIB-1), Pulmonary, Quantitative Immunohistochemistry, Automated

Specimen Requirements:

Supplies: Pathology Packaging Kit (T554)

Specimen Type:

Preferred: Formalin-fixed, paraffin-embedded tissue block containing carcinoid/atypical carcinoid tumor of the lung including metastases.

Acceptable: 2 Unstained sections containing carcinoid/atypical carcinoid tumor of the lung including metastases on charged slides cut at 4 microns <1 month ago. Tissue on the slides should have been fixed in 10% neutral buffered formalin.

Container/Tube: Pathology Packaging Kit

Collection Instructions: Submit formalin-fixed, paraffin-embedded tissue block.

Additional Information: Paraffin block will be returned with the final report.

Transport Temperature:

Specimen Type	Temperature	Time
Special	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88361

KIPM 72131 Ki-67(MIB-1), Pulmonary, Quantitative Immunohistochemistry, Manual (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see KI67P / Ki-67 (MIB-1), Pulmonary, Quantitative Immunohistochemistry, Automated.

Supplies: Pathology Packaging Kit (T554)

Specimen Type:

Preferred: Formalin-fixed, paraffin-embedded tissue block containing carcinoid/atypical carcinoid of the lung including metastases.

Acceptable: 2 Unstained sections on charged slides cut at 4 microns <1 month ago. Tissue on the slides should have been fixed in 10% neutral buffered formalin.

Container/Tube: Pathology Packaging Kit

Collection Instructions:

1. Submit formalin-fixed, paraffin-embedded tissue block.

2. Attach the green pathology address label included in the kit to the outside of the transport container.

Additional Information: Paraffin block will be returned with the final report.

Transport Temperature:

Specimen Type	Temperature	Time
Special	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88360

KIMEL
70483

Ki67 + Melan A Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88344-TC

KIDBN
82619

Kidney Bean (Red), IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

KIDST
605761

Kidney Stone Analysis

Specimen Requirements:

Supplies: Stone Analysis Collection Kit (T550)

Sources: Bladder, kidney, prostatic, renal, or urinary

Specimen Volume: Entire dried calculi specimen

Collection Instructions:

1. Have patient collect specimen using the Patient Collection Instructions for Kidney Stones.

2. For detailed instructions on kidney stone preparation and packaging, see Kidney Stone Packaging Instructions.

3. Prepare stone by cleaning any blood or foreign material from the stone with deionized water.

4. Place stone on a clean filter or paper towel and let dry at ambient temperature for a minimum of 24 hours.

5. **Do not place stone directly in a bag.** If specimen is received in a bag, either transfer stone into a screw-capped, plastic container or place bag containing stone in a screw-capped, plastic container.

6. Indicate source of specimen on the outside of the container (eg, left kidney, bladder, right ureter).

7. Repeat steps 2 through 5 for each stone received.

Specimen Stability Information: Ambient (preferred) 2 years/Refrigerated 1 year/Frozen 1 year

Specimen Minimum Volume: Entire stone

Transport Temperature:

Specimen Type	Temperature	Time
Stone	Ambient (preferred)	
	Frozen	365 days
	Refrigerated	365 days

CPT Code Information:

82365

KKBRP 65202 Kingella kingae, Molecular Detection, PCR, Blood

Specimen Requirements:

The high sensitivity of amplification by polymerase chain reaction requires the specimen to be processed in an environment in which contamination of the specimen by *Kingella kingae* DNA is unlikely.

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Royal blue top (EDTA), pink top (EDTA), or sterile vial containing EDTA-derived aliquot

Specimen Volume: 1 mL

Collection Instructions: Send specimen in original tube (preferred).

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798

KKRP 65201 Kingella kingae, Molecular Detection, PCR, Varies

Specimen Requirements:

The high sensitivity of amplification by polymerase chain reaction requires the specimen to be processed in an environment in which contamination of the specimen by *Kingella kingae* DNA is unlikely.

Submit only 1 of the following specimens:

Specimen Type: Synovial fluid

Preferred: Lavender top (EDTA)

Acceptable: Pink top (EDTA), royal blue top (EDTA), sterile vial containing EDTA-derived aliquot, red clot tube (no anticoagulant), or sterile container

Specimen Volume: 0.5 mL

Collection Instructions: Send specimen in original tube (preferred).

Specimen Stability Information: Refrigerated (preferred) <7 days /Frozen <7 days

Specimen Type: Fresh tissue or biopsy

Sources: Bone, joint, synovium, heart valve, aorta, or endocardium

Container/Tube: Sterile container

Specimen Volume: Entire collection or 5 mm(3)- approximately the size of a pencil eraser

Collection Instructions:

1. Collect fresh tissue specimen.

2. Submit tissue only, do not add fluid to tissue

3. Refrigerate or freeze specimen.

Specimen Stability Information: Refrigerated (preferred) <7 days/ Frozen <7 days

Preferred Paraffin-embedded tissue block:

Specimen Type: Formalin-fixed, paraffin-embedded tissue block (FFPE)

Sources: Bone, joint, synovium, heart valve, aorta, or endocardium

Supplies: Tissue Block Container (T553)

Container/Tube: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block to be cut and returned.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Acceptable Paraffin-embedded tissue block:

Specimen Type: Formalin-fixed, paraffin-embedded tissue block (FFPE)

Sources: Bone, joint, synovium, heart valve, aorta, or endocardium

Container/Tube: Sterile container for each individual cut section (scroll).

Collection Instructions: Perform microtomy and prepare five separate 10-micron sections. **Each section (scroll) must be placed in a separate sterile container for submission.**

Specimen Stability Information: Ambient (preferred)/Refrigerated

Specimen Minimum Volume: Fluid/fresh tissue or biopsy: See Specimen Required
Paraffin-embedded tissue block: Two 10-micron sections

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

87798

KIT
70485

KIT Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

KITQ
620901

KIT p.Asp816Val Variant Analysis, Quantitative, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix.

2. Send specimen in original tube. **Do not aliquot.**

3. Label specimen as blood.

Specimen Stability Information: Ambient (preferred) 14 days/Refrigerate 14 days

Specimen Type: Bone marrow aspirate

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix.

2. Send specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Stability Information: Ambient (preferred) 14 days/Refrigerate 14 days

Specimen Type: Extracted DNA

Container/Tube: 1.5- to 2-mL tube

Specimen Volume: Entire specimen

Collection Instructions:

1. Label specimen as extracted DNA

2. Provide indication of volume of DNA.

Specimen Stability Information: Frozen (preferred)/ Refrigerate/Ambient

Additional Information: We cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed, and a charge will be applied.

Specimen Type: Paraffin-embedded bone marrow aspirate clot

Container/Tube: Paraffin block

Specimen Stability Information: Ambient

Specimen Type: Tissue (FFPE)

Container/Tube: Paraffin block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block.

Specimen Stability Information: Ambient

Additional Information: Decalcified core biopsies are not accepted.

Specimen Minimum Volume: Blood, bone marrow: 1 mL; Extracted DNA: 50 mcL at 10 ng/mcL concentration; Paraffin block: 1 block

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81273

KIWI
82761

Kiwi Fruit, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

XYMF 35307 Known 45,X, Mosaicism Reflex Analysis, FISH, Whole Blood

Specimen Requirements:

Container/Tube:

Preferred: Green top (sodium heparin)

Acceptable: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 4 mL

Collection Instructions

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Other anticoagulants are not recommended and are harmful to the viability of the cells.

4. Cord blood is acceptable

Additional Information:

1. If the specimen does not grow in culture, you will be notified within 7 days of receipt.

2. Specimen cannot be frozen.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2, 88291-DNA probe, each (first probe set), Interpretation and report

88271x2-DNA probe, each; each additional probe set (if appropriate)

88271x1-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271x2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271x3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

GALC
622624

Krabbe Disease, GALC Gene Sequencing with Deletion/Duplication, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Tissue biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's

solution, or normal saline

Specimen Volume: 0.5 to 3 cm(3) or larger

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test

may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Prenatal cultured fibroblasts (eg, products of conception), amniocytes, or other confluent cultured cells. This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81406

88233-Tissue culture, skin or solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

KRASW 616452 KRAS Somatic Mutation Analysis, Peritoneal Fluid

Specimen Requirements:

Container/Tube: 50-mL Falcon tube

Preferred: Fresh, peritoneal washing; no fixatives added to wash

Specimen Volume: Two 50-mL Falcon tubes

Collection Instructions: Containers must be labeled with two unique patient identifiers.

Specimen Minimum Volume: 100 mL of peritoneal washing

Transport Temperature:

Specimen Type	Temperature	Time
Peritoneal	Refrigerated (preferred)	10 days
	Ambient	5 days

CPT Code Information:

81275-KRAS (v-Ki-ras2 Kirsten rat sarcoma viral oncogene) (eg, carcinoma) gene analysis, variants in codons 12 and 13

81276-KRAS (v-Ki-ras2 Kirsten rat sarcoma viral oncogene) (eg, carcinoma) gene analysis, additional variants

KRASP 610679 KRAS Somatic Mutation Analysis, Tumor

Specimen Requirements:

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens.

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides

: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears or ThinPrep)

Slides: 1 to 3 Slides

Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a total of 5000 nucleated cells (preferred), or at least 3000 nucleated cells (minimum).

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

81275-KRAS (v-Ki-ras2 Kirsten rat sarcoma viral oncogene) (eg, carcinoma) gene analysis, variants in codons 12 and 13

81276-KRAS additional variant(s)

88381-Microdissection, manual

LBCS 620895 Labile Bound Copper, Serum

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for at least 96 hours**.

Supplies:

-Metal Free Specimen Vial (T173)

-Metal Free B-D Tube (No Additive), 6 mL (T184)

Collection Container/Tube: 6-mL Plain, royal blue-top Vacutainer plastic trace element blood collection tube

Submission Container/Tube: 7-mL Metal-free, screw-capped, polypropylene vial

Specimen Volume: 0.75 mL serum

Collection Instructions:

1. For complete instructions, see Metals Analysis Specimen Collection and Transport.

2. Allow the specimen to clot for 30 minutes; then centrifuge the specimen to separate serum from the cellular fraction.

3. Remove the stopper. Carefully pour specimen into metal-free, polypropylene vial, avoiding transfer of the cellular components of blood. **Do not** insert a pipet into the serum to accomplish transfer, and **do not** ream the specimen with a wooden stick to assist with serum transfer.

4. Freeze sample on dry ice immediately after pouring off the serum.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum	Frozen	28 days	METAL FREE

CPT Code Information:

82525

LACO
62905

Lacosamide, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Collect specimen immediately before next scheduled dose.

2. For sustained-release formulations ONLY, collect specimen a minimum of 12 hours after last dose.

3. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80235

LDBF
606612

Lactate Dehydrogenase (LDH), Body Fluid

Specimen Requirements:

Specimen Type: Body fluid

Preferred Source:

-Peritoneal fluid (peritoneal, abdominal, ascites, paracentesis)

-Pleural fluid (pleural, chest, thoracentesis)

-Drain fluid (drainage, Jackson Pratt [JP] drain)

-Pericardial

-Synovial

-Cerebral spinal fluid

Acceptable Source: Write in source name with source location (if appropriate)

Collection Container/Tube: Sterile container

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Centrifuge to remove any cellular material and transfer into a plastic vial.

2. Indicate the specimen source and source location on label.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Body Fluid	Ambient (preferred)	7 days
	Refrigerated	48 hours

CPT Code Information:

83615

LD
8344

Lactate Dehydrogenase (LDH), Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Ambient (preferred)	7 days
	Frozen	30 days
	Refrigerated	48 hours

CPT Code Information:

83615

LAPYP
616609

Lactate Pyruvate Panel, Plasma

Specimen Requirements:

Collection Container/Tube:

Preferred: Green top (Sodium heparin)

Acceptable: Green top (Lithium heparin)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot plasma into a plastic vial.

Specimen Minimum Volume: 0.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Frozen (preferred)	90 days
	Ambient	7 days
	Refrigerated	7 days

CPT Code Information:

83605

84210

LACS1 Lactate, Plasma
601685

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Gray top (potassium oxalate/sodium fluoride)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.625 mL Plasma

Collection Instructions:

1. Draw at least 1 mL whole blood in a 2-mL collection tube or at least 2 mL whole blood in a 4-mL collection tube.

2. Centrifuge and aliquot plasma into a plastic vial.

Specimen Minimum Volume: Plasma: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma NaFl-KOx	Refrigerated (preferred)	14 days
	Ambient	8 hours

CPT Code Information:

83605

LASF1 Lactic Acid, Spinal Fluid
601821

Specimen Requirements:

Specimen Type: Spinal fluid

Container/Tube: Sterile container

Specimen Volume: 1 mL

Collection Instructions: Centrifuge to remove any cellular material.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Frozen (preferred)	60 days
	Refrigerated	24 hours
	Ambient	3 hours

CPT Code Information:

83605

LCTF
623285

Lactoferrin, Feces

Specimen Requirements:

Supplies: Stool container, Small (Random), 4 oz (T288)

Container/Tube: Stool container

Specimen Volume: 5 g

Collection Instructions:

1. Collect a fresh random fecal specimen, no preservatives.

2. If specimen is sent refrigerate, send immediately after collection.

3. If specimen cannot be sent immediately, freeze specimen and send frozen.

Additional Information:

1. Separate specimens must be submitted when multiple tests are ordered, with the exception of ELASF / Pancreatic Elastase, Feces and CALPR / Calprotectin, Feces. If only a single specimen is collected, it must be split prior to transport.

2. Testing cannot be added on to a previously collected specimen.

Specimen Minimum Volume: 1 g

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Frozen (preferred)	7 days
	Ambient	72 hours
	Refrigerated	72 hours

CPT Code Information:

83631

LACTO Lactotransferrin Immunostain, Technical Component Only
70625

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

LAMQ Lamb's Quarter, IgE, Serum
82682

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

LAMB
82699

Lamb, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

LAIHC
70499

Lambda Light Chain Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

LAMO
80999

Lamotrigine, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Collect blood immediately before next scheduled dose.

2. **Only** for sustained-release formulations, collect blood a minimum of 12 hours after last dose.

3. Within 2 hours of collection, centrifuge.

4. For red-top tubes, immediately aliquot serum into a plastic vial.

5. For serum gel tubes, **the serum must be aliquoted into a plastic vial within 24 hours of collection**.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80175

LANGR
70496

Langerin Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

LANG
82349

Langust (Lobster), IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

LATX
82787

Latex, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FLDLD
75160

LDL Cholesterol, Direct

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 1 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	365 days

CPT Code Information:

83721

PBUOE
608898

Lead Occupational Exposure, Random, Urine

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to interfere with most metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours.**

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine container with no metal cap or glued insert

Submission Container/Tube: Plastic vial or clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 3 mL

Collection Instructions:

1. Collect a random urine specimen.

2. See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

83655

82570

PBOU
608894

Lead Occupational Exposure, Random, Urine (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

PBUOE / Lead Occupational Exposure, Random, Urine

HMUOE / Heavy Metal Occupational Exposure, with Reflex, Random, Urine

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

83655

PBPFD
619722

Lead Profile Occupational Exposure, Blood

Specimen Requirements:

Both EDTA whole blood and heparin whole blood specimens are required.

Patient Preparation:

1. High concentrations of gadolinium and iodine are known to potentially interfere with most inductively couple plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

2. Patient should abstain from alcohol for 24 hours prior to specimen collection.

Specimen Type: EDTA Whole blood

Supplies:

-Metal Free B-D Tube (EDTA), 6 mL (T183)

-Metal Free EDTA 3 mL Tube (T989)

-Microtainer (EDTA) Tube, 0.5 mL (T174)

Container/Tube:

Preferred: Royal blue-top BD vacutainer with EDTA blood collection tube (3 mL) (BD catalog no. 36777) (T989)

Acceptable: Royal blue-top BD Vacutainer Plus with EDTA blood collection tube (6 mL) (BD catalog no. 368381) (T183) or BD Microtainer with EDTA (0.5 mL) (T174)

Specimen Volume: 2 mL

Collection Instructions:

1. See Metal Analysis Specimen Collection and Transport for complete instructions.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Refrigerate specimen as soon as possible after collection.

Specimen Type: Heparin Whole blood

Container/Tube:

Preferred: Green top (Sodium heparin)

Acceptable: Dark blue top (metal free heparin), green top (lithium heparin), or lavender top (EDTA)

Specimen Volume: 4 mL

Collection Instructions: Refrigerate specimen as soon as possible after collection.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated	7 days

CPT Code Information:

83655

82542

Lead, 24 Hour, Urine

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium or iodine containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine container with no metal cap or glued insert

Submission Container/Tube: Plastic, 10-mL urine tube or a clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 3 mL

Collection Instructions:

1. Collect urine for 24 hours.

2. Refrigerate specimen within 4 hours of completion of 24-hour collection.

3. See Metals Analysis Specimen Collection and Transport for complete instructions.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

83655

PBDC
113400

Lead, Capillary, with Demographics, Blood

Specimen Requirements:

*****This specimen container cannot be opened or used for any other testing before shipping.*****

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal testing. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

Supplies: Microtainer (EDTA) Tube, 0.5 mL (T174)

Collection Container/Tube: BD Microtainer MAP (microtube for automated process) with EDTA

Specimen Volume: 0.4 mL

Collection Instructions:

1. See Metals Analysis Specimen Collection and Transport for complete instructions.

2. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 0.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

83655

PBHA
8495

Lead, Hair

Specimen Requirements:

Supplies: Hair and Nails Collection Kit (T565)

Source: Head (preferred), beard, mustache, chest, pubic

Specimen Volume: 0.2 g

Collection Instructions: Prepare and transport specimen per the instructions in the kit or see Collecting Hair and Nails for Metals Testing.

Specimen Minimum Volume: 0.05 g

Transport Temperature:

Specimen Type	Temperature	Time

Hair	Ambient (preferred)
	Frozen
	Refrigerated

CPT Code Information:

83655

PBNA
89857

Lead, Nails

Specimen Requirements:

Supplies: Hair and Nails Collection Kit (T565)

Source: Fingernails or toenails

Specimen Volume: 0.2 g

Collection Instructions:

1. Prepare and transport specimen per the instructions in the kit or see Collecting Hair and Nails for Metals Testing.

2. Clippings should be taken from all 10 fingernails or toenails.

Specimen Minimum Volume: 0.05 g

Transport Temperature:

Specimen Type	Temperature	Time
Nail	Ambient (preferred)	
	Frozen	
	Refrigerated	

CPT Code Information:

83655

PBDV
113401

Lead, Venous, with Demographics, Blood

Specimen Requirements:

*****This specimen container cannot be opened or used for any other testing before shipping.*****

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based

metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

Supplies:

-Metal Free EDTA 3 mL Tube (T989)

-Metal Free B-D Tube (EDTA), 6 mL (T183)

Container/Tube:

Preferred: Royal blue-top BD vacutainer with EDTA blood collection tube (3 mL) (BD catalog no. 367777) (T989)

Acceptable: Royal blue-top BD Vacutainer Plus with EDTA blood collection tube (6 mL) (BD catalog no. 368381) (T183)

Specimen Volume: 2 mL

Collection Instructions:

1. See Metals Analysis Specimen Collection and Transport for complete instructions.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 0.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

83655

PBUCR 608908 Lead/Creatinine Ratio, Random, Urine

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to interfere with most metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours.**

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine container with no metal cap or glued insert

Submission Container/Tube: Plastic, 10-mL urine tube or clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 3 mL

Collection Instructions:

1. Collect a random urine specimen.

2. See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

83655

82570

PBCU
608904

Lead/Creatinine Ratio, Urine (Non-Orderable)

Specimen Requirements:

Only orderable as part of profile. For more information see:

PBUCR / Lead/Creatinine Ratio, Random, Urine

HMUCR / Heavy Metal/Creatinine Ratio, with Reflex, Random, Urine

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

83655

LEFLU 60292 Leflunomide Metabolite (Teriflunomide), Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL serum

Collection Instructions:

1. Draw blood no sooner than 12 hours (trough value) after last dose.

2. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Ambient (preferred)	28 days
	Frozen	28 days
	Refrigerated	28 days

CPT Code Information:

80193

LAGU 81268 Legionella Antigen, Random, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 0.5 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

3. Excessively bloody or very turbid specimens containing protein, cells, or particulates will be canceled. They can inhibit the function of the test.

4. Centrifuging to remove particulates is not approved.

5. Specimens with any dyes or unnatural color are not acceptable and will be canceled.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Frozen	14 days
	Ambient	24 hours

CPT Code Information:

87899

LEGI 8204 Legionella Culture, Varies

Specimen Requirements:

Specimen Type: Lower respiratory

Sources: Bronchoalveolar lavage, bronchial aspirate/brushing/lavage/washing, tracheal/endotracheal secretions/aspirate, sputum

Container/Tube: Sterile container

Specimen Volume: Entire collection

Specimen Type: Fresh tissue or biopsy

Sources: Lung, pleura, heart valve, pericardium

Container/Tube: Sterile container

Specimen Volume: 5 mm(3)

Collection Instructions: Aseptically collect a 1 to 2 cm(3) piece of tissue whenever possible

Specimen Type: Fluid

Sources: Pericardial, pleural, chest, chest tube drainage, thoracentesis, empyema

Container/Tube: Sterile container

Specimen Volume: 2 mL

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated	48 hours

CPT Code Information:

87081-*Legionella* culture

87077-Ident by MALDI-TOF mass spec (if appropriate)

87153-Aerobe ident by sequencing (if appropriate)

87176-Tissue processing (if appropriate)

SLEG 8122 Legionella pneumophila (Legionnaires Disease), Antibody, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86713

LEGRP 89564 Legionella species, Molecular Detection, PCR, Varies

Specimen Requirements:

The high sensitivity of amplification by polymerase chain reaction requires the specimen to be processed in an environment in which contamination of the specimen by *Legionella* DNA is unlikely.

Specimen Type: Lower respiratory

Sources: Bronchoalveolar lavage, bronchial aspirate/brushing/lavage/washing, tracheal/endotracheal secretions/aspirate, sputum

Container/Tube: Sterile container

Specimen Volume: 1 mL

Specimen Type: Fresh tissue or biopsy

Sources: Lung, pleura, heart valve, pericardium

Container/Tube: Sterile container

Specimen Volume: Entire collection or 5 mm(3) - approximately the size of a pencil eraser

Collection Instructions: Aseptically collect a 1 to 2 cm(3) piece of tissue whenever possible

Specimen type: Fluid

Sources: Pericardial, pleural, chest, chest tube drainage, thoracentesis, empyema

Container/Tube: Sterile container

Specimen Volume: 1 mL

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87801

LEIS
86219

Leishmaniasis (Visceral) Antibody, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.2 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86717

FLEMG
57643

Lemon IgG

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), a serum gel tube(s) is acceptable.

2. Centrifuge and aliquot 0.5 mL of serum in a plastic vial.

3 Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

LEM
82678

Lemon, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FLENG
57685

Lentil IgG

Specimen Requirements:

Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

LEN
82885

Lentil, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

LEPD
82849

Lepidoglyphus destructor, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FLEP
91339

Leptin

Specimen Requirements:

Patient Preparation:

Fasting: Recommended

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Within 45 minutes of collection, centrifuge and aliquot serum into a plastic vial.

2. Send frozen.

Specimen Minimum Volume: Serum: 0.3 mL NOTE: Minimum volume does not allow for repeat analysis.

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Frozen (preferred)	
	Refrigerated	14 days
	Ambient	8 days

CPT Code Information:

83520

LEPDT
65183

Leptospira, IgM, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.3 mL Serum

Collection Instructions: Centrifuge and aliquot serum into plastic vial.

Additional Information: If acute and convalescent specimens are being obtained to determine seroconversion, they should be collected 2 or more weeks apart.

Specimen Minimum Volume: Serum 0.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	30 days
	Frozen	30 days

CPT Code Information:

86720

FLETG
57639

Lettuce IgG

Specimen Requirements:

Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

LETT
82805

Lettuce, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

LLTOT 621505 Leukemia and Lymphoma Immunophenotyping, Technical Only, Tissue

Specimen Requirements:

Submit 1 of the following specimens:

Preferred

Specimen Type: Tissue

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with 15 mL of tissue culture medium (eg, Hank's balanced salt solution, RPMI, or equivalent)

Specimen Volume: 5 mm(3) or larger biopsy

Collection Instructions:

1. Place tissue into a sterile container with 15 mL of tissue culture medium (eg, Hank's balanced salt solution, RPMI, or equivalent).

2. Send intact specimen **(do not mince)**.

3. Specimen **cannot** be fixed.

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days

Acceptable

Specimen Type: Fine needle aspirate (FNA)

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with 15 mL of tissue culture medium (eg, Hank's balanced salt solution, RPMI, or equivalent)

Specimen Volume: Entire collection

Collection Instructions:

1. Collect FNA and transfer entire collection into a sterile container with 15 mL of tissue culture medium (eg, Hank's balanced salt solution, RPMI, or equivalent).

2. Send intact specimen **(do not mince)**.

3. Specimen **cannot** be fixed.

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days

Specimen Minimum Volume: Tissue: 1 mm(3); Fine needle aspirate: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Refrigerated (preferred)	
	Ambient	

CPT Code Information:

88184-Flow cytometry; first cell surface, cytoplasmic or nuclear marker x 1

88185-Flow cytometry; additional cell surface, cytoplasmic or nuclear marker (each)

Additional CPTs may be added if consultative help is needed with the case, or algorithm dictates Mayo consultant involvement.

88187-Flow Cytometry Interpretation, 2 to 8 Markers (if appropriate)

88188-Flow Cytometry Interpretation, 9 to 15 Markers (if appropriate)

88189-Flow Cytometry Interpretation, 16 or More Markers (if appropriate)

LLTOF 34488 Leukemia and Lymphoma Phenotyping, Technical Only, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Yellow top (ACD solution A or B)

Acceptable: Lavender top (EDTA) or green top (sodium heparin)

Specimen Volume: 10 mL

Collection Instructions:

1. Send whole blood specimen in original tube. **Do not aliquot.**

2. Label specimen as blood.

3. Include 1 to 2 unstained blood smears labeled with 2 unique identifiers.

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days

Specimen Type: Bone marrow

Container/Tube:

Preferred: Yellow top (ACD solution A or B)

Acceptable: Lavender top (EDTA) or green top (sodium heparin)

Specimen Volume: 1 to 5 mL

Collection Instructions:

1. Submission of bilateral specimens is not required.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

4. Include 1 to 2 unstained bone marrow smears labeled with 2 unique identifiers.

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days

Note: A fresh (less than 4 days post-collection), unfixed, nonembedded bone marrow core biopsy, bone or bone lesion is acceptable as an equivalent source for bone marrow aspirate for this test **only in the event of a dry tap** during the bone marrow harvesting

procedure. Indicate "dry tap" in performing lab notes or paperwork when submitting this specimen type.

Specimen Type: Body fluid

Sources: Serous effusions, pleural, pericardial, or abdominal (peritoneal fluid)

Container/Tube: Body fluid container

Specimen Volume: 20 mL

Collection Instructions:

1. If possible, body fluids other than spinal fluid should be anticoagulated with heparin (1 U/mL of fluid).

2. Label specimen with body fluid type.

Additional Information: The volume of serous effusion necessary to phenotype lymphocytes or blasts depends upon the cell count in the specimen. Usually, 20 mL of pleural or peritoneal fluid is sufficient. Smaller volumes can be used if there is a high cell count.

Specimen Stability Information: Refrigerated 4 days/Ambient 4 days

Specimen Type: Spinal fluid

Container/Tube: Sterile vial

Specimen Volume: 1 to 1.5 mL

Collection Instructions:

1. An original cytospin preparation (preferably unstained) should be included with the spinal fluid specimen so correlative morphologic evaluation can occur.

2. Label specimen as spinal fluid.

Specimen Stability Information: Refrigerated /Ambient 4 days

Additional Information: The volume of spinal fluid necessary to phenotype the lymphocytes or blasts depends upon the cell count in the specimen. A cell count should be determined and submitted with the specimen. Usually, 1 to 1.5 mL of spinal fluid is sufficient. Smaller volumes can be used if there is a high cell count. If the cell count is less than 10 cells/mcL, a larger volume of spinal fluid may be required. When cell counts drop below 5 cells/mcL, the immunophenotypic analysis may not be successful.

Specimen Minimum Volume: Whole blood: 3 mL; Bone marrow: 0.5 mL; Spinal fluid: 1 mL; Fluid from serous effusions: 5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

88184-Flow cytometry; first cell surface, cytoplasmic or nuclear marker

88185-Flow cytometry; additional cell surface, cytoplasmic or nuclear marker (each)

Additional CPTs may be added if consultative help is needed with the case, or algorithm dictates Mayo consultant involvement.

88187-Flow cytometry interpretation, 2 to 8 markers (if appropriate)

88188-Flow cytometry interpretation, 9 to 15 markers (if appropriate)

88189-Flow cytometry interpretation, 16 or more markers (if appropriate)

LLPB 800433 Leukemia/Lymphoma Immunophenotyping, Flow Cytometry, Blood

Specimen Requirements:

Specimen Type: Whole blood

Container/Tube:

Preferred: Yellow top (ACD solution A or B)

Acceptable: Lavender top (EDTA)

Specimen Volume: 6 mL

Slides: If possible, include 5 to 10 unstained blood smears labeled with **two unique identifiers**

Collection Instructions:

1. Send whole blood specimen in original tube. **Do not aliquot**.

2. Label specimen as blood.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Ambient	4 days

CPT Code Information:

88184-Flow cytometry; first cell surface, cytoplasmic or nuclear marker x 1

88185-Flow cytometry; additional cell surface, cytoplasmic or nuclear marker (each)

88187-Flow Cytometry Interpretation, 2 to 8 Markers (if appropriate)

88188-Flow Cytometry Interpretation, 9 to 15 Markers (if appropriate)

88189-Flow Cytometry Interpretation, 16 or More Markers (if appropriate)

LLPT 19499 Leukemia/Lymphoma Immunophenotyping, Flow Cytometry, Tissue

Specimen Requirements:

Submit 1 of the following specimens:

Preferred

Specimen Type: Tissue

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with 15 mL of tissue culture medium (eg, Hank's balanced salt solution, RPMI, or equivalent)

Specimen Volume: 5 mm(3) or larger biopsy

Collection Instructions:

1. Place tissue into a sterile container with 15 mL of tissue culture medium (eg, Hank's balanced salt solution, RPMI, or equivalent).

2. Send intact specimen (**do not mince**)

3. Specimen **cannot** be fixed.

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days

Acceptable

Specimen Type: Fine needle aspirate (FNA)

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with 15 mL of tissue culture medium (eg, Hank's balanced salt solution, RPMI, or equivalent)

Specimen Volume: Entire collection

Collection Instructions:

1. Collect FNA and transfer entire collection into a sterile container with 15 mL of tissue culture medium (eg, Hank's balanced salt solution, RPMI, or equivalent).

2. Send intact specimen (**do not mince**)

3. Specimen **cannot** be fixed.

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days

Specimen Minimum Volume: 1 mm(3)

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Refrigerated (preferred)	
	Ambient	

CPT Code Information:

88184-Flow cytometry; first cell surface, cytoplasmic or nuclear marker x 1

88185-Flow cytometry; additional cell surface, cytoplasmic or nuclear marker (each)

88187-Flow Cytometry Interpretation, 2 to 8 Markers (if appropriate)

88188-Flow Cytometry Interpretation, 9 to 15 Markers (if appropriate)

88189-Flow Cytometry Interpretation, 16 or More Markers (if appropriate)

LCMS 3287 Leukemia/Lymphoma Immunophenotyping, Flow Cytometry, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Yellow top (ACD solution A or B)

Acceptable: Lavender top (EDTA) or green top (sodium heparin)

Specimen Volume: 6 mL

Collection Instructions:

1. Send whole blood specimen in original tube. **Do not aliquot**.

2. Label specimen as blood.

3. Include 1 to 2 unstained blood smears labeled with 2 unique identifiers.

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days

Specimen Type: Bone marrow

Container/Tube:

Preferred: Yellow top (ACD solution A or B)

Acceptable: Lavender top (EDTA) or green top (sodium heparin)

Specimen Volume: 1 to 5 mL

Collection Instructions:

1. Submission of bilateral specimens is not required.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

4. Include 1 to 2 unstained bone marrow smears labeled with 2 unique identifiers.

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days

Note: A fresh (less than 4 days post-collection), unfixed, nonembedded bone marrow core biopsy, bone or bone lesion is acceptable as an equivalent source for bone marrow aspirate for this test **only in the event of a dry tap** during the bone marrow harvesting procedure. Indicate "dry tap" in performing lab notes or paperwork when submitting this specimen type.

Specimen Type: Body fluid

Sources: Serous effusions, pleural fluid, pericardial fluid, abdominal (peritoneal) fluid,

bronchoalveolar lavage (BAL)

Container/Tube: Body fluid container

Specimen Volume: 20 mL

Collection Instructions:

1. If possible, body fluids other than spinal fluid should be anticoagulated with heparin (1 U/mL of fluid).

2. Label specimen with body fluid type.

Specimen Stability Information: Refrigerated 4 days/Ambient 4 days

Additional Information: The volume of serous effusions necessary to phenotype lymphocytes or blasts depends upon the cell count in the specimen. Usually, 20 mL of pleural or peritoneal fluid is sufficient. Smaller volumes can be used if there is a high cell count.

Specimen Type: Spinal fluid

Container/Tube: Sterile vial

Specimen Volume: 1 to 1.5 mL

Collection Instructions:

1. An original cytospin preparation (preferably unstained) should be included with the spinal fluid specimen so correlative morphologic evaluation can occur.

2. Label specimen as spinal fluid.

Specimen Stability Information: Refrigerated 4 days/Ambient 4 days

Additional Information: The volume of spinal fluid necessary to phenotype lymphocytes or blasts depends upon the cell count in the specimen. A cell count should be determined and submitted with the specimen. Usually, 1 to 1.5 mL of spinal fluid is sufficient. Smaller volumes can be used if there is a high cell count. If cell count is less than 10 cells/mcL, a larger volume of spinal fluid may be required. When cell counts drop below 5 cells/mcL, the immunophenotypic analysis may not be successful.

Specimen Minimum Volume: Blood: 3 mL; Bone marrow: 0.5mL; Spinal fluid: 1 mL; Fluid from serous effusions: 5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

88184-Flow cytometry; first cell surface, cytoplasmic or nuclear marker x 1

88185-Flow cytometry; additional cell surface, cytoplasmic or nuclear marker (each)

88187-Flow Cytometry Interpretation, 2 to 8 Markers (if appropriate)

88188-Flow Cytometry Interpretation, 9 to 15 Markers (if appropriate)

88189-Flow Cytometry Interpretation, 16 or More Markers (if appropriate)

LAD1
81155

Leukocyte Adhesion Deficiency Type 1, CD11a/CD18 and CD11b/CD18 Complex Immunophenotyping, Blood

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 5 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Additional Information: For serial monitoring, it is recommended that specimens are collected at the same time of day.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole Blood EDTA	Ambient	72 hours	PURPLE OR PINK TOP/EDTA

CPT Code Information:

86356 x 3

LECT2
70497

Leukocyte Cell-Derived Chemotaxin 2 (LECT2), Immunostains Without Interpretation

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

TLTE4
606355

Leukotriene E4, 24 Hour, Urine

Specimen Requirements:

Patient Preparation: Patients taking 5-lipoxygenase inhibitor zileuton (Zyflo) may have decreased concentrations of leukotriene E4. If medically feasible, the patient **should not** take zileuton for 48 hours before specimen collection.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube: Plastic vial

Specimen Volume: 5 mL

Collection Instructions:

1. Start collection within a few hours of symptom onset and collect urine for 24 hours

2. No preservative preferred.

3. Aliquot urine into a plastic vial and send frozen.

Additional Information: See Urine Preservatives-Collection and Transportation for

24-Hour Urine Specimens for multiple collection.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen (preferred)	28 days
	Refrigerated	72 hours
	Ambient	8 hours

CPT Code Information:

82542

RLTE4 Leukotriene E4, Random, Urine
606354

Specimen Requirements:

Patient Preparation: Patients taking 5-lipoxygenase inhibitor zileuton (Zyflo) may have decreased concentrations of leukotriene E4. If medically feasible, the patient **should not** take zileuton for 48 hours before specimen collection.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube: Plastic vial

Specimen Volume: 5 mL

Collection Instructions:

1. Within a few hours of symptom onset, collect a random urine specimen.

2. No preservative.

3. Aliquot urine into a plastic vial and send frozen.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen (preferred)	28 days
	Refrigerated	72 hours
	Ambient	8 hours

CPT Code Information:

82542

82570

FLEVA
75401

Levamisole, Urine

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Collection Container: Plastic, preservative-free urine container

Specimen Volume: 1 mL

Collection Instructions:

1. Collect 1 mL random urine specimen in a plastic preservative-free urine container.

2. Ship refrigerate.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	30 days
	Ambient	30 days
	Frozen	30 days

CPT Code Information:

80375

LEV1P
113309

Level 1 Gross only (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time

Varies	Ambient (preferred)
	Refrigerated

CPT Code Information:

88300

LEV2P 113310 Level 2 Gross and microscopic (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88302

LEV3P 113311 Level 3 Gross and microscopic (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88304

LEV4P 113312

Level 4 Gross and microscopic (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88305

LV4RP
113313

Level 4 Gross and Microscopic, RB (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88305

LEV5P
113314

Level 5 Gross and microscopic (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88307

LEV6P
113315

Level 6 Gross and microscopic (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	
	Ambient	

CPT Code Information:

88309

LEVE
83140

Levetiracetam, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Draw blood immediately before next scheduled dose.

2. **For sustained-release formulations only**, draw blood a minimum of 12 hours after last dose.

3. Within 2 hours of collection, centrifuge, and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80177

LID
8382

Lidocaine, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	14 days
	Ambient	14 days
	Frozen	14 days

CPT Code Information:

80176

LMO2
70501

LIM Domain Only 2 (LMO2) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

LIME
82360

Lime, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

ALBLI 603181 Limited Bleeding Diathesis Profile Interpretation (Non-Orderable)

Specimen Requirements:

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85390-26 Special Coagulation Interpretation

LDPU 615289 Limited Drug Profile, 3 Drug Classes, Immunoassay, Random, Urine (Non-Orderable)

Specimen Requirements:

Only orderable as part of profile. For more information see CSMHU / Controlled Substance Monitoring Hybrid Drug Profile, 20 Drug Classes, High-Resolution Mass Spectrometry and Immunoassay Screen, Random, Urine.

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: Plastic, 10 mL tube

Specimen Volume: 5 mL

Collection Instructions:

1. Collect a random urine specimen.

2. Submit 5 mL in 10 plastic tube.

3. No preservative

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	14 days
	Ambient	72 hours

CPT Code Information:

80307

FLLYS 75928 Limulus Amebocyte Lysate, Endotoxin

Specimen Requirements:

Specimen Type: Aqueous solutions

Container/Tube: Nonpyrogenic, plastic, leak-proof container

Specimen Volume: 5 mL

Collection Instructions:

1.Collect 5 mL aqueous solutions in a nonpyrogenic, plastic, leak-proof container used in patient management.

2. Send frozen.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time

Varies	Frozen	30 days

CPT Code Information:

87999

LIND
82862

Linden, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

LINS
86311

Linseed, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

LPSBF
606615

Lipase, Body Fluid

Specimen Requirements:

Specimen Type: Body fluid

Preferred Sources:

-Peritoneal fluid (peritoneal, abdominal, ascites, paracentesis)

-Pleural fluid (pleural, chest, thoracentesis)

-Drain fluid (drainage, Jackson Pratt [JP] drain)

-Pericardial

Acceptable Source: Write in source name with source location (if appropriate)

Collection Container/Tube: Sterile container

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Centrifuge to remove any cellular material and transfer into a plastic vial.

2. Indicate the specimen source and source location on label.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Body Fluid	Refrigerated (preferred)	7 days
	Frozen	30 days
	Ambient	24 hours

CPT Code Information:

83690

FLIPR
90347

Lipase, Random Urine

Specimen Requirements:

Container/Tube: Plastic urine container

Specimen Volume: 2 mL

Collection Instructions:

1. Collect random urine without preservative.
2. Ship 2 mL urine in a sterile screw capped plastic container ship ambient.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Ambient (preferred)	5 days
	Frozen	21 days
	Refrigerated	5 days

CPT Code Information:

83690

LPS
8328

Lipase, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Serum gel tube must be centrifuged within 2 hours of collection.

2. Red-top tube must be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

83690

BFLA1 Lipid Analysis, Body Fluid

614164

Specimen Requirements:

Specimen Type: Body fluid

Preferred Sources:

-Peritoneal fluid (peritoneal, abdominal, ascites, paracentesis)

-Pleural fluid (pleural, chest, thoracentesis)

-Drain fluid (drainage, JP drain)

-Pericardial

Acceptable Source:

Collection Container/Tube: Sterile container, no additive

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL

Collection Instructions:

1. Centrifuge to remove any cellular material and transfer into a plastic vial.

2. Indicate the specimen source and source location on label.

Specimen Minimum Volume: 2.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Body Fluid	Frozen (preferred)	30 days
	Refrigerated	7 days
	Ambient	24 hours

CPT Code Information:

82664-Electrophoretic technique, not elsewhere specified (Chylomicrons and lipoproteins)

84311-Spectrophotometry, analyte not specified (Cholesterol)

84478-Triglycerides

LPSC1 616696 Lipid Panel, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions:

1. Serum gel tube must be centrifuged within 2 hours of collection.

2. Red-top tube must be centrifuged and the serum aliquoted into a plastic vial within 2 hours of collection.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days

	Frozen	30 days

CPT Code Information:

80061-Lipid panel (if all 3 performed)

82465-Cholesterol, total (if all 3 are not performed)

84478-Triglycerides (if all 3 are not performed)

83718-Cholesterol, HDL (if all 3 are not performed)

LIPOG 617337 Lipodystrophy Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81406 x2

81408

81479

81479 (if appropriate for government payers)

LPALD 610738 Lipoprotein (a) and Low-Density Lipoprotein Cholesterol, Serum

Specimen Requirements:

Patient Preparation:

1. **Fasting: 8 hours, required**

2. Patient must abstain from alcohol for 24 hours before collection.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 4 mL serum

Collection Instructions:

1. Centrifuge and aliquot serum into a plastic vial.

2. Send refrigerated.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	60 days

CPT Code Information:

83700

83701

LMPP
83673

Lipoprotein Metabolism Profile, Serum

Specimen Requirements:

Patient Preparation:

1**. Fasting: 12 hours, required**

2. Patient **must not** consume any alcohol for 24 hours before the specimen is collected.

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	60 days

CPT Code Information:

80061-Lipid panel (includes: HDL [CPT Code 83718], total cholesterol [CPT Code 82465], and triglycerides [CPT Code 84478] if all performed)

82172-Apolipoprotein B

83700-Lp(a) cholesterol electrophoresis

LIPA1 Lipoprotein(a), Serum

624437

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	20 days
	Frozen	91 days
	Ambient	5 days

CPT Code Information:

83695

LUCSF LiquidHALLMARK CSF

622814

Specimen Requirements:

Supplies: Sterile Specimen Tube, 6 mL (T485)

Container/Tube: Sterile tube

Specimen Volume: 5 mL

Collection Instructions:

1. Perform lumbar puncture and discard the first 1 mL to 2 mL of cerebrospinal fluid (CSF).

2. Collect CSF directly into a sterile tube.

3. Inspect specimen for visible discoloration. **Specimen must be clear and colorless to perform testing.**

4. Freeze sample upright prior to placing in transport container.

5. Send frozen.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Frozen	30 days

CPT Code Information:

81455

LUCHM 622811 LiquidHALLMARK ctDNA and ctRNA

Specimen Requirements:

Specimen Type: Whole blood

Supplies: Streck Tan Top Tube Kit (T715)

Container/Tube: Two 10-mL Streck cell-free DNA (cfDNA) blood collection tubes

Specimen Volume: 20 mL; 10 mL in two Streck tubes

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not centrifuge or aliquot.**

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
WB Streck	Ambient	4 days	Streck Black/Tan top

CPT Code Information:

0571U

FLISD
75638

Lisdexamfetamine as Metabolite, Urine

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Container/Tube: Plastic preservative-free urine container

Specimen Volume: 1 mL

Collection Instructions:

1. Send 1 mL random urine specimen in a plastic preservative-free urine container.

2. Ship refrigerate.

Specimen Minimum Volume: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	180 days
	Frozen	180 days
	Ambient	7 days

CPT Code Information:

80324

LITH
37046

Lithium, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood 8 to 12 hours after last dose (trough specimen).

2. Serum gel tubes should be centrifuged within 2 hours of collection.

3. Red-top tubes should be centrifuged, and the serum aliquoted into a plastic vial within 2 hours of collection.

Additional Information: Peak serum concentrations do not correlate with symptoms.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

80178

LFABP
70429

Liver Fatty Acid-Binding Protein (L-FABP) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

LIVPR
113633

Liver Profile, Serum

Specimen Requirements:

Supplies: Amber Frosted Tube, 5 mL (T915)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Amber vial

Specimen Volume: 0.6 mL

Collection Instructions:

1. Serum gel tubes should be centrifuged within 2 hours of collection.

2. Red-top tubes should be centrifuged, and the serum aliquoted into an amber vial within 2 hours of collection.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Refrigerated	7 days

CPT Code Information:

82247

82248

84450

84460

84075

82040

84155

80076 (if appropriate if all analytes performed)

LKM
80387

Liver/Kidney Microsome Type 1 Antibodies, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86376

LOB
82744

Lobster, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

LQTSG
617351

Long QT Syndrome Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from blood spots, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

2. Due to lower concentration of DNA yielded from blood spot, it is possible that additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81403

81406 x2

81407

81479

81479 (if appropriate for government payers)

LORAZ
80459

Lorazepam (Ativan), Serum

Specimen Requirements:

Submit only 1 of the following specimens:

Plasma:

Specimen Type: Plasma

Collection Container/Tube: Green top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL

Collection Instructions: Draw blood in a green-top sodium heparin tube(s), **plasma gel tube is not acceptable.** Centrifuge and send 2 mL of plasma refrigerated in a plastic vial.

Serum:

Specimen Type: Serum

Collection Container/Tube: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL

Collection Instructions: Draw blood in a plain, red-top tube(s), **serum gel tube is not acceptable.** Centrifuge and send 2 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	14 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80346

G0480 (if appropriate)

LDLD 606898 Low-Density Lipoprotein (LDL) Cholesterol, Beta-Quantification, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL

Collection Instructions: Centrifuge and aliquot serum into plastic vial. Send refrigerated.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	10 days
	Frozen	60 days

CPT Code Information:

83701

FUSF 35304 Low-Grade Fibromyxoid Sarcoma (LGFMS), 16p11.2 (FUS or TLS) Rearrangement, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Tissue

Preferred: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded (FFPE) tumor tissue block. Blocks prepared with alternative fixation methods may be acceptable; provide fixation method used.

Acceptable: Slides

Collection Instructions: Four consecutive, unstained, 5 micron-thick sections placed on positively charged slides, and 1 hematoxylin and eosin-stained slide.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Two consecutive, unstained, 5 micron-thick sections placed on positively charged slides, and 1 hematoxylin and eosin-stained slide.

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2, 88291-DNA probe, each (first probe set), Interpretation and report

88271x2-DNA probe, each; each additional probe set (if appropriate)

88271x1-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271x2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271x3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

FLXPN Loxapine, Plasma

75950

Specimen Requirements:

Must submit one specimen per order. Specimens cannot be shared between multiple orders.

Collection Container/Tube: Lavender or pink top (EDTA)

Submission Container/Tube: Plastic screw-capped vial

Specimen Volume: 1 mL plasma

Collection Instructions:

1. Promptly centrifuge and aliquot 1 mL plasma into a plastic screw-capped vial.

2. Send refrigerated.

Specimen Stability Information: Refrigerated (preferred) 30 days/Ambient 30 days/Frozen 15 months

Specimen Minimum Volume: Plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Refrigerated (preferred)	30 days
	Ambient	30 days
	Frozen	

CPT Code Information:

80342

FLSDA
75680

LSD Trace Analysis, Urine

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Specimen Type: Urine

Container/Tube: Plastic preservative-free urine container

Specimen Volume: 1 mL

Collection Instructions: Collect 1 mL random urine without preservative. Send specimen refrigerated in a plastic, preservative-free urine container.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	30 days
	Frozen	30 days

CPT Code Information:

80323

LCAF
35282

Lung Cancer, ALK (2p23) Rearrangement, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue block (fresh tissue is **not acceptable**)

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block.

2. Blocks prepared with alternative fixation methods (eg, Prefer, Bouin's) will be attempted but are less favorable for successful results. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Decalcified paraffin-embedded specimens will have testing attempted; however, the success rate is approximately 50%. **Testing may be canceled** if sufficient tumor tissue is not present.

3. **Submitted fresh tissue specimens will be canceled upon receipt**. If only fresh tissue is available, embed in paraffin prior to sending.

Acceptable:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained and 4 unstained

Collection Instructions: Submit 1 slide stained with hematoxylin and eosin and 4 consecutive unstained, positively charged, unbaked slides with 5-micron-thick sections of the tumor tissue.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Slides: 1 Hematoxylin and eosin-stained and 2 unstained

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	

	Refrigerated

CPT Code Information:

88271x2, 88291-DNA probe, each (first probe set), Interpretation and report

88271x2-DNA probe, each; each additional probe set (if appropriate)

88271x1-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271x2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271x3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

RETF 35846 Lung Cancer, RET (10q11) Rearrangement, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue block (fresh tissue is **not acceptable**)

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block.

2. Blocks prepared with alternative fixation methods (eg, Prefer, Bouin's) will be attempted but are less favorable for successful results. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Decalcified paraffin-embedded specimens will have testing attempted; however, the success rate is approximately 50%. **Testing may be canceled** if sufficient tumor tissue is not present.

3. **Submitted fresh tissue specimens will be canceled upon receipt**. If only fresh tissue is available, embed in paraffin prior to sending.

Acceptable:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained and 4 unstained

Collection Instructions: Submit 1 slide stained with hematoxylin and eosin and 4 consecutive unstained, positively charged, unbaked slides with 5 micron-thick sections of the tumor tissue.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Slides: 1 Hematoxylin and eosin-stained and 2 unstained

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2, 88291-DNA probe, each (first probe set), Interpretation and report

88271x2-DNA probe, each; each additional probe set (if appropriate)

88271x1-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271x2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271x3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

ROS1F 35845 Lung Cancer, ROS1 (6q22) Rearrangement, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue block (fresh tissue is **not acceptable**)

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block.

2. Blocks prepared with alternative fixation methods (eg, Prefer, Bouin's) will be attempted but are less favorable for successful results. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Decalcified paraffin-embedded specimens will have testing attempted; however, the success rate is approximately 50%. **Testing may be canceled** if sufficient tumor tissue is not present.

3. **Submitted fresh tissue specimens will be canceled upon receipt**. If only fresh tissue is available, embed in paraffin prior to sending.

Acceptable:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained and 4 unstained

Collection Instructions: Submit 1 slide stained with hematoxylin and eosin and 4 consecutive unstained, positively charged, unbaked slides with 5-micron thick sections of the tumor tissue.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Slides: 1 Hematoxylin and eosin-stained and 2 unstained

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2, 88291-DNA probe, each (first probe set), Interpretation and report

88271x2-DNA probe, each; each additional probe set (if appropriate)

88271x1-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271x2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271x3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

LUPN
82613

Lupin, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FLUPV
91714

Lupus Anticoagulant Evaluation with Reflex

Specimen Requirements:

Collection Container/Tube: Light-blue top (3.2% sodium citrate)

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL platelet-poor plasma, in plastic vial

Collection Instructions: Draw blood into a 3.2% sodium citrate (light blue-top) tube.

Centrifuge light blue-top tube 15 minutes at approx. 1500 g within 60 minutes of collection. Using a plastic pipette, remove plasma, taking care to avoid the WBC/platelet buffy layer and place into a plastic vial. Centrifuge a second time and transfer platelet-poor plasma into a new plastic vial. Plasma must be free of platelets (<10,000/mcl).

Freeze immediately and ship on dry ice.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	30 days

CPT Code Information:

85613/ Russell viper venom time (includes venom); diluted

85730/ Thromboplastin time, partial (PTT); plasma

85598/ Platelet neutralization (Hexagonal Phase Confirm)-(if appropriate)

85597/ Platelet neutralization (dRVVT Confirm)-(if appropriate)

85613/dRVVT 1:1 Dilution (if appropriate)

85670/Thrombin Clotting Time (if appropriate)

ALUPO 603465 Lupus Anticoagulant Profile Interpretation (Non-Orderable)

Specimen Requirements:

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85390-26 Special Coagulation Interpretation

ALUPP 603463 Lupus Anticoagulant Profile, Plasma

Specimen Requirements:

Patient Preparation:

1. If possible, patient should not be receiving anticoagulant treatment (eg, warfarin, heparin): If not possible for medical reasons, note on request.

a. If medically feasible, for 4 to 6 hours before specimen collection, do not administer intravenous heparin.

b. If medically feasible, for 10 to 14 days before specimen collection, do not administer subcutaneous heparin or Coumadin (warfarin).

2. Patient should not be receiving fibrinolytic agents (streptokinase, urokinase, tissue plasminogen activator: tPA).

3. If patient has been recently transfused, it is best to perform this study pretransfusion, if possible.

Specimen Type: Platelet-poor plasma

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Light-blue top (citrate)

Submission Container/Tube: 13-mm x 75-mm, 5-mL, plastic vial (polypropylene preferred)

Specimen Volume: 4 mL in 4 plastic vials each containing 1 mL

Collection Instructions:

1. Specimen must be collected prior to factor replacement therapy.

2. For complete instructions, see Coagulation Guidelines for Specimen Handling and Processing.

3. Centrifuge, transfer all plasma into a plastic vial, and centrifuge plasma again.

4. Aliquot plasma (1-2 mL per aliquot) into 4 separate plastic vials, leaving 0.25 mL in the bottom of centrifuged vial.

5. Freeze plasma immediately (no longer than 4 hours after collection) at -20 degrees C or, ideally, -40 degrees C or below.

Additional Information:

1. Double-centrifuged specimen is critical for accurate results as platelet contamination may cause spurious results.

2. Each coagulation assay requested should have its own vial.

Specimen Minimum Volume: 3 mL in 3 plastic vials each containing 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85610

85730

85613

85390

85130 (if appropriate)

85130 (if appropriate)

85210 (if appropriate)

85220 (if appropriate)

85230 (if appropriate)

85240 (if appropriate)

85245 (if appropriate)

85246 (if appropriate)

85247 (if appropriate)

85250 (if appropriate)

85260 (if appropriate)

85270 (if appropriate)

85280 (if appropriate)

85335 (if appropriate)

85335 (if appropriate)

85335 (if appropriate)

85366 (if appropriate)

85379 (if appropriate)

85384 (if appropriate)

85385 (if appropriate)

85390-26 (if appropriate)

85397 (if appropriate)

85597 (if appropriate)

85598 (if appropriate)

85611 (if appropriate)

85613 (if appropriate)

85613 (if appropriate)

85635 (if appropriate)

85670 (if appropriate)

85732 (if appropriate)

ALUPI 603464 Lupus Anticoagulant Profile, Technical Interpretation (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see ALUPP / Lupus Anticoagulant Profile, Plasma.

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Cit	Frozen	14 days

CPT Code Information:

85390

LUTHI 70498 Luteinizing Hormone (LH) Beta Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

LHPED 62999 Luteinizing Hormone (LH), Pediatrics, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube. 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.25 mL

Collection Instructions:

1. Within 2 hours of collection, centrifuge the specimen.

2. For red top tubes, aliquot serum into a plastic vial immediately.

3. For serum gel tubes, serum may sit on gel refrigerated but must be aliquoted into a plastic vial within 4 hours.

Specimen Minimum Volume: 0.13 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

83002

LH
602752

Luteinizing Hormone (LH), Serum

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection, patient **should not** take multivitamins or dietary supplements (eg, hair, skin, and nail supplements) containing biotin (vitamin B7).

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days

	Frozen	180 days
	Ambient	24 hours

CPT Code Information:

83002

SLYME 602732 Lyme Antibody Modified 2-Tier with Reflex, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	10 days
	Frozen	30 days

CPT Code Information:

86618

86617 x2 (if appropriate)

LNBAB 63502 Lyme Central Nervous System Infection IgG with Antibody Index Reflex, Serum and Spinal Fluid

Specimen Requirements:

Both spinal fluid (CSF) and serum are required for this test. CSF and serum must be collected within 24 hours (maximum) of each other.

Specimen Type: Spinal fluid

Container/Tube: Sterile vial

Specimen Volume: 1.2 mL

Collection Instructions:

1. A CSF sample of 1.2 mL needs to be collected within 24 hours of the serum specimen, preferably at the same time.

2. Label vial as spinal fluid or CSF.

3. CSF aliquot should be from the second, third, or fourth CSF vial collected during the lumbar puncture.

Do not submit CSF from the first vial due to the possibility of blood contamination, which will cause specimen rejection.

4. Band specimens together.

Specimen Type: Serum

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.2 mL

Collection Instructions:

1. A serum sample of 1.2 mL needs to be collected within 24 hours of the spinal fluid specimen, preferably at the same time.

2. Centrifuge and aliquot serum into a plastic vial.

3. Label as serum.

4. Band specimens together.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	11 days
	Frozen	35 days
Serum	Refrigerated (preferred)	11 days
	Frozen	35 days

CPT Code Information:

86618

86618 x2-(if appropriate)

82040-(if appropriate)

82042-(if appropriate)

82784 x2-(if appropriate)

LNBAI 63249 Lyme Central Nervous System Infection IgG, Antibody Index, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see LNBAB / Lyme Central Nervous System Infection IgG with Antibody Index Reflex, Serum and Spinal Fluid.

Both cerebrospinal fluid (CSF) and serum are required for this test. CSF and serum must be collected within 24 hours maximum of each other. CSF specimens with blood contamination will be rejected.

Specimen Type: Spinal fluid

Container/Tube: Sterile vial

Specimen Volume: 1.2 mL

Collection Instructions:

1. A spinal fluid (CSF) sample of 1.2 mL needs to be collected within 24 hours of the serum specimen, preferably at the same time.

2. Label vial as spinal fluid or CSF.

3. CSF aliquot should be from the second, third, or fourth CSF vial collected during the

lumbar puncture.

Do not submit CSF from the first vial due to the possibility of blood contamination, which will cause specimen rejection.

4. Band specimens together.

Specimen Type: Serum

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.2 mL

Collection Instructions:

1. A serum sample of 1.2 mL needs to be collected within 24 hours of the spinal fluid specimen, preferably at the same time.

2. Centrifuge and aliquot serum into a plastic vial.

3. Label as serum.

4. Band specimens together.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	11 days
	Frozen	35 days

CPT Code Information:

86618 x 2

82040

82042

82784 x 2

LYWB
9535

Lyme Disease Antibody, Immunoblot, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.75 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	30 days

CPT Code Information:

86617 x 2

LYME
9129

Lyme Disease Serology, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Information: Centrifuge and aliquot serum into plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	10 days
	Frozen	30 days

CPT Code Information:

86618

86617 x 2-Lyme disease confirmation (if appropriate)

PBORB 87973 Lyme Disease, Molecular Detection, PCR, Blood

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 1 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87476

87798 x 2

87801 (if appropriate for government payers)

LYMPV 618310 Lyme Disease, Molecular Detection, PCR, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Spinal fluid

Container/Tube: Sterile vial

Specimen Volume: 1 mL

Collection Instructions: Label specimen as spinal fluid.

Specimen Type: Synovial fluid

Container/Tube: Sterile vial

Specimen Volume: 1 mL

Collection Instructions: Label specimen as synovial fluid.

Specimen Type: Tissue (fresh only)

Sources: Skin or synovial biopsy

Container/Tube: Sterile container with normal saline

Specimen Volume: Approximately 4 mm(3)

Collection Instructions:

1. Submit only fresh tissue.

2. Skin biopsies:

a. Wash biopsy site with an antiseptic soap. Thoroughly rinse area with sterile water. Do not use alcohol or iodine preparations. A local anesthetic may be used.

b. Biopsy specimens are best taken by punch biopsy to include full thickness of dermis.

3. Label specimen with source of tissue.

Specimen Minimum Volume: Spinal fluid: 0.3 mL; Synovial fluid: 0.5 mL; Tissue: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87476

87798 x 2

87801 (if appropriate for government payers)

TLYME 608104 Lyme IgM and IgG, Whole Cell Sonicate, ELISA, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.6 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	10 days
	Frozen	30 days

CPT Code Information:

86617 x 2

PM3CX 606164 Lymph3Cx Assay, Primary Mediastinal Large B-cell Lymphoma and Diffuse Large B-cell Lymphoma, mRNA Gene Expression, NanoString, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin stained and 6 to 10 unstained

Collection Instructions:

1. Submit 1 slide stained with hematoxylin and eosin.

2. If the formalin-fixed paraffin-embedded (FFPE) biopsy specimen is a core needle biopsy or fine needle aspirate, submit 10 unstained slides.

3. If the FFPE biopsy specimen is a resection, submit 6 unstained slides.

4. Submitted slides should be 4- to 5-micron thick sections placed on positively charged slides.

Additional Information: Paraffin embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

Acceptable:

Specimen Type: Tissue block

Collection Instructions: Submit an FFPE tumor tissue block.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Minimum 60% tumor with or without macrodissection. Minimum required unstained tissue section input: 0.12 mm(3); Slides: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Tissue, Paraffin	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

0120U

88381

LPA3P 62205 Lymphocyte Proliferation to Anti-CD3/Anti-CD28 and Anti-CD3/Interleukin-2 (IL-2), Flow Cytometry, Blood

Specimen Requirements:

Supplies: Ambient Shipping Box-Critical Specimens Only (T668)

Container/Tube: Green top (sodium heparin)

Specimen Volume: 20 mL

See table for information on recommended volume based on absolute lymphocyte count

Pediatric Volume:

<3 months: 1 mL

3-24 months: 3 mL

25 months-18 years: 5 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Additional Information: For serial monitoring, it is recommended that specimen collection be performed at the same time of day.

Table. Blood Volume Recommendations Based on Absolute Lymphocyte Count (ALC)

ALC	Blood volume for minimum aCD28 only	Blood volume for minimum of aCD3, aCD28, and IL-2	Blood volume for full assay
<0.5	>15 mL	>28 mL	>50 mL
0.5-1.0	15 mL	28 mL	50 mL
1.1-1.5	6.5 mL	12 mL	24 mL
1.6-2.0	4.5 mL	8.5 mL	16 mL
2.1-3.0	3.5 mL	6.5 mL	12 mL
3.1-4.0	2.5 mL	4.5 mL	8 mL
4.1-5.0	1.8 mL	3.5 mL	6 mL
>5.0	1.5 mL	2.5 mL	5 mL

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container

WB Sodium Heparin	Ambient	48 hours	GREEN TOP/HEP

CPT Code Information:

86353 x 2

86353 (as appropriate)

LPAGF 60592 Lymphocyte Proliferation to Antigens, Blood

Specimen Requirements:

Supplies: Ambient Shipping Box-Critical Specimens Only (T668)

Container/Tube: Green top (sodium heparin)

Specimen Volume: 20 mL

See tables for information on recommended volume based on absolute lymphocyte count

Pediatric Volume:

<3 months: 1 mL

3-24 months: 3 mL

25 months-18 years: 5 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Additional Information: For serial monitoring, it is recommended that specimen collection be performed at the same time of day.

Table. Blood Volume Recommendations Based on Absolute Lymphocyte Count (ALC)

Antigen only		
ALC x 10(9)/L	**Blood volume for minimum *Candida albicans* (CA) and tetanus toxoid (TT) Only**	**Blood volume for full assay**
<0.5	>18.5 mL	>40 mL
0.5-1.0	18.5 mL	40 mL
1.1-1.5	8.5 mL	20 mL
1.6-2.0	6.0 mL	12 mL

2.1-3.0	4.5 mL	10 mL
3.1-4.0	3.0 mL	6 mL
4.1-5.0	2.5 mL	5 mL
>5.0	2.0 mL	4 mL
Mitogen and antigen		
ALC x 10(9)/L	**Blood volume for minimum of each assay**	**Blood volume for full assay**
<0.5	>28 mL	>60 mL
0.5-1.0	28 mL	60 mL
1.1-1.5	12 mL	30 mL
1.6-2.0	8.5 mL	20 mL
2.1-3.0	6.5 mL	15 mL
3.1-4.0	4.5 mL	10 mL
4.1-5.0	3.5 mL	8 mL
>5.0	2.5 mL	6 mL

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
WB Sodium Heparin	Ambient	48 hours	GREEN TOP/HEP

CPT Code Information:

86353

86353 (if appropriate)

LPMGF 60591 Lymphocyte Proliferation to Mitogens, Blood

Specimen Requirements:
Supplies: Ambient Shipping Box-Critical Specimens Only (T668)

Container/Tube: Green top (sodium heparin)

Specimen Volume: 20 mL

See tables for information on recommended volume based on absolute lymphocyte count

Pediatric Volume:

<3 months: 1 mL

3-24 months: 3 mL

25 months-18 years: 5 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Additional Information: For serial monitoring, it is recommended that specimen collection be performed at the same time of day.

Table. Blood Volume Recommendations Based on Absolute Lymphocyte Count (ALC)

Mitogen only			
ALC x 10(9)/L	**Blood volume for minimum phytohemagglutinin (PHA) only**	**Blood volume for minimum PHA and pokeweed mitogen (PWM)**	**Blood volume for full assay**
<0.5	>6.5 mL	>8.5 mL	>22 mL
0.5-1.0	6.5 mL	8.5 mL	22 mL
1.1-1.5	3.0 mL	4.0 mL	10 mL
1.6-2.0	2.0 mL	2.5 mL	7 mL
2.1-3.0	1.5 mL	2.0 mL	6 mL
>3.1	1.0 mL	1.5 mL	4 mL

4.1-5.0	1.0 mL	1.0 mL	3 mL
>5.0	1.0 mL	1.0 mL	2 mL

Mitogen and antigen

ALC x 10(9)/L	Blood volume for minimum of each assay	Blood volume for full assay
<0.5	>28 mL	>60 mL
0.5-1.0	28 mL	60 mL
1.1-1.5	12 mL	30 mL
1.6-2.0	8.5 mL	20 mL
2.1-3.0	6.5 mL	15 mL
3.1-4.0	4.5 mL	10 mL
4.1-5.0	3.5 mL	8 mL
>5.0	2.5 mL	6 mL

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
WB Sodium Heparin	Ambient	48 hours	GREEN TOP/HEP

CPT Code Information:

86353

86353 (if appropriate)

LEF1 71356 Lymphoid Enhancer-Binding Factor 1(LEF1) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

LPLFX 61114 Lymphoplasmacytic Lymphoma/Waldenstrom Macroglobulinemia, MYD88 L265P with Reflex to CXCR4, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Bone marrow aspirate

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD), green top (sodium heparin)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Stability Information: Ambient (preferred) 10 days/Refrigerated 10 days

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD), green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Label specimen as blood.

Specimen Stability Information: Ambient (preferred) 10 days /Refrigerated 10 days

Specimen Type: Paraffin-embedded tissue

Container/ Tube: Paraffin block

Collection Instructions:

1. Decalcified specimens (eg, bone marrow core biopsies) are not acceptable.

2. Indicate specimen source.

Specimen Stability Information: Ambient

Additional Information: If the quality of the biopsy specimen is poor, testing should not

be ordered. Testing may be canceled if DNA requirements are inadequate.

Acceptable:

Specimen Type: Tissue slide

Slides: 20 unstained slides

Container/ Tube: Transport in plastic slide holders.

Collection Instructions:

1. Send 20 unstained, nonbaked slides with 5-micron thick sections of tissue.

2. Decalcified specimens (eg, bone marrow core biopsies) are not acceptable.

3. Indicate specimen source.

Specimen Stability Information: Ambient

Additional Information: Testing may be canceled if resultant extracted DNA does not meet concentration requirements.

Specimen Type: Frozen tissue

Container/Tube: Plastic container

Specimen Volume: 100 mg

Collection Instructions:

1. Freeze tissue within 1 hour of collection

2. Indicate specimen source.

Specimen Stability Information: Frozen

Specimen Type: Extracted DNA

Container/Tube: 1.5- to 2-mL tube

Specimen Volume: Entire specimen

Collection Instructions:

1. DNA must be extracted within 7 days of collection.

2. Label specimen as extracted DNA and source of specimen.

3. Provide volume and concentration of DNA on label.

Specimen Stability Information: Frozen (preferred)/Refrigerated/Ambient

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). We cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied.

Specimen Minimum Volume: Whole blood, bone marrow aspirate: 0.5 mL; Frozen tissue: 50 mg; Extracted DNA: 50 mcL at 20 ng/mcL; Tissue slides: 10 unstained slides

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	10 days

CPT Code Information:

81305

LYNCP
614572

Lynch Syndrome Panel, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a bone marrow transplant, call 800-533-1710.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for samples received after 4 days and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the preferred volume of blood must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Minimum Volume: Whole blood: 1 mL; Saliva: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81435

LPCBS 61766 Lysophosphatidylcholines, LC MS/MS, Blood Spot

Specimen Requirements:

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Blood Spot Collection Card

Acceptable: PerkinElmer 226 filter paper, Munktell filter paper, Whatman Protein Saver 903 Paper, local newborn screening card, or blood collected in tubes containing ACD or EDTA, and then spotted and dried on filter paper

Specimen Volume: 2 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Completely fill at least 2 circles on the filter paper card (approximately 100 microliters blood per circle).

3. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

4. Do not expose specimen to heat or direct sunlight.

5. Do not stack wet specimens.

6. Keep specimen dry.

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions.

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777).

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800.

Specimen Minimum Volume: 1 Blood spot

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole blood	Refrigerated (preferred)	90 days	FILTER PAPER
	Frozen	90 days	FILTER PAPER
	Ambient	28 days	FILTER PAPER

CPT Code Information:

82542

LALB
62954

Lysosomal Acid Lipase, Blood

Specimen Requirements:

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD) or green top (sodium heparin)

Specimen Volume: 2 mL

Collection Instructions: Send whole blood in original tube. **Do not aliquot**

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated (preferred)	7 days
	Ambient	7 days

CPT Code Information:

82657

LALBS
62955

Lysosomal Acid Lipase, Blood Spot

Specimen Requirements:

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Collection Container/Tube:

Preferred: Blood spot collection card

Acceptable: PerkinElmer 226 filter paper, Munktell TFN, or Whatman Protein Saver 903 Paper

Specimen Volume: 2 Blood spots

Collection Instructions:

1. Completely fill at least 2 circles on the filter paper card (approximately 100 microliters blood per circle).

2. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

3. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

4. Do not expose specimen to heat or direct sunlight.

5. Do not stack wet specimens.

6. Keep specimen dry.

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions.

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777).

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800).

Specimen Minimum Volume: 1 Blood spot

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole blood	Refrigerated (preferred)	28 days	FILTER PAPER
	Frozen	90 days	FILTER PAPER
	Ambient	7 days	FILTER PAPER

CPT Code Information:

82657

LDALD 64907 Lysosomal and Peroxisomal Disorders Newborn Screen, Blood Spot

Specimen Requirements:

Patient must be older than 24 hours and younger than 1 week.

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Blood Spot Collection Card

Acceptable: PerkinElmer 226 filter paper, Munktell filter paper, Whatman Protein Saver 903 Paper, local newborn screening card, or blood collected in tubes containing ACD, or EDTA and then spotted and dried on filter paper

Specimen Volume: 2 Blood spots

Collection Instructions:

1. Completely fill at least 2 circles on the filter paper card (approximately 100 microliters blood per circle).

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800

Specimen Minimum Volume: 1 Blood spot

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole blood	Refrigerated (preferred)	90 days	FILTER PAPER
	Frozen	90 days	FILTER PAPER
	Ambient	28 days	FILTER PAPER

CPT Code Information:

83789

PLSD
89678

Lysosomal and Peroxisomal Disorders Screen, Blood Spot

Specimen Requirements:

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Blood Spot Collection Card

Acceptable: PerkinElmer 226 filter paper, Munktell filter paper, Whatman Protein Saver 903 paper, local newborn screening card, or blood collected in tubes containing ACD or EDTA and dried on acceptable filter paper

Specimen Volume: 2 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Completely fill at least 2 circles on the filter paper card (approximately 100 microliters blood per circle).

3. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

4. Do not expose specimen to heat or direct sunlight.

5. Do not stack wet specimens.

6. Keep specimen dry.

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Minimum Volume: 1 Blood spot

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole blood	Refrigerated (preferred)	90 days	FILTER PAPER

	Frozen	90 days	FILTER PAPER
	Ambient	28 days	FILTER PAPER

CPT Code Information:

83789

LSDS
606771

Lysosomal Disorders Screen, Random, Urine

Specimen Requirements:

Patient Preparation:

1. Do not administer low-molecular weight heparin before specimen collection.

2. Baby wipes or wipes containing soaps or lotions should **not** be used before specimen collection because these may interfere with results.

Supplies: Urine Container, 60 mL (T313)

Container/Tube: Clean, plastic urine container with no metal cap or glued insert

Specimen Volume: 12 mL

Pediatric Volume: 3.5 mL

Collection Instructions:

1. Collect a random urine specimen **(early morning preferred)**.

2. No preservative.

Specimen Minimum Volume: 3.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	15 days
	Frozen	90 days

CPT Code Information:

83789

83864

84377

84275

LSDGP 608011 Lysosomal Storage Disease Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

Container/Tube: T-25 flask

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured amniocytes from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies

on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

MURA
607462

Lysozyme (Muramidase), Plasma

Specimen Requirements:

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL Plasma

Collection Instructions:

1. Within 2 hours of collection, centrifuge and aliquot plasma into a plastic vial.

2. Freeze immediately.

Specimen Minimum Volume: Plasma: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen	21 days

CPT Code Information:

85549

LYSOZ
70503

Lysozyme Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

LYSO
82398

Lysozyme, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86008

MA2ES
605970

Ma2 Antibody by ELISA, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

83516

MA2EC
605971

Ma2 Antibody, ELISA, Spinal Fluid

Specimen Requirements:

Container/Tube: Sterile vial

Preferred: Collection vial number 1

Acceptable: Any collection vial

Specimen Volume: 1 mL

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

83516

MACNT
65405

Macadamia Nut, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

MACE
82492

Mace, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

MACK
82342

Mackerel, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FMACR 57817 Macroamylase

Specimen Requirements:

Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 2 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	30 days

CPT Code Information:

82150

MCRPL 34643 Macroprolactin, Serum

Specimen Requirements:

Patient Preparation: For 12 hours before specimen collection do not take multivitamins or dietary supplements containing biotin (vitamin B7), which is commonly found in hair, skin, and nail supplements and multivitamins.

Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	90 days
	Ambient	24 hours

CPT Code Information:

84146 x 2

MAGU
610768

Magnesium, 24 Hour, Urine

Specimen Requirements:

Supplies: Sarstedt 5 mL Aliquot Tube (T914)

Collection Container/Tube: 24-hour graduated urine container with no metal cap or glued insert

Submission Container/Tube: Plastic, 5 mL tube or a clean, plastic urine container with no metal cap or glued insert

Specimen Volume: 4 mL

Collection Instructions:

1. Collect urine for 24 hours.

2. Refrigerate specimen within 4 hours of completion of 24-hour collection.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	72 hours

CPT Code Information:

83735

MGS
8448

Magnesium, Serum

Specimen Requirements:

Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Specimen Volume: 0.5 mL

Collection Instructions:

1. See Metals Analysis Specimen Collection and Transport for complete instructions.

2. Serum gel tubes should be centrifuged within 2 hours of collection.

3. Red-top tubes should be centrifuged and aliquoted within 2 hours of collection.

Additional Information: If other metal tests are also desired when drawing for a serum magnesium level; the specimen must be drawn in a plain, royal blue-top Vacutainer plastic trace element blood collection tube.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	7 days
	Frozen	365 days

CPT Code Information:

83735

MAGRU
621396

Magnesium/Creatinine Ratio, Random, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: Plastic, 5-mL tube or a clean, plastic aliquot container

with no metal cap or glued insert

Specimen Volume: 4 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	30 days
	Ambient	72 hours

CPT Code Information:

83735

82570

FMME
57924

Mahi Mahi IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

MOBRD 28339 Mail Out Research, Blood

Transport Temperature:

Specimen Type	Temperature	Time
Varies		

MALI 601988 MAL Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

MALP
618475

Malabsorption Evaluation Panel, Feces

Specimen Requirements:

Supplies: Malabsorption Panel (T920)

Container/Tube: Malabsorption kit or 2 small stool containers

Specimen Volume: 18 g split equally between 2 containers

Collection Instructions:

1. Collect a fresh, random fecal specimen, no preservatives.

2. Split specimen equally between 2 small containers.

3. Label one small container with the A1AF and UREDF sample collection labels. Label the other small container with the CALPR, ELASF sample collection label.

4. Freeze immediately

Additional Information:

1. Specimen must be split prior to transport.

2. Testing cannot be added to a previously collected specimen.

3. Specimen cannot be collected from a diaper.

Specimen Minimum Volume: 5 g

Transport Temperature:

Specimen Type	Temperature	Time
Fecal	Frozen	7 days

CPT Code Information:

0430U

LCMAL
87860

Malaria, Molecular Detection, PCR, Varies

Specimen Requirements:

Both blood specimens and slides are required.

Specimen Type: Blood

Container/Tube: Lavender top (EDTA)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send specimen in original tube. **Do not aliquot.**

Specimen Type: Blood films

Container/Tube: Clean, grease-free slides in plastic slide container

Specimen Volume: 2 thin blood films and 2 thick blood films

Collection Instructions:

1. For complete instructions, see Thick and Thin Blood Slide Preparation Instructions.

2. Ideally, blood films should be made directly from uncoagulated blood acquired via fingerstick. However, EDTA anticoagulated blood is also acceptable.

3. Prepare thin blood films as follows:

a. Prepare a thin film with a "feathered edge" that is no more than a single cell thick.

b. Allow the film to thoroughly air dry and then fix by briefly immersing in either absolute or 95% methyl alcohol.

c. Allow to air dry after fixation.

4. Prepare thick blood films as follows:

a. Place a large drop of blood (approximately the size of a dime and preferably from a fingerstick) on a slide.

b. Using a corner of a second slide, spread the drop in a circular motion while applying firm pressure to literally scratch the blood onto the carrier slide. This technique allows the blood to dry quickly and adhere well to the slide. Use approximately 20 circular sweeps with the second slide. The drop of blood should be about the size of a quarter when finished.

c. **Do not fix.** Air dry thoroughly (approximately 45 minutes) before placing in transport container.

Specimen Minimum Volume: Blood: 1 mL
Slides: See Specimen Required.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Ambient	7 days

CPT Code Information:

87798

MCMF
113355

Malignant Cells Cyto/Heme (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88108

MALT
82834

Malt, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

MAMLF 58105 MAML2 (11q21) Rearrangement, Mucoepidermoid Carcinoma (MEC), FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred

Specimen Type: Tissue block (fresh tissue is **not acceptable**)

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block.

2. Blocks prepared with alternative fixation methods (eg, Prefer, Bouin's) will be attempted but are less favorable for successful results. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Decalcified paraffin-embedded specimens will have testing attempted; however, the success rate is approximately 50%. **Testing may be canceled** if sufficient tumor tissue is not present.

3. **Submitted fresh tissue specimens will be canceled upon receipt**. If only fresh tissue is available, embed in paraffin prior to sending.

Acceptable

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained and 4 unstained

Collection Instructions: Submit 1 slide stained with hematoxylin and eosin and 4 consecutive unstained, positively charged, unbaked slides with 5 micron-thick sections of the tumor tissue.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Slides: 1 Hematoxylin and eosin-stained and 2 unstained

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2, 88291-DNA probe, each (first probe set), Interpretation and report

88271x2-DNA probe, each; each additional probe set (if appropriate)

88271x1-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271x2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271x3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

MGB
70507

Mammaglobin (MGB) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

MAND
82352

Mandarin, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

MNU 8080 Manganese, 24 Hour, Urine

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours.**

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine container with no metal cap or glued insert

Submission Container/Tube: Plastic, 10-mL urine tube or clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 0.8 mL

Collection Instructions:

1. Collect urine for 24 hours.

2. Within 4 hours of completion of 24-hour collection, refrigerate specimen.

3. See Metals Analysis Specimen Collection and Transport for complete instructions.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	7 days

CPT Code Information:

83785

MNB 89120 Manganese, Blood

Specimen Requirements:

*****This specimen container cannot be opened or used for any other testing before shipping.*****

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

Supplies:

-Metal Free EDTA 3 mL Tube (T989)

-Metal Free B-D Tube (EDTA), 6 mL (T183)

Container/Tube:

Preferred: Royal blue-top BD vacutainer with EDTA blood collection tube (3 mL) (BD catalog no. 367777) (T989)

Acceptable: Royal blue-top BD Vacutainer Plus with EDTA blood collection tube (6 mL) (BD catalog no. 368381) (T183)

Specimen Volume: 1 mL

Collection Instructions:

1. See Metals Analysis Specimen Collection and Transport for complete instructions.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

83785

FMNRB Manganese, RBC and Plasma

57196

Specimen Requirements:

Both plasma and RBCs are required for testing.

Specimen Type: Packed RBCs and Plasma

Container/Tube: Royal blue-top (EDTA metal-free tube; metal-free transfer tubes

Specimen Volume: 3 mL each of plasma and packed RBCs

Collection Instructions: Both packed cells and plasma are required. Plasma and red cells must be separated ASAP and shipped separately. Failure to separate may cause rejection due to hemolysis of red cells and contamination of the plasma. Transfer RBCs and plasma into separate plastic, acid-washed, trace metal-free tubes. Ship refrigerated.

NOTE: Specimen must be received at the testing facility within five days of collection.

Specimen Minimum Volume: 0.5 mL each plasma and RBCs

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Metal Free EDTA Plasma	Refrigerated (preferred)	5 days	METAL FREE
	Ambient	72 hours	METAL FREE
RBCS	Refrigerated (preferred)	5 days	METAL FREE
	Ambient	72 hours	METAL FREE

CPT Code Information:

83785

MNS
8413

Manganese, Serum

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours.**

Supplies:

-Metal Free B-D Tube (No Additive), 6 mL (T184)

-Metal Free Specimen Vial (T173)

Collection Container/Tube: 6 mL Plain, royal blue-top Vacutainer plastic trace element blood collection tube

Submission Container/Tube: 7-mL Mayo metal-free, screw-capped, polypropylene

vial

Specimen Volume: 1.6 mL Serum

Collection Instructions: See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: Serum: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum	Refrigerated (preferred)	28 days	METAL FREE
	Ambient	28 days	METAL FREE
	Frozen	28 days	METAL FREE

CPT Code Information:

83785

MNRCU 614993 Manganese/Creatinine Ratio, Random, Urine

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine collection container with no metal cap or glued insert

Submission Container/Tube: Plastic, 10-mL urine tube or clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 3 mL

Collection Instructions:

1. Collect a random urine specimen.

2. See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	7 days

CPT Code Information:

82570

83785

MNCU
614994

Manganese/Creatinine Ratio, Urine (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see MNRCU / Manganese/Creatinine Ratio, Random, Urine.

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine collection container with no metal cap or glued insert

Submission Container/Tube: Plastic, 10-mL urine tube or clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 3 mL

Collection Instructions:

1. Collect a random urine specimen.

2. See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	7 days

CPT Code Information:

83785

MANGO Mango, IgE, Serum
82811

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

MBLF Mannan Binding Lectin Complement Pathway, Functional, Serum
614985

Specimen Requirements:

Patient Preparation:

1. Fasting: 8 hours, preferred but not required

2. Do **not** collect a specimen for at least 48 hours following plasma exchange.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Immediately after specimen collection, place the tube on wet ice and allow specimen to clot.

2. Centrifuge at 4 degrees C and aliquot serum into plastic vial.

3. Within 30 minutes of centrifugation, freeze specimen. Specimen must be placed on dry ice if not frozen immediately.

Note: If a refrigerated centrifuge is not available, it is acceptable to use a room temperature centrifuge, provided the specimen is kept on ice before centrifugation, and immediately afterward, the serum aliquoted and frozen.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Frozen	14 days

CPT Code Information:

86161

FMPRE 57535 Maple Red (Acer rubrum) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

MSUDP Maple Syrup Urine Disease Gene Panel, Varies
608031

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from blood spots, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin or tissue

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin or tissue biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Prenatal Specimens

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured amniocytes from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81405

81406 x 3

81479

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

81479 (if appropriate for government payers)

MARE
82141

Mare's Milk, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

MFRGG
617379

Marfan, Loeys-Dietz, and Aortopathy Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a bone marrow transplant.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Any anticoagulant

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred)/Refrigerated

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81410

MARJ
82605

Marjoram, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

MSPTC
113288

Mass Spectrometry (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

82542-Liquid Chromatography-Tandem Mass Spectrometry (LC-MS/MS)

MSMNT
621774

Mass Spectrometry, Membranous Nephropathy, Technical Component (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

82542

MCM24
608378

Mast Cell Mediators, 24 Hour, Urine

Specimen Requirements:

Patient Preparation:

1. Patients taking monoamine oxidase inhibitors (MAOIs) or aminoguanidine may have increased levels of *N*-methylhistamine. If medically feasible, patient **must not** take MAOIs or aminoguanidine for 3 weeks before specimen collection. For more information see Cautions.

2. Patients taking aspirin or nonsteroidal anti-inflammatory drugs (NSAIDs) may have decreased concentrations of prostaglandin F2 alpha. If medically feasible, the patient **should not** take aspirin for 2 weeks or NSAIDs for 72 hours before specimen collection.

3. Patients taking 5-lipoxygenase inhibitor zileuton (Zyflo) may have decreased concentrations of leukotriene E4. If medically feasible, the patient **should not** take zileuton for 48 hours before specimen collection.

Supplies: Urine Container, 60 mL (T313)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: Plastic, 60-mL urine bottle

Specimen Volume: 20 mL

Collection Instructions:

1. Collect urine for 24 hours.

2. No preservative.

3. Aliquot urine into plastic tube and send to lab.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 10 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen	28 days

CPT Code Information:

82542

84150

MCMRU 608379 Mast Cell Mediators, Random, Urine

Specimen Requirements:

Patient Preparation:

1. Patients taking monoamine oxidase inhibitors (MAOI) or aminoguanidine may have increased N-methylhistamine levels. If medically feasible, patient **must not** take MAOI or aminoguanidine for 3 weeks before specimen collection. For more information see Cautions.

2. Patients taking aspirin or nonsteroidal anti-inflammatory drugs (NSAIDs) may have decreased concentrations of prostaglandin F2 alpha. If medically feasible, the patient **should not** take aspirin for 2 weeks or NSAIDs for 72 hours before specimen collection.

3. Patients taking 5-lipoxygenase inhibitor zileuton (Zyflo) may have decreased concentrations of leukotriene E4. If medically feasible, the patient **should not** take zileuton for 48 hours before specimen collection.

Supplies: Urine Container, 60 mL (T313)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: Plastic, 60-mL urine bottle

Specimen Volume: 20 mL

Collection Instructions:

1. Within a few hours of symptom onset, collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 10 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen	28 days

CPT Code Information:

82570

84150

82542

MATCC 35479 Maternal Cell Contamination, Molecular Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a bone marrow transplant.

Specimen Type: Maternal blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Any anticoagulant

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send specimen in original tube.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Prenatal Specimens:

Submit only 1 of the following specimens:

Specimen Type: Cord blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Any anticoagulant

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send specimen in original tube.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Additional Information: A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid.

Specimen Stability Information: Refrigerated (preferred)/Ambient

Specimen Type: Cultured amniocytes

Container/Tube: T-25 flask

Specimen Volume: 2 full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Additional Information: A separate culture charge will be assessed under CULFB / Fibroblast Culture for Genetic Testing, Tissue.

Specimen Stability Information: Refrigerated

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Specimen Minimum Volume: Amniotic Fluid: 10 mL
Blood, Cord Blood: 0.5 mL
Chorionic Villus: 5 mg

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Varies	Varies		

CPT Code Information:

81265

Fibroblast Culture for Genetic Test

88233-(if appropriate)

88240-(if appropriate)

Amniotic Fluid Culture/Genetic Test

88235-(if appropriate)

88240-(if appropriate)

Each additional specimen

81266

Specimen Requirements:

Specimen # 1 collection must occur between 10 weeks, 0 days and 13 weeks, 6 days gestation. If gestational age is based on Crown-Rump length (CRL), the specimen must be collected when the CRL is between 32.4 - 83.9 mm.

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable.

2. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

3. Ship refrigerate.

Note:

Submit with order: Patient's date of birth, current weight, number of fetuses present, patient's race, if the patient was diabetic at the time of conception, if there is a known family history of neural tube defects, if the patient has had a previous pregnancy with a trisomy, if the patient is currently smoking, if the patient is taking valproic acid or carbamazepine (Tegretol), if this is a repeat sample, and the age of the egg donor if in vitro fertilization.

In addition to the above:

If an NT measurement is performed: the date of ultrasound, the CRL measurement, the nuchal translucency (NT) measurement and the name and certification number of the sonographer are required. NT must be measured when the CRL is between 38-83.9mm. The NT measurement must be performed by an ultrasonographer that is certified by the Fetal Medicine Foundation (FMF).

OR

If no NT measurement is performed: a due date or CRL measurement with the date of ultrasound is required.

See the PATIENT HISTORY FOR MATERNAL SERUM TESTING form.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	365 days
	Ambient	72 hours

CPT Code Information:

84163

FMSS2
75351

Maternal Serum Screening, Integrated, Specimen #2, Alpha Fetoprotein, hCG, Estriol, and Inhibin A

Specimen Requirements:

Specimen collection must occur between 14 weeks, 0 days and 24 weeks, 6 days gestation. The recommended time for maternal serum screening is 16 to 18 weeks gestation. Acceptable date ranges to collect the second sample is provided in the Integrated-1 report.

Collection Container/Tube: Serum gel or red top

Submission Container/Tube: Plastic vial

Specimen Volume: 3 mL Serum

Collection Instructions:

1. As soon as possible or within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

2. Send refrigerate.

Note: This test requires that a previous first trimester specimen, FFMSS / Maternal Serum Screening, Integrated, Specimen #1, PAPP-A, NT has been performed.

Specimen Minimum Volume: Serum: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Refrigerated (preferred)	14 days
	Frozen	365 days
	Ambient	72 hours

CPT Code Information:

81511

FMT21 MaterniT21 Plus

57852

Specimen Requirements:

NOTE: Completed Sequenom Test Requisition form is required

A Core Option must be marked on TRF under MaterniT 21 PLUS test

If nothing indicated by client, mark option- Core (chr 21, 18, 13, sex)

Preferred evacuated tube: (1)10 mL Streck tube kit (MCL supply number T715).

Absolute minimum collection for analysis: (1) 10 mL in Streck tube

Collection instructions: Draw 1 tube of blood, 10 mL in special Streck tube kit (MCL supply number T715). Ship ambient.

REQUIRED:

1. Specimen MUST be received at MCL within 72 hours of collection.

2. Specimen collected NOT less than 9 weeks of gestation

3. Sequenom collection kit (MCL Supply T715)

4. Completed Sequenom Test Requisition form

5. Maternal Height (inches)

6. Maternal Weight (pounds)

7. Gestational Age (weeks)

8. Gestational Age (days)

9. Gestation (Number of fetuses)

10. Increased risk due to

Specimen Minimum Volume: 10 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
WB Streck	Ambient	7 days	Streck Black/Tan top

CPT Code Information:

81420

MSMRT 64946 Mayo Algorithmic Approach for Stratification of Myeloma and Risk-Adapted Therapy Report, Bone Marrow

Specimen Requirements:

Specimen Type: Redirected bone marrow

Preferred: Yellow top (ACD solution A or B)

Acceptable: Lavender top (EDTA) or green top (heparin)

Specimen Volume: 4 mL

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Bone Marrow	Ambient (preferred)	72 hours
	Refrigerated	72 hours

CPT Code Information:

88182-Flow cytometry, cell cycle or DNA analysis

88184-Flow cytometry; first cell surface, cytoplasmic or nuclear marker

88185 x 5-Flow cytometry; additional cell surface, cytoplasmic or nuclear marker (each)

88187-Flow cytometry interpretation, 2 to 8 Markers

NGAML 65089 MayoComplete Acute Myeloid Leukemia, 11-Gene Panel, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Bone marrow aspirate

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Stability Information: Ambient (preferred) 14 days/Refrigerate 14 days

Additional Information: To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Acceptable:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Label specimen as blood.

Specimen Stability Information: Ambient (preferred) 14 days/Refrigerate 14 days

Additional Information: To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Extracted DNA from blood or bone marrow

Container/Tube: 1.5 to 2 mL tube

Specimen Volume: Entire specimen

Collection Instructions:

1. DNA must be extracted within 14 days after collection.

2. Label specimen as extracted DNA and source of specimen.

3. Provide volume and concentration of the DNA.

Specimen Stability Information: Frozen (preferred)/Refrigerated/Ambient

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). We cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied.

Specimen Minimum Volume: Whole blood, one marrow: 1 mL; Extracted DNA: 100 mcL at 20 ng/mcL concentration

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	14 days

CPT Code Information:

81450

NGAMT 65661 MayoComplete Acute Myeloid Leukemia, Therapeutic Gene Mutation Panel (FLT3, IDH1, IDH2, TP53), Next-Generation Sequencing, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Bone marrow aspirate

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Stability Information: Ambient (preferred) 14 days/Refrigerate 14 days

Additional Information: To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Acceptable:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Label specimen as blood.

Specimen Stability Information: Ambient (preferred) 14 days/Refrigerate 14 days

Additional Information: To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Extracted DNA from blood or bone marrow

Container/Tube: 1.5- to 2-mL tube

Specimen Volume: Entire specimen

Collection Instructions:

1. DNA must be extracted within 14 days after collection.

2. Label specimen as extracted DNA and source of specimen.

3. Provide volume and concentration of the DNA.

Specimen Stability Information: Frozen (preferred)/Refrigerate/Ambient

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). We cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied.

Specimen Minimum Volume: Blood, Bone marrow: 1 mL
Extracted DNA: 100 mcL at 20 ng/mcL concentration

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	14 days

CPT Code Information:

81120

81121

81245

81246

81352

81479 (if appropriate for government payers)

NGBCL 618477 MayoComplete B-Cell Lymphoma, Next-Generation Sequencing, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Bone marrow aspirate

Container/Tube:

Preferred: Lavender or pink top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Stability Information: Ambient (preferred) 14 days/Refrigerate 14 days

Additional Information: To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender or pink top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Label specimen as blood.

Specimen Stability Information: Ambient (preferred) 14 days/Refrigerate 14 days

Additional Information: To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Paraffin-embedded tissue

Container/ Tube: Paraffin block

Collection Instructions:

1. If available, send 1 representative hematoxylin and eosin-stained slide.

2. Minimum amount of tumor nuclei is 20%

3. Required amount of tissue area is at least 25 mm(2)

4. Tissue should be fixed in 10% neutral-buffered formalin. Other fixatives are not acceptable.

5. Decalcified specimens (eg, bone marrow core biopsies) are not acceptable.

Specimen Stability Information: Ambient

Additional Information: If the quality of the biopsy specimen is poor or the target tumor cell population is below 20%, testing should not be ordered. Testing may be canceled if DNA requirements are inadequate.

Acceptable:

Specimen Type: Tissue slide

Slides: 20 unstained slides

Container/ Tube: Transport in plastic slide holders.

Collection Instructions:

1. Send 20 unstained, nonbaked slides with 5-micron thick sections of tissue.

2. If available, also send 1 representative hematoxylin and eosin-stained slide.

3. Minimum amount of tumor nuclei is 20%

4. Required amount of tissue area is at least 25 mm(2)

5. Tissue should be fixed in 10% neutral-buffered formalin. Other fixatives are not acceptable.

6. Decalcified specimens (eg, bone marrow core biopsies) are not acceptable.

Specimen Stability Information: Ambient

Additional Information:

1. Testing may be canceled if resultant extracted DNA does not meet concentration requirements.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Frozen tissue

Container/Tube: Plastic container

Specimen Volume: 100 mg

Collection Instructions: Freeze tissue within 1 hour of collection

Specimen Stability Information: Frozen

Additional Information: Testing may be canceled if resultant extracted DNA does not meet concentration requirements.

Specimen Type: Body fluid

Container/Tube: Sterile container

Specimen Volume: 5 mL

Collection Instructions: Specify the type of fluid being submitted.

Specimen Stability Information: Refrigerated 14 days/Frozen 14 days

Additional Information: Testing may be canceled if resultant extracted DNA does not meet concentration requirements.

Specimen Type: Extracted DNA

Container/Tube: 1.5- to 2-mL tube

Specimen Volume: Entire specimen

Collection Instructions:

1. DNA must be extracted within 14 days after collection.

2. Label specimen as extracted DNA and source of specimen.

3. Indicate volume and concentration of DNA on label.

Specimen Stability Information: Frozen (preferred)/Refrigerated/Ambient

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). We cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied.

Specimen Minimum Volume: Whole blood, bone marrow aspirate, body fluid: 0.5 mL; Frozen tissue: 50 mg; Extracted DNA: 100 microliters (mcL) at 20 ng/mcL; Tissue slides: 10 unstained slides

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	14 days

CPT Code Information:

81450

MCBPP 619604 MayoComplete Bladder and Prostate Cancer Panel, Next-Generation Sequencing, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 216 mm(2)

-Minimum amount of tumor area: tissue 36 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirement for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4 mm x 10 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 10 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens.

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears or ThinPrep)

Slides: 1 to 3 Slides

Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a total of

5000 nucleated cells (preferred), or at least 3000 nucleated cells (minimum).

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88381-Microdissection, manual
81457

NGCLN 618476 MayoComplete Chronic Lymphoid Neoplasms, Next-Generation Sequencing, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Bone marrow aspirate

Container/Tube:

Preferred: Lavender or pink top EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Stability Information: Ambient (preferred) 14 days/Refrigerate 14 days

Additional Information: To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender or pink top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Label specimen as blood.

Specimen Stability Information: Ambient (preferred) 14 days/Refrigerate 14 days

Additional Information: To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Paraffin-embedded tissue

Container/Tube: Paraffin block

Collection Instructions:

1. If available, send 1 representative hematoxylin and eosin-stained slide.

2. Minimum amount of tumor nuclei is 20%

3. Required amount of tissue area is at least 25 mm(2)

4. Tissue should be fixed in 10% neutral-buffered formalin. Other fixatives are not

acceptable.

5. Decalcified specimens (eg, bone marrow core biopsies) are not acceptable.

Specimen Stability Information: Ambient

Additional Information: If the quality of the biopsy specimen is poor or the target tumor cell population is below 20%, testing should not be ordered. Testing may be canceled if DNA requirements are inadequate.

Acceptable:

Specimen Type: Tissue slide

Slides: 20 unstained slides

Container/ Tube: Transport in plastic slide holders.

Collection Instructions:

1. Send 20 unstained, nonbaked slides with 5-micron thick sections of tissue.

2. If available, also send 1 representative hematoxylin and eosin-stained slide.

3. Minimum amount of tumor nuclei is 20%

4. Required amount of tissue area is at least 25 mm(2)

5. Tissue should be fixed in 10% neutral-buffered formalin. Other fixatives are not acceptable.

6. Decalcified specimens (eg, bone marrow core biopsies) are not acceptable.

Specimen Stability Information: Ambient

Additional Information:

1. Testing may be canceled if resultant extracted DNA does not meet concentration requirements.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Frozen tissue

Container/Tube: Plastic container

Specimen Volume: 100 mg

Collection Instructions: Freeze tissue within 1 hour of collection

Specimen Stability Information: Frozen

Additional Information: Testing may be canceled if resultant extracted DNA does not meet concentration requirements.

Specimen Type: Body fluid

Container/Tube: Sterile container

Specimen Volume: 5 mL

Collection Instructions: Specify the type of fluid being submitted.

Specimen Stability Information: Refrigerated 14 days/Frozen 14 days

Additional Information: Testing may be canceled if resultant extracted DNA does not meet concentration requirements.

Specimen Type: Extracted DNA

Container/Tube: 1.5- to 2-mL tube

Specimen Volume: Entire specimen

Collection Instructions:

1. DNA must be extracted within 14 days after collection.

2. Label specimen as extracted DNA and source of specimen.

3. Indicate volume and concentration of DNA on label.

Specimen Stability Information: Frozen (preferred)/Refrigerated/Ambient

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). We cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied.

Forms

Specimen Minimum Volume: Whole blood, bone marrow aspirate, body fluid: 0.5 mL; Frozen tissue:

50 mg; Extracted DNA: 100 microliters (mcL) at 20 ng/mcL; Tissue slides: 10 unstained slides

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	14 days

CPT Code Information:

81450

MCCRC 616489 MayoComplete Colorectal Cancer Panel, Next-Generation Sequencing, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 216 mm(2)

-Minimum amount of tumor area: tissue 36 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirements for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4 mm x 10 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 10 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens.

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears or ThinPrep)

Slides: 1 to 3 Slides

Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a preferred total of 5000 nucleated cells, or a minimum of at least 3000 nucleated cells.

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88381 - Microdissection, manual

81457

MCSRC 616492 MayoComplete Comprehensive Sarcoma Panel, Next-Generation Sequencing, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 360 mm(2)

-Minimum amount of tumor area: tissue 72 mm(2)

-These amounts are cumulative over up to 15 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirements for Sold Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4 mm x 10 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 10 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens**.**

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 15 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

15 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (Diff-Quik or Pap stained direct smears or ThinPrep)

Slides: 2 to 6 Slides

Collection Instructions: Submit 2 to 6 slides, stained and coverslipped, with a total of 10,000 nucleated cells (preferred) or 2 slides with at least 2000 nucleated cells per slide (minimum).

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

81457

81456

MCECP 619631 MayoComplete Endometrial Carcinoma Panel, Next-Generation Sequencing, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 216 mm(2)

-Minimum amount of tumor area: tissue 36 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirement for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4 mm x 10 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 10 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens.

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears or ThinPrep)

Slides: 1 to 3 Slides

Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a preferred total of 5000 nucleated cells, or a minimum of at least 3000 nucleated cells.

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.
Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88381-Microdissection, manual

81457

MCGST 616490 MayoComplete Gastrointestinal Stromal Tumor (GIST) Panel, Next-Generation Sequencing, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 216 mm(2)

-Minimum amount of tumor area: tissue 36 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirement for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4 mm x 10 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 10 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens..

Acceptable: Submit **at least one** of the following specimens**.**

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting,

digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears or ThinPrep)

Slides: 1 to 3 Slides

Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a total of 5000 nucleated cells (preferred) or at least 3000 nucleated cells (minimum).

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88381-Microdissection, manual
81457

MCGYN 619622 MayoComplete Gynecological Cancer Panel, Next-Generation Sequencing, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 216 mm(2)

-Minimum amount of tumor area: tissue 36 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirement for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4 mm x 10 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 10 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens**.**

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears or ThinPrep)

Slides: 1 to 3 Slides

Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a total of 5000 nucleated cells (preferred) or at least 3000 nucleated cells (minimum).

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88381-Microdissection, manual

81457

NGHIS 618480 MayoComplete Histiocytic Neoplasms, Next-Generation Sequencing, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Bone marrow aspirate

Container/Tube:

Preferred: Lavender or pink top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Stability Information: Ambient (preferred) 14 days/Refrigerate 14 days

Additional Information: To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender or pink top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Label specimen as blood.

Specimen Stability Information: Ambient (preferred) 14 days/Refrigerate 14 days

Additional Information: To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Paraffin-embedded tissue

Container/Tube: Paraffin block

Collection Instructions:

1. If available, send 1 representative hematoxylin and eosin-stained slide.

2. Minimum amount of tumor nuclei is 20%.

3. Required amount of tissue area is at least 25 mm(2).

4. Tissue should be fixed in 10% neutral-buffered formalin. Other fixatives are not acceptable.

5. Decalcified specimens (eg, bone marrow core biopsies) are not acceptable.

Specimen Stability Information: Ambient

Additional Information: If the quality of the biopsy specimen is poor or the target tumor cell population is below 20%, testing should not be ordered. Testing may be canceled if DNA requirements are inadequate.

Acceptable:

Specimen Type: Tissue slide

Slides: 20 unstained slides

Container/ Tube: Transport in plastic slide holders.

Collection Instructions:

1. Send 20 unstained, nonbaked slides with 5-micron thick sections of tissue.

2. If available, also send 1 representative hematoxylin and eosin-stained slide.

3. Minimum amount of tumor nuclei is 20%

4. Required amount of tissue area is at least 25 mm(2)

5. Tissue should be fixed in 10% neutral-buffered formalin. Other fixatives are not acceptable.

6. Decalcified specimens (eg, bone marrow core biopsies) are not acceptable.

Specimen Stability Information: Ambient

Additional Information:

1. Testing may be canceled if resultant extracted DNA does not meet concentration requirements.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Frozen tissue

Container/Tube: Plastic container

Specimen Volume: 100 mg

Collection Instructions: Freeze tissue within 1 hour of collection

Specimen Stability Information: Frozen

Additional Information: Testing may be canceled if resultant extracted DNA does not meet concentration requirements.

Specimen Type: Body fluid

Container/Tube: Sterile container

Specimen Volume: 5 mL

Collection Instructions: Specify the type of fluid being submitted.

Specimen Stability Information: Refrigerated 14 days/Frozen 14 days

Additional Information: Testing may be canceled if resultant extracted DNA does not meet concentration requirements.

Specimen Type: Extracted DNA

Container/Tube: 1.5- to 2-mL tube

Specimen Volume: Entire specimen

Collection Instructions:

1. Label specimen as extracted DNA and source of specimen

2. Indicate volume and concentration of DNA on label

Specimen Stability Information: Frozen (preferred)/Refrigerated/Ambient

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). We cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied.

Specimen Minimum Volume: Whole blood, bone marrow aspirate, body fluid: 0.5 mL; Frozen tissue: 50 mg; Extracted DNA: 100 microliters (mcL) at 20 ng/mcL

Transport Temperature:

Specimen Type	Temperature	Time

Varies	Varies	14 days

CPT Code Information:

81450

MCKCP 619595 MayoComplete Kidney Cancer Panel, Next-Generation Sequencing, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 216 mm(2)

-Minimum amount of tumor area: tissue 36 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirement for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4 mm x 10 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 10 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens**.**

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears or ThinPrep)

Slides: 1 to 3 Slides

Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a total of 5000 nucleated cells (preferred) or at least 3000 nucleated cells (minimum).

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88381-Microdissection, manual

81457

MCLBP 614939 MayoComplete Liquid Biopsy Panel, Next-Generation Sequencing, Cell-Free DNA

Specimen Requirements:

Supplies: Streck Black/Tan Top Tube Kit (T715)

Container/Tube: Streck Cell-Free DNA (cfDNA) blood collection kit

Specimen Volume: Two 10-mL Streck Cell-Free DNA blood collection tubes

Additional Information: Only blood collected in Streck Cell-Free DNA BCT tubes will be accepted for analysis. Whole blood will be processed to produce platelet-poor plasma before cfDNA isolation.

Specimen Minimum Volume: One 10-mL Streck tube

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole blood	Ambient (preferred)	7 days	Streck Black/Tan top
	Refrigerated	7 days	Streck Black/Tan top

CPT Code Information:

81463

MCLNM 616487 MayoComplete Lung Cancer Mutations, Next-Generation Sequencing, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 216 mm(2)

-Minimum amount of tumor area: tissue 36 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

- For specimen preparation guidance, see Tissue Requirements for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4 mm x 10 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 10 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens.

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears or ThinPrep)

Slides: 1 to 3 Slides

Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a total of 5000 nucleated cells (preferred) or at least 3000 nucleated cells (minimum).

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88381–Microdissection, manual

81457

MCLNG 616486 MayoComplete Lung Cancer-Targeted Gene Panel with Rearrangement, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 360 mm(2)

-Minimum amount of tumor area: tissue 72 mm(2)

-These amounts are cumulative over up to 20 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirements for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4 mm x 20 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 20 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens.

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 20 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

20 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 20 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (Diff-Quik, Pap-stained direct smears, or ThinPrep)

Slides: 2 to 3 Slides

Collection Instructions: Submit at least 2 slides stained and coverslipped with a total of 10,000 nucleated cells (preferred) or at least 2000 nucleated cells (minimum) per slide.

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will

not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88381-Microdissection, manual

81457

MCLNR 616488 MayoComplete Lung Rearrangements, Rapid Test, Tumor

Specimen Requirements:

This assay requires at least 10% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 36 mm(2)

-Minimum amount of tumor area: tissue 18 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens.

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears or ThinPrep)

Slides: 1 to 3 Slides

Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a total of 5000 nucleated cells (preferred) or at least 3000 nucleated cells (minimum).

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

81449

MCMLN 616491 MayoComplete Melanoma Panel, Next-Generation Sequencing, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 216 mm(2)

-Minimum amount of tumor area: tissue 36 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirements for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4 mm x 10 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 10 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens**.**

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears or ThinPrep)

Slides: 1 to 3 Slides

Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a total of 5000 nucleated cells (preferred) or at least 3000 nucleated cells (minimum).

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88381-Microdissection, manual

81457

OncoHeme Next-Generation Sequencing, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred

Specimen Type: Bone marrow aspirate

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Stability Information: Ambient (preferred) 14 days/Refrigerate 14 days

Additional Information: To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Acceptable:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Label specimen as blood.

Specimen Stability Information: Ambient (preferred) 14 days/Refrigerate 14 days

Additional Information: To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Extracted DNA from blood or bone marrow

Container/Tube: 1.5- to 2-mL tube

Specimen Volume: Entire specimen

Collection Instructions:

1. DNA must be extracted within 14 days after collection.

2. Label specimen as extracted DNA and source of specimen.

3. Provide volume and concentration of the DNA.

Specimen Stability Information: Frozen (preferred)/Refrigerate/Ambient

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). We cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied.

Specimen Minimum Volume: Blood, Bone marrow: 1 mL
Extracted DNA: 100 mcL at 20 ng/mcL concentration

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	14 days

CPT Code Information:

81450

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 216 mm(2)

-Minimum amount of tumor area: tissue 36 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirement for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4 mm x 10 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 10 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens**.**

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears or ThinPrep)

Slides: 1 to 3 Slides

Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a total of 5000 nucleated cells (preferred) or at least 3000 nucleated cells (minimum).

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88381-Microdissection, manual

81457

NGPCM 618479 MayoComplete Plasma Cell Myeloma, Next-Generation Sequencing, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Bone marrow aspirate

Container/Tube: Lavender or pink top (EDTA) or yellow top (ACD)

Specimen Volume: 2 mL

Collection Instructions:

1. Minimum plasma cell percentage is 5%.

2. Invert several times to mix bone marrow.

3. Send bone marrow specimen in original tube. **Do not aliquot.**

4. Label specimen as bone marrow.

5. **Fresh specimen is required for this test**, as testing is performed on sorted cells.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerate

Additional Information: To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Paraffin-embedded bone marrow clot

Container/Tube: Paraffin block

1. Send 1 hematoxylin and eosin-stained slide in addition to the paraffin block.

2. Minimum plasma cell percentage is 20%.

3. Required amount of tissue area is at least 25 mm(2).

4. Tissue should be fixed in 10% neutral-buffered formalin. Other fixatives are not acceptable.

5. Decalcified specimens (eg, bone marrow core biopsies) are not acceptable.

Specimen Stability Information: Ambient

Additional Information: If the quality of the specimen is poor or the plasma cell population is below 20%, testing should not be ordered. Testing may be canceled if DNA requirements are inadequate.

Acceptable:

Specimen Type: Tissue slide, bone marrow clot

Slides: 20 unstained slides

Container/Tube: Transport in plastic slide holders.

Collection Instructions:

1. Send 20 unstained, nonbaked slides with 5-micron thick sections of tissue and 1 hematoxylin and eosin-stained slide.

2. Minimum amount of plasma cell percentage is 20%.

4. Required amount of tissue area is at least 25 mm(2).

5. Tissue should be fixed in 10% neutral-buffered formalin. Other fixatives are not acceptable.

6. Decalcified specimens (eg, bone marrow core biopsies) are not acceptable.

Specimen Stability Information: Ambient

Additional Information:

1. Testing may be canceled if resultant extracted DNA does not meet concentration requirements.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Bone marrow aspirate: 1 mL; Tissue slides: 10 unstained slides

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81450

MCFRC 619613 MayoComplete Renal Cell Carcinoma with Fibromyomatous Stroma Panel, Next-Generation Sequencing, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 216 mm(2)

-Minimum amount of tumor area: tissue 36 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirement for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4 mm x 10 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 10 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens.

Acceptable:

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.
Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears or ThinPrep)
Slides: 1 to 3 Slides
Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a preferred total of 5000 nucleated cells, or a minimum of at least 3000 nucleated cells.
Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.
Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88381-Microdissection, manual

81479

81406

81407

81404

81479 (if appropriate for government payers)

MCSMP 616493 MayoComplete Sarcoma Mutation Panel, Next-Generation Sequencing, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 216 mm(2)

-Minimum amount of tumor area: tissue 36 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirement for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4 mm x 10 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 10 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens**.**

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (direct smears or ThinPrep)

Slides: 1 to 3 Slides

Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a total of 5000 nucleated cells (preferred) or at least 3000 nucleated cells (minimum).

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

81457

MCSTP 606162 MayoComplete Solid Tumor Panel, Next-Generation Sequencing, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei. However, 40% tumor is preferred.

-Preferred amount of tumor area: 432 mm(2) tissue on up to 20 unstained slides

-Minimum amount of tumor area: 72 mm(2) tissue on up to 20 unstained slides

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For this test, at least 4.6 mm x 4.6 mm areas on 20 unstained slides is preferred: this is approximately equivalent to 432 mm(2). The minimum acceptable area is 1.9 mm x 1.9 mm on 20 unstained slides: approximately equivalent to 72 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens**.**

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 20 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

20 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 20 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slide (Diff-Quik, Pap-stained direct smears, or ThinPrep)

Slides: 2 to 6 Slides

Collection Instructions: Submit at least 2 slides, stained and coverslipped, with a total of 10,000 nucleated cells (preferred) or at least 2000 nucleated cells (minimum) per slide.

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

81459

88381-Microdissection, manual

NGTCL 618478 MayoComplete T-Cell Lymphoma, Next-Generation Sequencing, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Bone marrow aspirate

Container/Tube:

Preferred: Lavender or pink top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Stability Information: Ambient (preferred) 14 days/Refrigerate 14 days

Additional Information: To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender or pink top (EDTA) or yellow top (ACD)

Acceptable: Green top (sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Label specimen as blood.

Specimen Stability Information: Ambient (preferred) 14 days/Refrigerate 14 days

Additional Information: To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Paraffin-embedded tissue

Container/Tube: Paraffin block

Collection Instructions:

1. If available, send 1 representative hematoxylin and eosin-stained slide.

2. Minimum amount of tumor nuclei is 20%

3. Required amount of tissue area is at least 25 mm(2)

4. Tissue should be fixed in 10% neutral-buffered formalin. Other fixatives are not acceptable.

5. Decalcified specimens (eg, bone marrow core biopsies) are not acceptable.

Specimen Stability Information: Ambient

Additional Information: If the quality of the biopsy specimen is poor or the target tumor cell population is below 20%, testing should not be ordered. Testing may be canceled if DNA requirements are inadequate.

Acceptable:

Specimen Type: Tissue slide

Slides: 20 unstained slides

Container/ Tube: Transport in plastic slide holders.

Collection Instructions:

1. Send 20 unstained, nonbaked slides with 5-micron thick sections of tissue.

2. If available, also send 1 representative hematoxylin and eosin-stained slide.

3. Minimum amount of tumor nuclei is 20%

4. Required amount of tissue area is at least 25 mm(2)

5. Tissue should be fixed in 10% neutral-buffered formalin. Other fixatives are not acceptable.

6. Decalcified specimens (eg, bone marrow core biopsies) are not acceptable.

Specimen Stability Information: Ambient

Additional Information:

1. Testing may be canceled if resultant extracted DNA does not meet concentration requirements.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Frozen tissue

Container/Tube: Plastic container

Specimen Volume: 100 mg

Collection Instructions: Freeze tissue within 1 hour of collection

Specimen Stability Information: Frozen

Additional Information: Testing may be canceled if resultant extracted DNA does not meet concentration requirements.

Specimen Type: Body fluid

Container/Tube: Sterile container

Specimen Volume: 5 mL

Collection Instructions: Specify the type of fluid being submitted

Specimen Stability Information: Refrigerated 14 days/Frozen 14 days

Additional Information: Testing may be canceled if resultant extracted DNA does not meet concentration requirements.

Specimen Type: Extracted DNA

Container/Tube: 1.5- to 2-mL tube

Specimen Volume: Entire specimen

Collection Instructions:

1. Label specimen as extracted DNA and source of specimen

2. Indicate volume and concentration of DNA on label

Specimen Stability Information: Frozen (preferred)/Refrigerated/Ambient

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). We cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied.

Specimen Minimum Volume: Whole blood, bone marrow aspirate, body fluid: 0.5 mL; Frozen tissue: 50 mg; Extracted DNA: 100 microliters (mcL) at 20 ng/mcL; Tissue slides: 10 unstained slides

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	14 days

CPT Code Information:

81450

MCRSP 622969 MayoComplete Targeted RNA Sequencing Panel, Next-Generation Sequencing, Tumor

Specimen Requirements:

This assay requires at least 10% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 144 mm(2)

-Minimum amount of tumor area: tissue 36 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirements for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4 mm x 10 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 10 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 3, if available, or 2 of the following specimens.

Acceptable: Submit **at least one** of the following specimens.

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Cytology slides (direct smears or ThinPrep)

Slides: 1 to 3 Slides

Collection Instructions: Submit 1 to 3 slides stained and coverslipped with a total of 5000 nucleated cells (preferred) or at least 3000 nucleated cells (minimum).

Note: Glass coverslips are preferred; plastic coverslips are acceptable but will result in longer turnaround times.

Additional Information:

1. Cytology slides will not be returned. An image of the slides will be stored per regulatory requirements.

2. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

81456

JMDMF
616053

MDM2 (12q15) Amplification, Well-Differentiated Liposarcoma/Atypical Lipomatous Tumor, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tumor tissue block. Blocks prepared with alternative fixation methods are not accepted; provide fixation method used.

Acceptable:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained and 2 unstained

Collection Instructions: Submit 1 slide stained with hematoxylin and eosin and 2 consecutive, unstained, positively charged, unbaked slides with 4 to 5-micron-thick sections of the tumor tissue.

Specimen Minimum Volume: Slides: 1 hematoxylin and eosin-stained and 1 unstained

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88377

MDM2F
63049

MDM2 (12q15) Amplification, Well-Differentiated Liposarcoma/Atypical Lipomatous Tumor, FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue block (fresh tissue is **not acceptable**)

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block.

2. Blocks prepared with alternative fixation methods (eg, Prefer, Bouin's) will be attempted but are less favorable for successful results. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Decalcified paraffin-embedded specimens will have testing attempted; however, the success rate is approximately 50%. **Testing may be canceled** if sufficient tumor tissue is not present.

3. **Submitted fresh tissue specimens will be canceled upon receipt**. If only fresh tissue is available, embed in paraffin prior to sending.

Acceptable:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained and 4 unstained

Collection Instructions: Submit 1 slide stained with hematoxylin and eosin and 4 consecutive unstained, positively charged, unbaked slides with 5-micron-thick sections of the tumor tissue.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Slides: 1 Hematoxylin and eosin-stained and 2 unstained

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2, 88291-DNA probe, each (first probe set), Interpretation and report

88271x2-DNA probe, each; each additional probe set (if appropriate)

88271-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271x2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271x3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

88275-Interphase in situ hybridization, 100 to 300 cells, each probe set (if appropriate)

MEAD
82890

Meadow Fescue, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

MFOX
82914

Meadow Foxtail, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

ROPG
34941

Measles (Rubeola) Antibodies, IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Instructions: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86765

ROM
80979

Measles (Rubeola) Antibodies, IgM, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86765

ROGM
62066

Measles (Rubeola) Virus Antibody, IgM and IgG, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86765 x2

MEASU 617822 Measles Virus, Molecular Detection, PCR, Random, Urine

Specimen Requirements:

Container/Tube: Sterile tube

Specimen Volume: 1 mL

Collection Instructions:

1. Collect a random urine specimen during the 6-to-14-day period after rash onset in a sterile urine tube.

2. No preservative.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated	7 days

CPT Code Information:

87638

MEASR 617821 Measles Virus, Molecular Detection, PCR, Throat

Specimen Requirements:

Specimen Type: Throat Swab

Supplies: Culturette (BBL Culture Swab) (T092)

Container/Tube: Sterile container with transport media

Specimen Volume: Entire collection

Collection Instructions:

1. Apply swab tip back and forth firmly over mucosal surface of pharynx (throat) to maximize recovery of cells.

2. Place swab in viral transport media (eg, Liquid Stuart, M4-RT, M4, M5, Bartels FlexTrans Transport Media) and refrigerate for transport to the testing laboratory.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87638

MMRV
61853

Measles, Mumps, Rubella, and Varicella (MMRV) Immune Status Profile, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Instructions: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	14 days

CPT Code Information:

86735

86762

86765

86787

FFMTH 75544 Meconium Methadone Screen with Reflex Confirmation

Specimen Requirements:

Container/Tube: Sterile collection container

Specimen Volume: 3 g

Collection Instructions: Collect specimen into the same sterile collection container until 3 g of meconium have been collected or until the first milk stool appears. When at least 3 g of meconium has been collected, tightly screw on the cap of the collection vial and send specimen ambient.

Note: Specimens from different voids may be pooled if necessary.

Specimen Minimum Volume: 1 g

Transport Temperature:

Specimen Type	Temperature	Time
Meconium	Ambient (preferred)	14 days
	Frozen	365 days
	Refrigerated	14 days

CPT Code Information:

80307

MECP2 Gene, Full Gene Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a bone marrow transplant.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Any anticoagulant

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Stability Information: Ambient (preferred)/Refrigerated

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/**Tube**: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm punch

Specimen Stability Information: Refrigerated (preferred)/Ambient

Additional Information: A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblast

Container/Tube: T-25 flask

Specimen Volume: 2 flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy from another laboratory. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred)/Refrigerated (<24 hours)

Additional Information: A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filtration Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 (formerly Ahlstrom 226) filter paper or blood spot collection card

Specimen Volume: 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect Dried Blood Spot Samples.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Due to lower concentration of DNA yielded from blood spot, it is possible that additional specimen may be required to complete testing.

2. For collection instructions, see Blood Spot Collection Instructions

3. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

4. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Minimum Volume: Blood: 1 mL; Skin biopsy, cultured fibroblasts, dried blood spots, or saliva: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81302

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

Medium-Chain Acyl-CoA Dehydrogenase Deficiency (MCAD), ACADM Gene Sequencing with Deletion/Duplication, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about for testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Tissue biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 0.5 to 3 cm (3) or larger

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed, and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Prenatal cultured fibroblasts (eg, products of conception), amniocytes, or other confluent cultured cells. This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81479

88233-Tissue culture, skin or solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

MEGR
82347

Megrim, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
More than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL dead space

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

MELAI
82724

Melaleuca leucadendron, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

MELAN
70504

Melan A (MART-1) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

FMARP 75449 Melanoma Associated Retinopathy MAR Panel by Immunoblot and IHC

Specimen Requirements:

Submit only one of the following specimens:

Serum:

Draw blood in a plain, red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 5 mL serum refrigerated in a plastic vial.

Plasma:

Draw blood in a lavender-top (EDTA) tube(s). Spin down and send 5 mL EDTA plasma refrigerated in a plastic vial.

Complete and submit with specimen:

1. Completed OHSU Ocular request form
2. Clinical history
3. Referring physician information (name & phone number)

-NOTE: Without this information, testing cannot be completed.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated	7 days

CPT Code Information:

84182 x 6

MAGE4 622925 Melanoma-Associated Antigen-A4 (MAGE-A4), Semi-Quantitative Immunohistochemistry, Manual

Specimen Requirements:

Specimen Type: Tissue

Supplies: Pathology Packaging Kit (T554)

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

3 Unstained glass, "positively charged" slides with 4-microns formalin-fixed, paraffin-embedded tissue

Additional Information:

1. One slide will be stained with hematoxylin and eosin and returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most

cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Special	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88360

FMELA
75386

Melatonin, Plasma

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Specimen Type: Plasma

Collection Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Pink top (EDTA)

Submission Container/Tube: Plastic vial

Specimen volume: 3 mL

Collection instructions:

1. Draw blood in EDTA (lavender top) tube(s).

2. Centrifuge and send 1 mL of plasma refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Refrigerated (preferred)	30 days
	Frozen	365 days
	Ambient	14 days

CPT Code Information:

80299

FMELG
57652

Melons IgG

Specimen Requirements:

Draw blood in a plain, red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

MELN
82762

Melons, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days

	Frozen	90 days

CPT Code Information:

86003

MSMN 620788 Membranous Nephropathy Target Antigen Identification, Mass Spectrometry, Tissue

Specimen Requirements:

Specimen Type: Kidney tissue

Supplies: Pathology Packaging Kit (T554)

Container/Tube: Formalin-fixed, paraffin-embedded tissue block

Additional Information: Do not send fixed tissue slides. Testing can only be done on paraffin-embedded tissue blocks.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Special	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

82542

88380

CSFME 67846 Meningitis/Encephalitis Pathogen Panel, PCR, Spinal Fluid

Specimen Requirements:

Specimen Type: Spinal fluid

Container/Tube: Sterile vial

Specimen Volume: 1 mL

Collection Instructions: Frozen specimens are **not** acceptable.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated	7 days

CPT Code Information:

87483

FMEP
90090

Meperidine (Demerol) and Normeperidine, serum

Specimen Requirements:

Submit only 1 of the following specimens:

Plasma

Draw blood in a green-top (sodium heparin) tube(s). Plasma gel tube is not acceptable. Spin down and send 3 mL of sodium heparin plasma refrigerated in a plastic vial.

Serum

Draw blood in a plain red-top tube(s). Serum gel tube is not acceptable. Spin down and send 3 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.6 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	14 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80362

FMMM
57766

Mephedrone, MDPV and Methylone, Urine

Specimen Requirements:

Container: Preservative-free plastic urine container

Specimen volume: 10 mL

Collection instructions

1. collect a random urine specimen
2. No preservative

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80307-single drug

80371-(if appropriate)

FMERC 91120 Mercaptopurine (6-MP, Purinethol)

Specimen Requirements:

Submit only 1 of the following specimens:

Plasma

Draw blood in a green-top (sodium heparin) tube(s), **plasma gel tube is not acceptable.** Spin down and send 2 mL sodium heparin plasma refrigerated in a plastic vial.

Serum

Draw blood in a plain, red-top tube(s), **serum gel tube is not acceptable.** Spin down and send 2 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80299

HGUOE
608897

Mercury Occupational Exposure, Random, Urine

Specimen Requirements:

Patient Preparation:

1. For 48 hours before collection, patient **should not** eat seafood.

2. High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours.**

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine container with no metal cap or glued insert

Submission Container/Tube: Plastic, 10-mL urine tube or clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 3 mL

Collection Instructions:

1. Collect a random urine specimen.

2. See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

83825

82570

HGOU 608893 Mercury Occupational Exposure, Random, Urine (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

HGUOE/ Mercury Occupational Exposure, Random, Urine

HMUOE / Heavy Metal Occupational Exposure, with Reflex, Random, Urine

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

83825

HGU 8592 Mercury, 24 Hour, Urine

Specimen Requirements:

Patient Preparation:

1. For the 48-hour period prior to the start of collection, as well as during the collection, patient **should not** eat seafood.

2. High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours.**

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine container with no metal cap or glued insert

Submission Container/Tube: Plastic, 10-mL urine tube or a clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 3 mL

Collection Instructions:

1. Collect urine for 24 hours.

2. Refrigerate specimen within 4 hours of completion of 24-hour collection.

3. See Metals Analysis Specimen Collection and Transport for complete instructions.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

83825

HG
8618

Mercury, Blood

Specimen Requirements:

*****This specimen container cannot be opened or used for any other testing before shipping.*****

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours**.

Supplies:

-Metal Free EDTA 3 mL Tube (T989)

-Metal Free B-D Tube (EDTA), 6 mL (T183)

Collection Container/Tube:

Preferred: Royal blue-top BD vacutainer with EDTA blood collection tube (3 mL) (BD catalog no. 367777) (T989)

Acceptable: Royal blue-top BD Vacutainer Plus with EDTA blood collection tube (6 mL) (BD catalog no. 368381) (T183)

Specimen Volume: 1 mL

Collection Instructions:

1. See Metals Analysis Specimen Collection and Transport for complete instructions.

2. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

83825

HGHAR 8498 Mercury, Hair

Specimen Requirements:

Supplies: Hair and Nails Collection Kit (T565)

Source: Head, beard, mustache, chest, pubic

Specimen Volume: 0.2 g

Collection Instructions: Prepare and transport specimen per the instructions in the kit or see Collecting Hair and Nails for Metals Testing.

Specimen Minimum Volume: 0.05 g

Transport Temperature:

Specimen Type	Temperature	Time
Hair	Ambient (preferred)	
	Frozen	
	Refrigerated	

CPT Code Information:

83825

HGNA
89856

Mercury, Nails

Specimen Requirements:

Supplies: Hair and Nails Collection Kit (T565)

Source: Fingernails or toenails

Specimen Volume: 0.2 g

Collection Instructions:

1. Prepare and transport specimen per the instructions in the kit or see Collecting Hair and Nails for Metals Testing.

2. Clippings should be taken from all 10 fingernails or toenails.

Specimen Minimum Volume: 0.05 g

Transport Temperature:

Specimen Type	Temperature	Time
Nail	Ambient (preferred)	
	Frozen	
	Refrigerated	

CPT Code Information:

83825

HGUCR
608907

Mercury/Creatinine Ratio, Random, Urine

Specimen Requirements:

Patient Preparation:

1. For 48 hours before collection, patient **should not** eat seafood.

2. High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for 96 hours.**

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Clean, plastic urine container with no metal cap or glued

insert

Submission Container/Tube: Plastic, 10-mL urine tube or clean, plastic aliquot container with no metal cap or glued insert

Specimen Volume: 3 mL

Collection Instructions:

1. Collect a random urine specimen.

2. See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

83825

82570

HGCU
608903

Mercury/Creatinine Ratio, Random, Urine (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

HGUCR / Mercury/Creatinine Ratio, Random, Urine

HMUCR / Heavy Metal/Creatinine Ratio, with Reflex, Random Urine.

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

83825

MERKC
71538

Merkel CC (MCPyV) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

NF2
622010

Merlin (NF2) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

HBME 70452 Mesothelial Cell (HBME-1) Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

MESOF 609714 Mesothelioma, CDKN2A FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Tissue block (fresh tissue is **not acceptable**)

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block.

2. Blocks prepared with alternative fixation methods (eg, Prefer, Bouin's) will be attempted but are less favorable for successful results. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Decalcified paraffin-embedded specimens will have testing attempted; however, the success rate is approximately 50%. **Testing may be canceled** if sufficient tumor tissue is not present.

3. **Submitted fresh tissue specimens will be canceled upon receipt**. If only fresh tissue is available, embed in paraffin prior to sending.

Acceptable:

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin-stained and 4 unstained

Collection Instructions: Submit 1 slide stained with hematoxylin and eosin and 4 consecutive unstained, positively charged, unbaked slides with 5-micron thick sections of the tumor tissue.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Slides: 1 Hematoxylin and eosin-stained and 2 unstained

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2, 88291 - DNA probe, each (first probe set), Interpretation and report

88271x2 - DNA probe, each; each additional probe set (if appropriate)

88271x1 - DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271x2 - DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271x3 - DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52 - Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274 - Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

88275 - Interphase in situ hybridization, 100 to 300 cells, each probe set (if appropriate)

MESQ
82776

Mesquite, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

METF
58123

MET (7q31), FISH, Tissue

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred

Specimen Type: Tissue block (fresh tissue is **not acceptable**)

Collection Instructions:

1. Submit a formalin-fixed, paraffin-embedded tumor tissue block.

2. Blocks prepared with alternative fixation methods (eg, Prefer, Bouin's) will be attempted but are less favorable for successful results. Provide fixation method used.

Additional Information:

1. Paraffin-embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Decalcified paraffin-embedded specimens will have testing attempted; however, the success rate is approximately 50%.Testing **may be** canceled if sufficient tumor tissue is not present.

3. **Submitted fresh tissue specimens will be canceled upon receipt**. If only fresh tissue is available, embed in paraffin prior to sending.

Acceptable

Specimen Type: Tissue slides

Slides: 1 Hematoxylin and eosin stained and 3 unstained

Collection Instructions: Submit 3 consecutive unstained, positively charged, unbaked slides with 5 micron-thick sections of the tumor tissue and 1 slide stained with hematoxylin and eosin.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: Slides: 1 Hematoxylin and eosin stained and 2 unstained

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2, 88291-DNA probe, each (first probe set), Interpretation and report

88271x2-DNA probe, each; each additional probe set (if appropriate)

88271x1-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271x2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271x3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

METSP 623179 MET (SP44), Semi-Quantitative Immunohistochemistry, Manual

Specimen Requirements:

Specimen Type: Tissue

Supplies: Pathology Packaging Kit (T554)

Submit:

Formalin-fixed, paraffin-embedded tissue block

OR

3 Unstained glass, "positively charged" slides with 4-microns formalin-fixed, paraffin-embedded tissue

Collection Instructions: Store slides at 2 to 8 degrees C and send within 45 days.

Additional Information:

1. One slide will be stained with hematoxylin and eosin and returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
Special	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88360

GL5CS 623397 Metabotropic Glutamate Receptor-5 (mGluR5) IgG, Cell Based Assay, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-DMS2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Serum

-ENS2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Serum

-EPS2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Serum

-MDS2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Serum

-PCDES / Pediatric Autoimmune Encephalopathy/CNS Disorder Evaluation, Serum

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255

GL5CC 623396 Metabotropic Glutamate Receptor-5 (mGluR5) IgG, Cell Based Assay, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-DMC2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-ENC2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-EPC2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-MDC2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-PCDEC / Pediatric Autoimmune Encephalopathy/CNS Disorder Evaluation, Spinal Fluid

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255

GL5TS 623488 Metabotropic Glutamate Receptor-5 (mGluR5) IgG, Tissue Immunofluorescence Titer, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-DMS2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Serum

-ENS2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Serum

-EPS2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Serum

-MDS2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Serum

-PCDES / Pediatric Autoimmune Encephalopathy/CNS Disorder Evaluation, Serum

Specimen Minimum Volume: Serum: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

GL5TC 623487 Metabotropic Glutamate Receptor-5 (mGluR5) IgG, Tissue Immunofluorescence Titer, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-DMC2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-ENC2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-EPC2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-MDC2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-PCDEC / Pediatric Autoimmune Encephalopathy/CNS Disorder Evaluation, Spinal Fluid

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86256

GL5IS 623395 Metabotropic Glutamate Receptor-5 (mGluR5) IgG, Tissue Immunofluorescence, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

-DMS2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Serum

-ENS2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Serum

-EPS2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Serum

-MDS2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Serum

-PCDES / Pediatric Autoimmune Encephalopathy/CNS Disorder Evaluation, Serum

Specimen Minimum Volume: Serum: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255

GL5IC 623394 Metabotropic Glutamate Receptor-5 (mGluR5) IgG, Tissue Immunofluorescence, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see:

-DMC2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-ENC2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-EPC2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid
-MDC2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-PCDEC / Pediatric Autoimmune Encephalopathy/CNS Disorder Evaluation, Spinal Fluid

Specimen Minimum Volume: 1.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255

MT 619972 Metallothionein Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

MHBRP 65129 Metamycoplasma hominis, Molecular Detection, PCR, Blood

Specimen Requirements:

The high sensitivity of amplification by polymerase chain reaction requires the specimen to be processed in an environment in which contamination of the specimen by *Metamycoplasma hominis* DNA is unlikely.

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Royal blue top (EDTA), pink top (EDTA), or sterile vial containing EDTA-derived aliquot

Specimen Volume: 1 mL

Collection Instructions: Send whole blood specimen in original tube (preferred). **Do not aliquot.**

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798

MHPRP 65134 Metamycoplasma hominis, Molecular Detection, PCR, Plasma

Specimen Requirements:

The high sensitivity of amplification by polymerase chain reaction requires the specimen to be processed in an environment in which contamination of the specimen by

Metamycoplasma hominis DNA is unlikely.

Collection Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Royal blue top (EDTA), pink top (EDTA)

Submission Container/Tube: Screw-capped, sterile container

Specimen Volume: 1 mL

Collection Instructions: Centrifuge and aliquot plasma into a sterile vial within 24 hours of collection.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798

MHRP 60756 Metamycoplasma hominis, Molecular Detection, PCR, Varies

Specimen Requirements:

The high sensitivity of amplification by polymerase chain reaction (PCR) requires the specimen to be processed in an environment in which contamination of the specimen by *Mycoplasma hominis* DNA is not likely.

Submit only 1 of the following specimens:

Specimen Type: Swab

Supplies:

-Culturette (BBL Culture Swab) (T092)

-BD E-Swab (T853)

-M4-RT (T605)

Sources: Vaginal, cervix, urethra, urogenital, chest/mediastinal; bronchus or lung (donor swab); or upper respiratory sources (only infants <3 months: nasopharynx, nose, throat)

Container/Tube:

Preferred: Culture swab transport system (Dacron or rayon swab with aluminum or plastic shaft with either Stuart or Amies liquid medium)

Acceptable: Swab in transport media: M4, M4-RT, M5, M6, universal transport media, or ESwab

Specimen Volume: 1 swab

Collection Instructions:

Vaginal:

1. Collect specimen by swabbing back and forth over mucosa surface to maximize recovery of cells.

2. Place swab back into swab cylinder.

Urethra or cervical:

1. Collect specimen by inserting swab 1 to 3 cm and rotating 360 degrees.

2. Place swab back into swab cylinder.

Wound:

1. Collect specimen by swabbing back and forth over wound surface to maximize recovery of cells.

2. Place swab back into swab cylinder.

Specimen Type: Fluid

Sources: Pelvic, peritoneal, amniotic, prostatic secretions, semen, reproductive drainage or fluid, pleural/chest, chest tube, pericardial

Container/Tube: Sterile container

Specimen Volume: 1 to 2 mL

Specimen Type: Respiratory

Sources: sputum, tracheal secretions, bronchial washings, bronchoalveolar lavage, lung; or nasal washings (Note: Nasal washings may only be submitted for infants <3 months of age.)

Container/Tube: Sterile container

Specimen Volume: 1 to 2 mL

Specimen Type: Synovial fluid

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Pink top (EDTA), royal blue top (EDTA), sterile vial containing EDTA-derived aliquot, red clot tube (no anticoagulant), or sterile container

Specimen Volume: 0.5 mL

Collection Instructions: Send specimen in original tube.

Specimen Type: Urine (first void), kidney/bladder stone, or ureter

Container/Tube: Sterile container

Specimen Volume: 10 mL or entire specimen

Collection instructions: Urine first void: Specimen can be collected at any time during the day. The patient should not have urinated for at least 1 hour prior to specimen collection. The first voided portion is the initial 20 to 30 mL of the urine stream obtained **without** cleaning the external urethra.

Specimen Type: Tissue

Sources: Placenta, products of conception, urogenital, respiratory, bronchus, chest/mediastinal, bone, spine, or joint

Container/Tube: Sterile container

Specimen Volume: 5 mm(3)

Collection Instructions:

1. Collect fresh tissue specimen.

2. Submit fresh tissue only, do not add fluid to tissue

3. Refrigerate or freeze specimen.

Specimen Minimum Volume: Fluid: 1 mL
Urine, first void: 2 mL
Swab: 1 swab
Tissue: 5 mm(3)

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days
	Frozen	7 days

CPT Code Information:

87798

META3 Metanephrines with 3-Methoxytyramine, 24 Hour, Urine
65158

Specimen Requirements:

Patient Preparation: Tricyclic antidepressants, labetalol, and sotalol medications may elevate levels of metanephrines producing results that cannot be interpreted. If clinically feasible, it is optimal to discontinue these medications at least 1 week before collection. For advice on assessing the risk of removing patients from these medications and alternatives, consider consultation with a specialist in endocrinology or hypertension.

Supplies: Urine Tubes, 10 mL (T068)

Submission Container/Tube: Plastic urine tube

Specimen Volume: 10 mL

Collection Instructions:

1. Complete 24-hour urine collections are preferred, especially for patients with episodic hypertension; ideally the collection should begin at the onset of a "spell."

2. Add 10 g (pediatric: 3 g) of boric acid or 25 mL (pediatric: 15 mL) of 50% acetic acid as preservative **at start of collection**.

3. Collect urine for 24 hours.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	21 days

CPT Code Information:

82542

83835

METAF
83006

Metanephrines, Fractionated, 24 Hour, Urine

Specimen Requirements:

Patient Preparation: Tricyclic antidepressants, labetalol, and sotalol medications may elevate levels of metanephrines producing results that cannot be interpreted. If clinically feasible, it is optimal to discontinue these medications at least 1 week before collection. For advice on assessing the risk of removing patients from these medications and alternatives, consider consultation with a specialist in endocrinology or hypertension.

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic urine tube

Specimen Volume: 10 mL

Collection Instructions:

1. Complete 24-hour urine collections are preferred, especially for patients with episodic hypertension; ideally the collection should begin at the onset of a "spell."

2. Add 10 g (pediatric: 3 g) of boric acid or 25 mL (pediatric: 15 mL) of 50% acetic acid as preservative **at start of collection**.

3. Collect urine for 24 hours.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days

	Ambient	28 days
	Frozen	28 days

CPT Code Information:

83835

PMET
81609

Metanephrines, Fractionated, Free, Plasma

Specimen Requirements:

Patient Preparation: Use of an Epi-pen within the last 7 days may produce inaccurate results.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Within 2 hours of collection centrifuge and aliquot plasma into a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen (preferred)	28 days
	Refrigerated	14 days
	Ambient	7 days

CPT Code Information:

83835

FMETN
75389

Metformin, Plasma

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Specimen Type: Plasma

Collection Container/Tube: Lavender top or pink top (EDTA) (Plasma gel tubes are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen volume: 1 mL

Collection instructions:

1. Draw blood in an EDTA (lavender top or pink top) tube.

2. Centrifuge and aliquot 1 mL of plasma into a preservative-free plastic vial.

3. Send refrigerated (preferred)

Specimen Stability Information: Refrigerated (preferred) 30 days/Ambient 30 days/Frozen 2 years

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Refrigerated (preferred)	30 days
	Ambient	30 days
	Frozen	

CPT Code Information:

80299

MDNS 36309 Methadone and Metabolites, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (Serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.0 mL

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

G0480

80358 (if appropriate for select payers)

MTDNX
62734

Methadone Confirmation, Chain of Custody, Random, Urine

Specimen Requirements:

Supplies: Chain of Custody Kit (T282)

Container/Tube: Chain-of-custody kit containing the specimen containers, seals, and documentation required.

Specimen Volume: 10 mL

Collection Instructions: Collect urine specimen in the container provided, seal, and submit with the associated documentation to satisfy the legal requirements for chain-of-custody testing.

Additional Information: Submitting less than 10 mL will compromise our ability to perform all necessary testing.

Specimen Minimum Volume: 2.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

G0480

80358 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

MTDNU
83129

Methadone Confirmation, Random, Urine

Specimen Requirements:

Supplies: Urine Tubes, 10 mL (T068)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 10 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Specimen Minimum Volume: 2.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	14 days

CPT Code Information:

G0480

80358 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

FMETH
57996

Methaqualone Confirmation, Urine

Specimen Requirements:

Container/Tube: Plastic, preservative-free urine container

Specimen Volume: 20 mL

Collection Instructions:

1. Collect 20 mL random urine without preservative.

2. Transfer into a preservative-free plastic urine container.

3. Ship refrigerate.

Specimen Minimum Volume: 10 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	10 days
	Frozen	396 days
	Ambient	72 hours

CPT Code Information:

80375

MET 81032 Methemoglobin and Sulfhemoglobin, Blood

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: Full tube

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Additional Information: Patient's age is required.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated	72 hours

CPT Code Information:

83050-Methemoglobin

83060-Sulfhemoglobin

MEV0 608089 Methemoglobin Summary Interpretation (Non-Orderable)

Transport Temperature:

Specimen Type	Temperature	Time

Whole Blood EDTA	Refrigerated

MEV1 Methemoglobinemia Evaluation, Blood

607495

Specimen Requirements:

The following specimens are required for testing:

Whole blood ACD-B specimen

2 Whole blood EDTA specimens

Container/Tube: Lavender top (EDTA) **and** yellow top (ACD solution B)

Specimen Volume:

EDTA: Two 4 mL tubes

ACD: One 6 mL tube

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: EDTA blood: 3 mL; ACD blood: 2.7 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD-B	Refrigerated	72 hours
Whole Blood EDTA	Refrigerated	72 hours

CPT Code Information:

83020-26-Hemoglobinopathy Interpretation

83020-Hb Variant, A2 and F Quantitation

83021-HPLC Hb Variant

82657-Methemoglobin reductase

83050-Methemoglobin, quantitative

83060-Sulfhemoglobin, quantitative

82664 (if appropriate)

83068 (if appropriate)

83789 (if appropriate)

88184 (if appropriate)

MEVI 608086 Methemoglobinemia Interpretation (Non-Orderable)

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood EDTA	Refrigerated	72 hours

CPT Code Information:

83020-26

MECPI 621419 Methicillin Resistance Gene, Molecular Detection, PCR, Varies

Specimen Requirements:

Supplies: Infectious Container, Large (T146)

Container/Tube: Agar slant or other appropriate media

Specimen Volume: Organism in pure culture

Collection Instructions:

1. Perform isolation of bacteria.

2. Organism must be in pure culture, actively growing. **Do not submit mixed cultures.**

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

87641

MRSAP 609735 Methicillin Resistant Staphylococcus aureus, PCR, Nasal

Specimen Requirements:

Specimen Type: Nasal cavity swab

Supplies: BD E-Swab (T853)

Container/Tube:

Preferred: BD Liquid Amies Elution Swab (E-Swab)

Acceptable: Copan Liquid Amies Elution Swab (ESwab); Cepheid nasal sample collection device (dual swab in liquid Stuart media); or Copan swab and transport systems (LQ Stuart or LQ Amies) (scored swabs only)

Specimen Volume: Entire collection

Collection Instructions:

1. Ask the patient to blow their nose prior to collection.

2. Use one swab for both right and left nares.

3. Insert the white Dacron swab tip (do not insert any further in) into the anterior nares.

4. Rotate the swab for 3 seconds against the nasal mucosa. Apply slight pressure on the outside of the nose with your finger to ensure good contact.

5. Repeat the process on the other nostril.

6. Insert the swab back into the carrier. The swabs should go all the way into the transport container. Tightly cap the container.

7. Label the transport container with patient label and send refrigerate.

Specimen Minimum Volume: See Specimen Required.

Transport Temperature:

Specimen Type	Temperature	Time
Swab	Refrigerated (preferred)	5 days
	Ambient	24 hours

CPT Code Information:

87641

MTXSG 62580 Methotrexate Post Glucarpidase, Serum

Specimen Requirements:

Supplies: Amber Frosted Tube, 5 mL (T915)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Amber vial

Specimen Volume: 1 mL serum

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into an amber vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum	Refrigerated (preferred)	7 days	LIGHT PROTECTED
	Frozen	14 days	LIGHT PROTECTED
	Ambient	7 days	LIGHT PROTECTED

CPT Code Information:

80204

MTHX
37047

Methotrexate, Serum

Specimen Requirements:

Supplies: Amber Frosted Tube, 5 mL (T915)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Amber vial

Specimen Volume: 0.5 mL

Collection Instructions:

1. Methotrexate is sensitive to fluorescent light; avoid prolonged exposure of specimen to direct light.

2. Within 2 hours of collection, centrifuge the specimen.

3. For serum gel tubes, aliquot serum into a plastic vial within 2 hours of collection. Keep protected from light.

4. For red-top tubes, aliquot serum into a plastic vial immediately. Keep protected from light.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum	Refrigerated (preferred)	14 days	LIGHT PROTECTED
	Frozen	28 days	LIGHT PROTECTED
	Ambient	72 hours	LIGHT PROTECTED

CPT Code Information:

80204

FMETX 91822 Methsuximide (Celontin) as Desmethylmethsuximide

Specimen Requirements:

Submit only 1 of the following specimens:

Plasma

Draw blood in a green-top (sodium heparin) tube(s), **plasma gel tube is not acceptable.** Spin down and send 2 mL sodium heparin plasma refrigerated in a plastic vial.

Serum

Draw blood in a plain red-top tube(s), **serum gel tube is not acceptable.** Spin down and send 2 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days

	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80339

MMAP 31927 Methylmalonic Acid, Quantitative, Plasma

Specimen Requirements:

Collection Container/Tube:

Preferred: Green top (sodium heparin)

Acceptable: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL plasma

Collection Instructions: Centrifuge and aliquot plasma into a plastic vial.

Specimen Minimum Volume: Plasma: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Refrigerated (preferred)	48 days
	Ambient	48 days
	Frozen	48 days

CPT Code Information:

83921

MMAS 80289 Methylmalonic Acid, Quantitative, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	48 days
	Ambient	48 days
	Frozen	48 days

CPT Code Information:

83921

MMAU
80290

Methylmalonic Acid, Quantitative, Urine

Specimen Requirements:

Patient Preparation:

Fasting: 8 hours, required

Supplies: Urine Tubes, 10 mL (T068)

Container/Tube: Plastic, 10-mL urine tube

Specimen Volume: 4 mL

Collection Instructions:

1. After fasting at least 8 hours overnight, patient should void first-morning urine as normal. **Do not collect**.

2. Patient should continue to fast and then collect the next (second morning) urine voided.

Specimen Minimum Volume: 1.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Frozen (preferred)	28 days
	Refrigerated	28 days
	Ambient	21 days

CPT Code Information:

83921

MMAGP 608021 Methylmalonic Aciduria Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies.

Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry

Send: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from blood spots, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

2. Due to lower concentration of DNA yielded from blood spot, it is possible that additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for

Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and

will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required

to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be

applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order

MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

MPAGP 608022 Methylmalonic Aciduria-Propionic Aciduria Combined Gene Panel, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a hematopoietic stem cell transplant.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See

Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight..

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy. Cultured cells from a prenatal specimen will not be accepted.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Prenatal cultured amniocytes

This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 Flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81443

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

MPHNU 608882 Methylphenidate and Metabolite, Random, Urine

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube: Plastic urine container

Submission Container/Tube: Plastic, 5-mL tube

Specimen Volume: 5 mL

Collection Instructions:

1. Collect a random urine specimen.

2. No preservative.

Additional Information:

1. No specimen substitutions.

2. STAT requests are **not accepted** for this test.

3. Submitting less than 1 mL will compromise the ability to perform all necessary testing.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	10 days
	Frozen	28 days

CPT Code Information:

G0480

80360 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

MPHNS 624101 Methylphenidate and Metabolite, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (Serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Frozen	28 days

CPT Code Information:

G0480

80360 (if appropriate for select payers)

MEX 9245 Mexiletine, Serum

Specimen Requirements:

Patient Preparation: Specimens should only be collected after patient has been receiving mexiletine for at least 3 days. Trough concentrations should be collected just before administration of the next dose.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1.5 mL

Collection Instructions:

1. Draw blood immediately before next scheduled dose.

2. Within 2 hours of collection centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80299

MGMTD 621858 MGMT Promoter Methylation Analysis, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 144 mm(2) tissue (4 x 6 mm x 6 mm areas)

-Minimum amount of tumor area: 36 mm(2) tissue (1 x 6 mm x 6 mm area)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: formalin-fixed paraffin-embedded (FFPE), non-decalcified

Submit only 1 of the following specimens:

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed non-decalcified, paraffin-embedded tissue block.

Specimen Type: Tissue slide

Slides: 1 Stained and 10 unstained

Collection Instructions: Submit 1 slide stained with hematoxylin and eosin and 10 unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: 5 unstained slides at 5-microns thickness

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

81287

CERAM
606777

MI-Heart Ceramides, Plasma

Specimen Requirements:

Patient Preparation: Patients should not be receiving Intralipid because it may cause false elevations in measured ceramides.

Collection Container/Tube: Lavender top (EDTA)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Centrifuge, aliquot at least 1 mL of plasma into a plastic vial and freeze within 8 hours.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma EDTA	Frozen (preferred)	30 days
	Refrigerated	24 hours
	Ambient	8 hours

CPT Code Information:

0119U

MADON
622523

MLCPC
113370

Microbiology Add-On
Microdissection, Laser Capture (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88380-Microdissection; laser capture

MNLCP 621773 Microdissection, Laser Capture, Membranous Nephropathy (Bill Only)

Specimen Requirements:

This test is for billing purposes only.

This is not an orderable test.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88380

TMSI 609364 Microsatellite Instability, Tumor

Specimen Requirements:

This assay requires at least 40% tumor nuclei for endometrial specimens and at least 20% tumor nuclei for colorectal specimens.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 72 mm(2)

-Minimum amount of tumor area: 18 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: formalin-fixed paraffin-embedded, non-decalcified

Preferred: Submit 2, if available, of the following specimens.

Acceptable: Submit **at least one** of the following specimens**.**

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Frozen	
	Refrigerated	

CPT Code Information:

81301

88381-Microdissection, manual

LCMSP 63097 Microsporidia species, Molecular Detection, PCR, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Preferred:

Specimen Type: Unpreserved feces

Supplies:

-Stool container, Small (Random), 4 oz (T288)

-Stool Collection Kit, Random (T635)

Container/Tube: Fecal container

Specimen Volume: 5 g

Specimen Type: Preserved feces

Supplies:

-ECOFIX Stool Transport Vial (Kit) (T219)

-Stool Collection Kit, Random (T635)

Container/Tube: ECOFIX preservative

Specimen Volume: 5 g

Specimen Type: Urine

Container/Tube: Sterile container

Specimen Volume: 5 mL

Collection Instructions: Mid-stream, clean-catch, suprapubic aspirates and catheterization collections are acceptable. Submit in a clean, sterile container free from preservatives. The first portion of the voided urine (first void) is also acceptable.

Specimen Minimum Volume: Feces: 1 g
Urine: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	7 days

	Ambient	7 days
	Frozen	7 days

CPT Code Information:

87798

MTBS 81507 Microsporidia Stain, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Duodenal aspirate (small intestinal aspirate, jejunal aspirate, small bowel aspirate)

Container/Tube: Sterile container

Specimen Volume: 0.5 mL

Additional Information: ECOFIX and 10% formalin are acceptable preservatives.

Specimen Stability Information: Preserved Ambient (preferred) <10 days/Refrigerated <3 days

Specimen Type: Respiratory secretions (bronchoalveolar lavage [BAL], sputum, bronchial wash, pleural fluid)

Container/Tube: Sterile container

Specimen Volume: 0.5 mL

Specimen Stability Information: Refrigerated <3 days (preferred)/Frozen <10 days

Specimen Type: Eye (vitreous fluid, ocular fluid)

Container/Tube: Sterile container

Specimen Volume: 0.5 mL

Specimen Stability: Refrigerated <3 days

Specimen Type: Corneal scraping

Container/Tube: Sterile container

Specimen Volume: 0.5mL in sterile saline

Additional Information: Place scrapings in sterile saline for shipping.

Specimen Stability: Refrigerated <3 days

Specimen Type: Fresh tissue (lung, eye, bladder, rectal, intestinal, colon, skin, muscle, kidney)

Container/Tube: Sterile container

Specimen Volume: 3-mm biopsy in 0.1-mL sterile saline

Specimen Stability: Refrigerated <3 days

Specimen Type: Gallbladder aspirate/Bile aspirate

Container/Tube: Sterile container

Specimen Volume: 0.5 mL

Specimen Stability: Refrigerated <3 days/Frozen <10 days

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

87015-Concentration

87207-Stain

MP1BS 621406 Microtubule-Associated Protein 1B (MAP1B) Antibody, Immunoblot, Serum (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-AIAES / Axonal Neuropathy, Autoimmune/Paraneoplastic Evaluation, Serum

-DMS2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Serum

-DYS2 / Dysautonomia, Autoimmune/Paraneoplastic Evaluation, Serum

-ENS2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Serum

-EPS2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Serum

-GID2 / Gastrointestinal Dysmotility, Autoimmune/Paraneoplastic Evaluation, Serum

-MAS1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Serum

-MDS2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Serum

Specimen Minimum Volume: Serum:1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

84182

MP1BC 621405 Microtubule-Associated Protein 1B (MAP1B) Antibody, Immunoblot, Spinal Fluid (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see:

-DMC2 / Dementia, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-ENC2 / Encephalopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-EPC2 / Epilepsy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-MAC1 / Myelopathy, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

-MDC2 / Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

84182

FMIDZ Midazolam (Versed), serum

90112

Specimen Requirements:

Submit only 1 of the following specimens:

Plasma

Draw blood in a green-top (sodium heparin) tube(s), plasma gel tube is not acceptable. Spin down and send 2 mL sodium heparin plasma refrigerated in a plastic vial.

Serum

Draw blood in a plain red-top tube(s), serum gel tube is not acceptable. Spin down and send 2 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	14 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80346

G0480 (if appropriate)

FMCG4 Milk Cow IgG4

57536

Specimen Requirements:

Draw blood in a plain red-top tube(s), serum gel tube(s) is acceptable. Spin down and send 0.5 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

86001

MILK
82871

Milk, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

PMLK
82827

Milk, Processed, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FMINT
57885

Mint (Mentha Piperita) IgE

Specimen Requirements:

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Centrifuge and aliquot 0.5 mL of serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

86003

MIRI
623669

Mirikizumab, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions:

1. Draw blood immediately before next scheduled dose (trough specimen).

2. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.25 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Ambient	28 days
	Frozen	28 days

CPT Code Information:

80299

FMIRT
57749

Mirtazapine (Remeron)

Specimen Requirements:

Submit only 1 of the following specimens:

Plasma

Draw blood in a green-top (sodium heparin) tube(s), **plasma gel tube is not acceptable.** Spin down and send 2 mL sodium heparin plasma refrigerated in a plastic vial.

Serum

Draw blood in a plain red-top tube(s), **serum gel tube is not acceptable.** Spin down and send 2 mL of serum refrigerated in a plastic vial.

Specimen Minimum Volume: 0.3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Refrigerated (preferred)	14 days
	Frozen	180 days
	Ambient	72 hours

CPT Code Information:

80335

ZW153
91213

Misc Center for Human Genetics

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW204 Misc Cincinnati Childrens Hospital Medical Center

91801

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW59 Misc Esoterix Endocrinology

90498

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW179 Misc Medical Neurogenetics, LLC

91403

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo

Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW149 Misc Monogram Biosciences, Inc.

91170

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie,

consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW196 Misc National Jewish Health Mycobacteriology Test

91643

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW146
90585

Misc Prometheus Laboratories Test

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW305
75411

Misc Washington University Neuromuscular Clinical Lab

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW206 Misc. Seattle Children's Hospital Laboratories

91803

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW185 Miscellaneous Ambry Genetics
91516

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW242 Miscellaneous ARUP Testing
57378

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo

Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW127 Miscellaneous Athena Testing

90566

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie,

consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW221 Miscellaneous Baylor Medical Genetics Laboratories

57103

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW155
91272

Miscellaneous BioAgilytix Diagnostics

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

SCT2
20521

Miscellaneous Biochemical Genetics Testing

Specimen Requirements:

Only orderable as part of the miscellaneous test process.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

ZW293
75252

Miscellaneous CBL Path Inc.

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW172 Miscellaneous Center for Genetic Testing at St. Francis
91302

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW10 Miscellaneous Chemistry Testing, Varies
8921

Specimen Requirements:

For specimen requirements, call Mayo Clinic Laboratories at 800-533-1710.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Frozen	30 days

ZW140 Miscellaneous Child Med Ctr Dallas
90579

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW163 Miscellaneous Children's Hospital of Philadelphia (CHOP)

91291

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW299 Miscellaneous Childrens Hospital Los Angeles

76271

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW246 Miscellaneous Childrens Hospital of Colorado Testing

57992

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW239 57180 Miscellaneous Cincinnati Children's Hospital Medical Center-Hematology/Oncology

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW166 Miscellaneous Cincinnati Childs Hosp Center

91294

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW346 Miscellaneous Cleveland Clinic

75850

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW58 Miscellaneous Duke University

90497

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW57 Miscellaneous Esoterix Coagulation

90496

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW69 Miscellaneous Eurofins Viracor

90508

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW168 Miscellaneous GeneDx, Inc. Testing

91296

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW68 Miscellaneous Greenwood Genetic Ctr

90507

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW193 Miscellaneous HNL Genomics

91601

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW143 Miscellaneous IMMCO Diagnostics

90582

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW71 Miscellaneous Inter Science Institute

90510

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW298 Miscellaneous IntrinsicDx

75278

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW175 Miscellaneous Kennedy Krieger Institute

91323

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW76 Miscellaneous LabCorp of America

90515

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW113 Miscellaneous LabCorp-RTP,NC

90552

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW266 Miscellaneous Machaon Diagnostics
58067

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW342 Miscellaneous MDX Health Inc.
75816

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW78 Miscellaneous Medical Coll of WI

90517

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW80 Miscellaneous Medtox Laboratories, Inc.

90519

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW111 Miscellaneous MiraVista Diagnostics
90550

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW1 Miscellaneous MML Referral Test 1
99991

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW2
99992

Miscellaneous MML Referral Test 2

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW3
99993

Miscellaneous MML Referral Test 3

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW227
57163

Miscellaneous National B Virus Resource Laboratory

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW85 Miscellaneous National Jewish Health

90524

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW333 Miscellaneous Nationwide Children's Hospital

75744

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW190 89123468 Miscellaneous Nemours Children's Health-Gastroenterology

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW359 624088 Miscellaneous Neurocode USA, Inc

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW335 Miscellaneous New York Blood Center

75641

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW86 Miscellaneous NMS Testing
90525

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW214 Miscellaneous Ocular Immunology Laboratory OHSU
91861

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW87 Miscellaneous Palo Alto Testing

90526

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW194 Miscellaneous Prevention Genetics Lab

91602

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW83 Miscellaneous Quest Diagnostics

90522

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW51 Miscellaneous Quest Testing

90490

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW131 Miscellaneous Quest/Nichols Testing

90570

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW224 Miscellaneous Revvity Omics

57157

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW91 Miscellaneous RFFIT Testing

90530

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW124 Miscellaneous Sequenom

75456

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

MISCF 35267 Miscellaneous Studies Using Chromosome-Specific Probes, FISH

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20-25 mL

Collection Instructions:

1. Optimal timing for specimen collection is during 14 to 18 weeks of gestation, but specimens collected at other weeks of gestation are also accepted. Provide gestational age at the time of amniocentesis.

2. Discard the first 2 mL of amniotic fluid.

Additional Information:

1. Unavoidably, about 1% to 2% of mailed-in specimens are not viable.

2. Bloody specimens are undesirable.

3. If the specimen does not grow in culture, you will be notified within 7 days of receipt.

4. Results will be reported and telephoned or faxed if requested.

Specimen Type: Blood (only accepted for Congenital/Hereditary [nonhematologic] testing)

Container/Tube:

Preferred: Yellow top (ACD)

Acceptable: Green top (heparin) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Other anticoagulants are not recommended and are harmful to the viability of the cells.

Specimen Type: Chorionic villi (CVS)

Supplies: CVS Media (RPMI) and Small Dish (T095)

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 to 25 mg

Collection Instructions:

1. Collect specimen by the transabdominal or transcervical method.

2. Transfer chorionic villi to a Petri dish containing transport medium (eg, CVS media (RPMI)).

3. Using a stereomicroscope and sterile forceps, assess the quality and quantity of the villi and remove any blood clots and maternal decidua.

Specimen Type: Lymph node

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 1 cm(3)

Specimen Type: Skin biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 1-cm(3) biopsy specimen of muscle/fascia from the thigh

Collection Instructions:

1. Wash biopsy site with an antiseptic soap.

2. Thoroughly rinse area with sterile water.

3. Do not use alcohol or iodine preparations.

4. A local anesthetic may be used.

5. Biopsy specimens are best taken by punch biopsy to include full thickness of dermis.

Specimen Type: Tissue

Preferred: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tumor tissue block. Blocks prepared with alternative fixation methods may be acceptable; provide fixation method used.

Additional Information:

1. Paraffin embedded specimens can be from any anatomic location (skin, soft tissue, lymph node, etc).

2. Bone specimens that have been decalcified will be attempted for testing, but the success rate is approximately 50%.

Acceptable: Slides

Collection Instructions: For each probe set ordered, 4 consecutive, unstained, 5 micron-thick sections placed on positively charged slides. Include 1 hematoxylin and eosin-stained slide for the entire test order.

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Type: Tumor

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 0.5 to 3 cm(3) or larger

Specimen Minimum Volume: Amniotic fluid: 5 mL; Blood: 2 mL; Chorionic villi: 5 mg; Lymph node: 0.5 cm(3); Solid tumor: 0.5 cm(3)

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271x2, 88291-DNA probe, each (first probe set), Interpretation and report

88271x2-DNA probe, each; each additional probe set (if appropriate)

88271x1-DNA probe, each; coverage for sets containing 3 probes (if appropriate)

88271x2-DNA probe, each; coverage for sets containing 4 probes (if appropriate)

88271x3-DNA probe, each; coverage for sets containing 5 probes (if appropriate)

88273 w/modifier 52-Chromosomal in situ hybridization, less than 10 cells (if appropriate)

88273-Chromosomal in situ hybridization, 10-30 cells (if appropriate)

88274 w/modifier 52-Interphase in situ hybridization, <25 cells, each probe set (if appropriate)

88274-Interphase in situ hybridization, 25 to 99 cells, each probe set (if appropriate)

88275-Interphase in situ hybridization, 100 to 300 cells, each probe set (if appropriate)

ZW102 Miscellaneous U of TX San Ant Test

90541

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW288 Miscellaneous UF Health Pathology Laboratories

75170

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW99 Miscellaneous Univ of AL-Metabolic Disease Lab

90538

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW210 Miscellaneous Univ of IA Molecular Otolaryngology

91857

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW128 Miscellaneous University Hospital Clinical Lab Test

90567

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW234 Miscellaneous University of Alabama at Birmingham

57175

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW186 Miscellaneous University of Chicago Genetics Services

91515

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW187 Miscellaneous University of Iowa Diagnostic Labs

91514

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW61 90500 Miscellaneous University of Minnesota Outreach Laboratory

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW238 57179 Miscellaneous University of Southern California Endocrine Laboratories

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW191 91599 Miscellaneous University of Texas Health Center at Tyler

Microbiology

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW280 58092 Miscellaneous University of Washington Medical Center (Molecular Diagnosis Clinical Microbiology Lab)

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW282 58088 Miscellaneous University of Washington Medical Center (UW Virology Dept of Lab Medicine)

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW278 58078 Miscellaneous University of Washington Medical Center-Clinical Immunology Lab

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW334 75750 Miscellaneous Variantyx, Inc.

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

ZW121 Miscellaneous Versiti Blood Center of Wisconsin

90560

Specimen Requirements:

This is a Miscellaneous Referral Test. For specific requirements, refer to the Referred Tests List under the Test Catalog tab on www.MayoClinicLabs.com. Internal Mayo Clinic providers: refer to the Referral Catalog in CRM. If unable to find the correct test, request forms, or specimen requirements, contact Customer Service at 800-533-1710 or 507-266-5700.

The following must be provided when ordering:

1. Test name

2. Performing lab code

3. Specimen Type

4. For required forms including requisitions and patient specific information forms, ie, consent forms, clinical information, family history, contact Customer Service at 800-533-1710 or 507-266-5700.

Specimen Minimum Volume: See Individual Test ID on the Referred Tests List

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

See Individual Test ID on the Referred Tests List

IHC 35466 Mismatch Repair (MMR) Protein Immunohistochemistry Only, Tumor

Specimen Requirements:

Tumor tissue is required.

Preferred: Submit both of the following specimens.

Acceptable: Submit **at least one** of the following specimens.

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88341 MLH1, MLH2, or MLH6 (if appropriate)

88342 PMS2 (if appropriate)

MITF
70509

MiTF Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

AMA
8176

Mitochondrial Antibodies (M2), Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.4 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	21 days
	Frozen	21 days

CPT Code Information:

86381

DMITO
618558

Mitochondrial DNA Deletion Heteroplasmy, ddPCR, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic

donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a tissue biopsy.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Tissue biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 0.5 to 3 cm(3) or larger

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Muscle tissue biopsy

Supplies: Muscle Biopsy Kit (T541)

Specimen Volume: 20 to 80 mg

Collection Instructions: Prepare and transport specimen per instructions in Muscle Biopsy Specimen Preparation Instructions.

Specimen Stability Information: Frozen (preferred) <24 hours/Ambient <24 hours/Refrigerated <24 hours

Additional Information: Specimens are preferred to be received within 24 hours of collection. Extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

Specimen Type: Snap frozen nerve tissue biopsy

Collection Instructions: Prepare snap frozen tissue biopsy per surgical procedure

Specimen Volume: 0.25 to 0.5 cm

Specimen Stability Information: Frozen

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will

be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured amniocytes from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81479

MITOP 62510 Mitochondrial Full Genome Analysis, Next-Generation Sequencing (NGS), Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA are met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A

maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and

extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a tissue biopsy.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Tissue biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 0.5 to 3 cm(3) or larger

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur

Specimen Type: Snap frozen nerve tissue biopsy

Collection Instructions: Prepare snap frozen tissue biopsy per surgical procedure

Specimen Volume: 0.25 to 0.5 cm

Specimen Stability Information: Frozen

Specimen Type: Muscle tissue biopsy

Supplies: Muscle Biopsy Kit (T541)

Specimen Volume: 20 to 80 mg

Collection Instructions: Prepare and transport specimen per instructions in Muscle Biopsy Specimen Preparation Instructions.

Specimen Stability Information: Frozen (preferred) <24 hours/Ambient <24 hours/Refrigerated <24 hours

Additional Information: Specimens are preferred to be received within 24 hours of collection. Extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Confluent cultured amniocytes

Container/Tube: T-25 flask

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured amniocytes from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies

on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81460-Whole Mitochondrial Genome

81465-Whole Mitochondrial Genome Large Deletion Analysis

88233-Tissue culture, skin, solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

MMPP 616610 Mitochondrial Metabolites, Plasma

Specimen Requirements:

Collection Container/Tube:

Preferred: Green top (Sodium heparin)

Acceptable: Green top (Lithium heparin)

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Collection Instructions: Centrifuge and aliquot plasma into a plastic vial.

Specimen Minimum Volume: 0.1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma	Frozen	7 days

CPT Code Information:

82542

MITAN 621811 Mitotane, Plasma

Specimen Requirements:

Collection Container/Tube: Green top (sodium heparin) (Lithium heparin and PST/plasma gel tubes **are not** acceptable.)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL

Collection Instructions: Within 2 hours of collection, centrifuge and aliquot plasma into plastic vial.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Plasma Na Heparin	Refrigerated (preferred)	21 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

80299

FMITU 75773 Mitragynine (Qualitative), Urine

Specimen Requirements:

*****Must submit one specimen per order. Specimens cannot be shared between multiple orders.*****

Container/Tube: Plastic preservative-free urine container

Specimen Volume: 2 mL

Collection Instructions:

1. Collect 2 mL random urine in a plastic, preservative-free urine container.

2. Send refrigerated.

Specimen Minimum Volume: 0.7 mL

Transport Temperature:

Specimen Type	Temperature	Time

Urine	Refrigerated (preferred)	30 days
	Ambient	30 days
	Frozen	30 days

CPT Code Information:

80323

FCDX2 75842 Mitragynine Confirmation (Qualitative), Umbilical Cord Tissue (Non-Orderable)

Specimen Requirements:

Only orderable as a profile. For more information see FCDSU / Comprehensive Drug Screen, Umbilical Cord Tissue.

Transport Temperature:

Specimen Type	Temperature	Time
Tissue	Refrigerated (preferred)	7 days
	Frozen	14 days

CPT Code Information:

80375

MLHPB 35500 MLH1 Hypermethylation Analysis, Blood

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. Call 800-533-1710 for instructions for testing patients who have received a bone marrow transplant.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Minimum Volume: See Specimen Requirements

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81288

ML1HM
35494

MLH1 Hypermethylation Analysis, Tumor

Specimen Requirements:

This assay requires at least 20% tumor nuclei.

-Preferred amount of tumor area with sufficient percent tumor nuclei: tissue 216 mm(2)

-Minimum amount of tumor area: tissue 36 mm(2)

-These amounts are cumulative over up to 10 unstained slides and must have adequate percent tumor nuclei.

-Tissue fixation: 10% neutral buffered formalin, not decalcified

-For specimen preparation guidance, see Tissue Requirement for Solid Tumor Next-Generation Sequencing. In this document, the sizes are given as 4 mm x 4 mm x 10 slides as preferred: approximate/equivalent to 144 mm(2) and the minimum as 3 mm x 1 mm x 10 slides: approximate/equivalent to 36 mm(2).

Preferred: Submit 2, if available, of the following specimens.

Acceptable: Submit **at least one** of the following specimens**.**

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Note: The total amount of required tumor nuclei can be obtained by scraping up to 10 slides from the same block.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Frozen	
	Refrigerated	

CPT Code Information:

81288

88381

BRMLH 35491 MLH1 Hypermethylation and BRAF Mutation Analysis, Tumor

Specimen Requirements:

Submit 1 of the following specimens:

Specimen Type: Tissue block

Collection Instructions: Submit a formalin-fixed, paraffin-embedded tissue block with acceptable amount of tumor tissue.

Specimen Type: Tissue slide

Slides: 1 Hematoxylin and eosin-stained and 10 unstained

Collection Instructions:

Submit the followings slides:

1 Slide stained with hematoxylin and eosin

AND

10 Unstained, nonbaked slides with 5-micron thick sections of the tumor tissue.

Additional Information:

1. Hematoxylin and eosin-stained and unstained slides will not be returned.

2. Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Ambient (preferred)	
	Frozen	
	Refrigerated	

CPT Code Information:

Slide Review

88381-Microdissection, manual

81210-*BRAF* (*v-raf murine sarcoma viral oncogene homolog B1*) (eg, colon cancer), gene analysis, V600E variant, if appropriate

81288-*MLH1* promoter methylation analysis, if appropriate

MLH1
70510

MLH1 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

MOC31 MOC-31 Immunostain, Technical Component Only

70505

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

MOLD1
81878

Mold Panel, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

MOLPS
89270

Molybdenum, Serum

Specimen Requirements:

Patient Preparation: High concentrations of gadolinium and iodine are known to potentially interfere with most inductively coupled plasma mass spectrometry-based metal tests. If either gadolinium- or iodine-containing contrast media has been administered, **a specimen should not be collected for at least 96 hours**.

Supplies:

-Metal Free B-D Tube (No Additive), 6 mL (T184)

-Metal Free Specimen Vial (T173)

Collection Container/Tube: Plain, royal blue-top Vacutainer plastic trace element blood collection tube

Submission Container/Tube: 7-mL Mayo metal-free, screw-capped, polypropylene vial

Specimen Volume: 0.25 mL Serum

Collection Instructions:

1. Allow specimen to clot for 30 minutes; then centrifuge the specimen to separate serum from the cellular fraction.

2. Remove the stopper. Carefully pour specimen into a Mayo metal-free, polypropylene vial, avoiding transfer of the cellular components of blood. **Do not** insert a pipet into the serum to accomplish transfer, and **do not** ream the specimen with a wooden stick to assist with serum transfer.

3. See Metals Analysis Specimen Collection and Transport for complete instructions.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Serum	Refrigerated (preferred)	14 days	METAL FREE
	Ambient	14 days	METAL FREE
	Frozen	14 days	METAL FREE

CPT Code Information:

83018

QMPTS 620169 **Monoclonal Protein Isotype, Quantitative, Serum (Non-Orderable)**

Specimen Requirements:

Only orderable as part of a profile. For more information see QMPSS / Monoclonal Protein Study, Quantitative, Serum.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	7 days

CPT Code Information:

0077U

MPQU
616907

Monoclonal Protein Quantitation, 24 Hour, Urine

Specimen Requirements:

Supplies: Urine Container, 60 mL (T313)

Submission Container/Tube: Plastic, 60-mL urine bottle

Specimen Volume: 50 mL

Collection Instructions:

1. Collect urine for 24 hours and refrigerate during collection.

2. Aliquot between 30 mL and 50 mL urine into plastic, 60-mL urine bottle.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 30 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	5 days
	Ambient	24 hours

CPT Code Information:

84156

84166

0077U (if appropriate)

RMPQU
616909

Monoclonal Protein Quantitation, Random, Urine

Specimen Requirements:

Supplies: Urine Container, 60 mL (T313)

Submission Container/Tube: Plastic, 60-mL urine bottle

Specimen Volume: 50 mL

Collection Instructions:

1. Collect random urine specimen.

2. Aliquot between 30 mL and 50 mL of urine into a plastic, 60-mL urine bottle and refrigerate.

Specimen Minimum Volume: 30 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	5 days
	Ambient	24 hours

CPT Code Information:

84156

82570

84166

0077U (if appropriate)

SMPU 616913 Monoclonal Protein Screen, 24 Hour, Urine

Specimen Requirements:

Supplies: Urine Container, 60 mL (T313)

Submission Container/Tube: Plastic, 60-mL urine bottle

Specimen Volume: 50 mL

Collection Instructions:

1. Collect urine for 24 hours and refrigerate specimen during collection.

2. Aliquot between 30 mL and 50 mL of urine into a plastic, 60-mL urine bottle.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 30 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	5 days
	Ambient	24 hours

CPT Code Information:

84156

84166 Electrophoresis, protein (if appropriate)

0077U

RSMPU 616915 Monoclonal Protein Screen, Random, Urine

Specimen Requirements:

Supplies: Urine Container, 60 mL (T313)

Submission Container/Tube: Plastic, 60-mL urine bottle

Specimen Volume: 50 mL

Collection Instructions:

1. Collect random urine specimen.

2. Aliquot between 30 mL and 50 mL of urine into a plastic, 60 mL urine bottle and refrigerate.

Specimen Minimum Volume: 30 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	5 days
	Ambient	24 hours

CPT Code Information:

84156

82570

84166 Electrophoresis, protein (if appropriate)

0077U

MPU 616911 Monoclonal Protein Studies, 24 Hour, Urine

Specimen Requirements:

Supplies: Urine Container, 60 mL (T313)

Submission Container/Tube: Plastic, 60-mL urine bottle

Specimen Volume: 50 mL

Collection Instructions:

1. Collect urine for 24 hours and refrigerate during collection.

2. Aliquot between 30 mL and 50 mL urine into plastic, 60-mL urine bottle.

Additional Information: See Urine Preservatives-Collection and Transportation for 24-Hour Urine Specimens for multiple collections.

Specimen Minimum Volume: 30 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	5 days
	Ambient	24 hours

CPT Code Information:

84156

84166

0077U

RMPU
616912

Monoclonal Protein Studies, Random, Urine

Specimen Requirements:

Supplies: Urine Container, 60 mL (T313)

Submission Container/Tube: Plastic, 60-mL urine bottle

Specimen Volume: 50 mL

Collection Instructions:

1. Collect a random urine specimen.

2. Aliquot between 30 mL and 50 mL of urine into a plastic, 60-mL urine bottle and refrigerate.

Specimen Minimum Volume: 30 mL

Transport Temperature:

Specimen Type	Temperature	Time
Urine	Refrigerated (preferred)	14 days
	Frozen	5 days
	Ambient	24 hours

CPT Code Information:

84156

82570

84166

0077U

QMPSS 620919 Monoclonal Protein Study, Quantitative, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 2 mL Serum total; 2 separate plastic vials, each containing 1 mL serum

Collection Instructions: Centrifuge and aliquot serum into 2 plastic vials, each containing 1 mL

Specimen Minimum Volume: Serum: 2 Plastic vials, each containing 0.75 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	7 days

CPT Code Information:

0077U

82784 x 3

MONOF 610018 Monocyte Repartition by CD14/CD16, Blood

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution A or B)

Acceptable: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Refrigerated	72 hours

CPT Code Information:

88184-Flow cytometry; first cell surface, cytoplasmic or nuclear marker

88185 x 7-Flow cytometry; additional cell surface, cytoplasmic or nuclear marker (each)

88187-Flow Cytometry Interpretation, 2 to 8 markers

MORS
623447

Morphine and Metabolite, Serum

Specimen Requirements:

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top (Serum gel/SST are **not acceptable**)

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	14 days
	Frozen	28 days
	Ambient	24 hours

CPT Code Information:

G0480

80361 (if appropriate for select payers)

Clinical Toxicology CPT Code Client Guidance

MSPP
82845

Mosquito Species, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

MOTH
82738

Moth, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

FMNPP
75811

Motor Neuropathy Panel

Specimen Requirements:

Collection Container/Tube: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 4 mL Serum

Collection Instructions:

1. As soon as possible or within 2 hours of specimen collection, centrifuge and aliquot 4 mL serum into a plastic vial.

2. Send refrigerate.

Specimen Minimum Volume: Serum: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum SST	Refrigerated (preferred)	7 days
	Frozen	30 days

CPT Code Information:

82784 x 3

83516 x 7

84155

84165

86334

CED
82668

Mountain Cedar, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

MOUS
82707

Mouse Epithelium, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

MOSP
82792

Mouse Serum Protein, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

MOUP
82795

Mouse Urine Protein, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

JAMOVATSS Movat Pentachrome Stain
102516

Specimen Requirements:

This is not an orderable test. Order CONFL / Pathology Consultation. The consultant will determine the need for special stains.

CPT Code Information:

88313

MDS2 Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Serum
606192

Specimen Requirements:

Patient Preparation: For optimal antibody detection, collect specimen before starting immunosuppressant medication or intravenous immunoglobulin treatment.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube:

Preferred: Red top

Acceptable: Serum gel

Submission Container/Tube: Plastic vial

Specimen Volume: 4 mL Serum

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: Serum: 3 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86596

86255 x 26

83516

86341

0432U

84182 AGNBS (if appropriate)

86256 AGNTS (if appropriate)

86256 AMPIS (if appropriate)

86256 APHTS (if appropriate)

84182 AMIBS (if appropriate)

84182 AN1BS (if appropriate)

86256 AN1TS (if appropriate)

84182 AN2BS (if appropriate)

86256 AN2TS (if appropriate)

86256 AN3TS (if appropriate)

86255 APBCS (if appropriate)

86256 APBTS (if appropriate)

86256 CRMTS (if appropriate)

84182 CRMWS (if appropriate)

86256 DPPTS (if appropriate)

86256 GABIS (if appropriate)

86255 GRFCS (if appropriate)

86256 GRFTS (if appropriate)

86256 IG5TS (if appropriate)

86255 ITPCS (if appropriate)

86256 ITPTS (if appropriate)

86256 K11TS (if appropriate)

84182 MP1BS (if appropriate)

86255 GL1CS (if appropriate)

86256 GL1TS (if appropriate)

86255 GL5CS (if appropriate)

86256 GL5TS (if appropriate)

86255 NCDCS (if appropriate)

86256 NCDTS (if appropriate)

86256 NMDIS (if appropriate)

84182 PC1BS (if appropriate)

86256 PC1TS (if appropriate)

86256 PC2TS (if appropriate)

84182 PCTBS (if appropriate)

86256 PCTTS (if appropriate)

86256 PDETS (if appropriate)

86255 SP5CS (if appropriate)

86256 SP5TS (if appropriate)

86255 SP7CS (if appropriate)

86256 SP7TS (if appropriate)

86255 T46CS (if appropriate)

86256 T46TS (if appropriate)

MDC2
606193

Movement Disorder, Autoimmune/Paraneoplastic Evaluation, Spinal Fluid

Specimen Requirements:

Patient Preparation: For optimal antibody detection, collect specimen before starting immunosuppressant medication, corticosteroids, or intravenous immunoglobulin treatment.

Container/Tube: Sterile vial

Preferred: Collection vial number 1

Acceptable: Any collection vial

Specimen Volume: 4 mL

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Refrigerated (preferred)	28 days
	Frozen	28 days
	Ambient	72 hours

CPT Code Information:

86255 x28

83516

86341

0432U

84182 AGNBC (if appropriate)

86256 AGNTC (if appropriate)

86255 AINCC (if appropriate)

86256 AMPIC (if appropriate)

86256 APHTC (if appropriate)

84182 AMIBC (if appropriate)

84182 AN1BC (if appropriate)

86256 AN1TC (if appropriate)

84182 AN2BC (if appropriate)

86256 AN2TC (if appropriate)

86256 AN3TC (if appropriate)

86255 APBCC (if appropriate)

86256 CRMTC (if appropriate)

84182 CRMWC (if appropriate)

86256 DPPTC (if appropriate)

86256 GABIC (if appropriate)

86255 GFACC (if appropriate)

86256 GFATC (if appropriate)

86255 GRFCC (if appropriate)

86256 GRFTC (if appropriate)

86256 IG5TC (if appropriate)

86255 ITPCC (if appropriate)

86256 ITPTC (if appropriate)

86256 K11TC (if appropriate)

84182 MP1BC (if appropriate)

86255 GL1CC (if appropriate)

86256 GL1TC (if appropriate)

86255 GL5CC (if appropriate)

86256 GL5TC (if appropriate)

86255 NCDCC (if appropriate)

86256 NCDTC (if appropriate)

86255 NFHCC (if appropriate)

86256 NIFTC (if appropriate)

86255 NFLCC (if appropriate)

86256 NMDIC (if appropriate)

84182 PC1BC (if appropriate)

86256 PC1TC (if appropriate)

86256 PC2TC (if appropriate)

84182 PCTBC (if appropriate)

86256 PCTTC (if appropriate)

86256 PDETC (if appropriate)

86255 SP5CC (if appropriate)

86256 SP5TC (if appropriate)

86255 SP7CC (if appropriate)

86256 SP7TC (if appropriate)

86255 T46CC (if appropriate)

86256 T46TC (if appropriate)

MPLFM MPL Exon 10 Mutation Detection, Bone Marrow

618661

Specimen Requirements:

Container/Tube: Lavender top (EDTA)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Bone Marrow	Ambient (preferred)	7 days
	Refrigerated	7 days

CPT Code Information:

81339

MPLR 36682 MPL Exon 10 Mutation Detection, Reflex, Varies (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see MPNR / Myeloproliferative Neoplasm, *JAK2* V617F with Reflex to *CALR* and *MPL*, Varies.

Submit only 1 of the following specimens:

Specimen Type: Whole Blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD solution B)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not** aliquot.

3. Label specimen as whole blood.

Specimen Stability Information: Ambient (preferred)/Refrigerate

Specimen Type: Bone marrow

Container/Tube: Lavender top (EDTA) or yellow top (ACD solution B)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not** aliquot.

3. Label specimen as bone marrow.

Specimen Stability Information: Ambient (preferred)/Refrigerate

Specimen Type: Extracted DNA from whole blood or bone marrow

Container/Tube: 1.5- to 2-mL tube

Specimen Volume: Entire specimen

Collection Instructions:

1. DNA must be extracted from blood or bone marrow within 7 days of collection.

2. Label specimen as extracted DNA and source of specimen

3. Provide volume and concentration of the DNA.

Specimen Stability Information: Frozen (preferred) 1 year/Refrigerate/Ambient

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). We cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	7 days

CPT Code Information:

81339-*MPL* (myeloproliferative leukemia virus oncogene, thrombopoietin receptor, TPOR) (eg, myeloproliferative disorder), exon 10 sequence

MPLVS 602599 MPL Exon 10 Mutation Detection, Varies

Specimen Requirements:

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collections Instructions:

1. Invert several times to mix blood.

2. Send specimen in original tube. **Do not aliquot.**

3. Label specimen as whole blood.

Specimen Stability: Ambient (preferred)/Refrigerate

Specimen Type: Bone marrow

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 2 mL

Collections Instructions:

1. Invert several times to mix bone marrow.

2. Send specimens in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Stability: Ambient (preferred)/Refrigerate

Specimen Type: Extracted DNA from whole blood or bone marrow

Container/Tube: 1.5- to 2- mL tube

Specimen Volume: Entire specimen

Collection Instructions:

1. DNA must be extracted from blood or bone marrow within 7 days of collection.

2. Label specimen as extracted DNA and source of specimen

3. Provide volume and concentration of the DNA.

Specimen Stability Information: Frozen (preferred) 1 year/Refrigerated/Ambient

Additional Information: DNA must be extracted in a CLIA-certified laboratory or

equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). We cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied.

Specimen Minimum Volume: Whole blood, Bone marrow: 0.5 mL; Extracted DNA: 50 mcL at 20 ng/mcL concentration

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	7 days

CPT Code Information:

81339-*MPL* (myeloproliferative leukemia virus oncogene, thrombopoietin receptor, TPOR) (eg, myeloproliferative disorder), exon 10 sequence

MPNML 44179 MPL Exon 10 Sequencing, Reflex, Varies (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see MPNCM / Myeloproliferative Neoplasm, *CALR* with Reflex to *MPL*, Varies.

Submit only 1 of the following specimens:

Specimen Type: Whole Blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD-B)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot.**

3. Label specimen as whole blood.

Specimen Stability Information: Ambient (preferred)/Refrigerate

Specimen Type: Bone marrow

Container/Tube: Lavender top (EDTA) or yellow top (ACD-B)

Specimen Volume: 2 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

3. Label specimen as bone marrow.

Specimen Stability Information: Ambient (preferred)/Refrigerate

Specimen Type: Extracted DNA from blood or bone marrow

Container/Tube: 1.5 to 2 mL tube

Specimen Volume: Entire specimen

Collection Instructions:

1. DNA must be extracted from blood or bone marrow within 7 days of collection.

2. Label specimen as extracted DNA and source of specimen.

3. Provide volume and concentration of the DNA.

Specimen Stability Information: Frozen (preferred) 1 year/Refrigerate/Ambient

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). We cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied.

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81339-MPL (myeloproliferative leukemia virus oncogene, thrombopoietin receptor, TPOR) (eg, myeloproliferative disorder), exon 10 sequence

Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see MPNJM / Myeloproliferative Neoplasm, *JAK2* V617F with Reflex to *CALR* and *MPL*, Bone Marrow.

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: None

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send specimen in original tube. Do not aliquot.

3. Label specimen as bone marrow.

Note: Extracted DNA from bone marrow is not acceptable.

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Bone Marrow	Ambient (preferred)	7 days
	Refrigerated	7 days

CPT Code Information:

81339

FMPVP
75817

Mpox Virus DNA, Qualitative Real-Time PCR

Specimen Requirements:

Specimen Type: Lesion swab

Container/Tube: Viral Culture Media (VCM), Viral Transport Media (VTM), Universal Transport Media (UTM), or equivalent

Specimen Volume: 1 Swab

Collection Instructions:

1. Swab the pustule/lesion vigorously and place the swab in 3 mL Viral Culture Media (VCM), Viral Transport Media (VTM), Universal Transport Media (UTM), or equivalent.

2. Include the following required information with the specimen: Patient race, ethnicity, specimen type (source), and anatomic location specimen was collected.

3. Send frozen.

Additional Information: Each individual specimen submitted should be transported in its own sealed bag. Multiple specimens collected on a single patient should be submitted separately.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Frozen (preferred)	30 days
	Refrigerated	7 days

CPT Code Information:

87593 x 2

MSH2
70512

MSH2 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

MSH6
70513

MSH6 Immunostain, Technical Component Only

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, Primary

88341-TC, If additional IHC

CSMRT 607626 mSMART Plasma Cell Proliferative Disorder, Pre-Analysis Cell Sorting, Bone Marrow (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see one of the following:

MSMRT / Mayo Algorithmic Approach for Stratification of Myeloma and Risk-Adapted Therapy Report, Bone Marrow

MSMRD / Myeloma Stratification and Risk-Adapted Therapy with Reflex to Minimal Residual Disease, Bone Marrow

Container/Tube:

Preferred: Yellow top (ACD solution A or B)

Acceptable: Lavender top (EDTA) or green top (heparin)

Specimen Volume: 4 mL

Collection Instructions:

1. Invert several times to mix bone marrow.

2. Send bone marrow specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time
Bone Marrow	Ambient (preferred)	4 days
	Refrigerated	4 days

CPT Code Information:

88184-Flow Cytometry; first cell surface, cytoplasmic or nuclear marker

88185 x 5-Flow Cytometry, additional cell surface, cytoplasmic or nuclear marker (each)

MPCDS 606090 mSMART, Plasma Cell Proliferative Disorder, FISH, Bone Marrow (Non-Orderable)

Specimen Requirements:

Only orderable as part of a profile. For more information see MSMRT / Mayo Algorithmic Approach for Stratification of Myeloma and Risk-Adapted Therapy Report, Bone Marrow.

Specimen Type: Redirected bone marrow

Preferred: Yellow top (ACD)

Acceptable: Lavender top (EDTA) or green top (heparin)

Specimen Volume: 4 mL

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Bone Marrow	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88271 x 2, 88274, 88291-FISH Probe, Analysis, Interpretation; 1 probe set

88271 x 2, 88274-FISH Probe, Analysis; each additional probe set (if appropriate)

MUCN2 Mucin 2, Immunostain, Technical Component Only

605116

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However,

due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

MUCN4 Mucin 4, Immunostain, Technical Component Only
601740

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

MUCN5 Mucin 5AC, Immunostain, Technical Component Only
605118

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

MUCN6 Mucin 6, Immunostain, Technical Component Only
605120

Specimen Requirements:

Specimen Type: Tissue

Supplies: Immunostain Technical Only Envelope (T693)

Container/Tube: Immunostain Technical Only Envelope

Submit:

-Formalin-fixed, paraffin-embedded tissue block

OR

-2 Unstained, positively charged glass slides (25- x 75- x 1-mm) per test ordered; sections 4-microns thick

Additional Information: Broken or cracked unstained slides will be rejected. Stained slides that were broken and repaired (ie, double mounted) at the client site will not be rejected, as, in most cases, they can still be reviewed under the microscope. However, due to the double mounting, digital scanning cannot be performed.

Transport Temperature:

Specimen Type	Temperature	Time
TECHONLY	Ambient (preferred)	
	Refrigerated	

CPT Code Information:

88342-TC, primary

88341-TC, if additional IHC

MPSQU 606299 Mucopolysaccharides Quantitative, Random, Urine

Specimen Requirements:

Patient Preparation: For 6 hours before specimen collection, patient **should not** receive heparin.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Container/Tube: Plastic, 5-mL urine tube

Specimen Volume: 2 mL

Pediatric Volume: 1 mL

Collection Instructions: Collect a random urine specimen **(early morning preferred).**

Specimen Minimum Volume: 1 mL

Transport Temperature:

Specimen Type	Temperature	Time

Urine	Refrigerated (preferred)	90 days
	Frozen	365 days
	Ambient	7 days

CPT Code Information:

83864

82570

MPSER 604905 Mucopolysaccharides Quantitative, Serum

Specimen Requirements:

Patient Preparation: For 6 hours before specimen collection, patient **should not** receive heparin.

Supplies: Sarstedt Aliquot Tube, 5 mL (T914)

Collection Container/Tube: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL

Pediatric: 0.2 mL

Collection Instructions:

1. Do not collect specimen from a line that may have been used to infuse heparin or has been flushed with heparin.

2. Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: 0.2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Serum Red	Refrigerated (preferred)	90 days
	Frozen	90 days
	Ambient	14 days

CPT Code Information:

83864

MPS3W 618293

Mucopolysaccharidosis III, Four-Enzyme Panel, Leukocytes

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution B)

Acceptable: Yellow top (ACD solution A) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Minimum Volume: 5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD	Refrigerated (preferred)	6 days
	Ambient	6 days

CPT Code Information:

82657

MPS3B 618292 Mucopolysaccharidosis III, Three-Enzyme Panel, Blood Spot

Specimen Requirements:

Submit only 1 of the following specimen types:

Preferred:

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Blood Spot Collection Card

Acceptable: Whatman Protein Saver 903 Paper, PerkinElmer 226 filter paper, Munktell filter paper, or blood collected in tubes containing ACD or EDTA and dried on filter paper.

Specimen Volume: 2 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. At least 2 spots should be complete (ie, unpunched).

3. Let blood dry on filter paper at room temperature in a horizontal position for a minimum of 3 hours.

4. Do not expose specimen to heat or direct sunlight.

5. Do not stack wet specimens.

6. Keep specimen dry.

Specimen Stability Information: Refrigerated (preferred) 60 days/Ambient 7 days/Frozen 60 days

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions.

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777).

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800).

Acceptable:

Specimen Type: Whole Blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 2 mL

Collection Instructions: Send whole blood specimen in original tube**. Do not aliquot.**

Specimen Stability Information: Refrigerate (preferred) 7 days/Ambient 48 hours

Specimen Minimum Volume: Blood spots: 1; Whole blood: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time

Whole blood	Varies

CPT Code Information:

82657

83864 (if appropriate)

MPS4B 618294 Mucopolysaccharidosis IV Enzyme Panel, Blood Spot

Specimen Requirements:

Submit only 1 of the following specimen types:

Preferred:

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Blood Spot Collection Card

Acceptable: Whatman Protein Saver 903 Paper, PerkinElmer 226 filter paper, Munktell filter paper, or blood collected in tubes containing ACD or EDTA and dried on filter paper.

Specimen Volume: 2 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. At least 2 spots should be complete, ie, unpunched.

3. Let blood dry on filter paper at room temperature in a horizontal position for a minimum of 3 hours.

4. Do not expose specimen to heat or direct sunlight.

5. Do not stack wet specimens.

6. Keep specimen dry.

Specimen Stability Information: Refrigerated (preferred) 60 days/Ambient 7 days/Frozen 60 days

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Acceptable:

Specimen Type: Whole Blood

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 2 mL

Collection Instructions: Send whole blood specimen in original tube**. Do not aliquot.**

Specimen Stability Information: Refrigerate (preferred) 7 days/Ambient 48 hours

Specimen Minimum Volume: Blood spot: 1; Whole blood: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Varies	

CPT Code Information:

82657

83864 (if appropriate)

MPS4W 618295 Mucopolysaccharidosis IV Enzyme Panel, Leukocytes

Specimen Requirements:

Container/Tube:

Preferred: Yellow top (ACD solution B)

Acceptable: Yellow top (ACD solution A) or lavender top (EDTA)

Specimen Volume: 6 mL

Collection Instructions: Send whole blood specimen in original tube. **Do not aliquot.**

Specimen Minimum Volume: 5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole Blood ACD	Refrigerated (preferred)	6 days
	Ambient	6 days

CPT Code Information:

82657

IDUA
622652

Mucopolysaccharidosis Type I, IDUA Gene Sequencing with Deletion/Duplication, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about for testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal

essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Tissue biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 0.5 to 3 cm(3) or larger

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed, and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry.

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed, and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Prenatal cultured fibroblasts (eg, products of conception), amniocytes, or other confluent cultured cells. This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24

hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks is required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81406 IDUA (iduronidase, alpha-L-) (eg, mucopolysaccharidosis type I), full gene sequence

88233-Tissue culture, skin or solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

IDS
622666

Mucopolysaccharidosis Type II, IDS Gene Sequencing with Deletion/Duplication, Varies

Specimen Requirements:

Patient Preparation: A previous hematopoietic stem cell transplant from an allogenic donor will interfere with testing. For information about testing patients who have received a hematopoietic stem cell transplant, call 800-533-1710.

Submit only 1 of the following specimens:

Specimen Type: Whole blood

Container/Tube: Lavender top (EDTA) or yellow top (ACD)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

3. Whole blood collected postnatal from an umbilical cord is also acceptable. See Additional Information.

Specimen Stability Information: Ambient (preferred) 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for specimens received after 4 days, and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the requested volume must be submitted. Testing may be canceled if DNA requirements are inadequate.

3. For postnatal umbilical cord whole blood specimens, maternal cell contamination studies are recommended to ensure test results reflect that of the patient tested. A maternal blood specimen is required to complete maternal cell contamination studies. Order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on both the cord blood and maternal blood specimens under separate order numbers.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific

gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Type: Skin biopsy

Supplies: Fibroblast Biopsy Transport Media (T115)

Container/Tube: Sterile container with any standard cell culture media (eg, minimal essential media, RPMI 1640). The solution should be supplemented with 1% penicillin and streptomycin.

Specimen Volume: 4-mm Punch

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Cultured fibroblasts

Source: Skin

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured fibroblast cells from a skin biopsy

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Tissue biopsy

Supplies: Hank's Solution (T132)

Container/Tube: Sterile container with sterile Hank's balanced salt solution, Ringer's solution, or normal saline

Specimen Volume: 0.5 to 3 cm(3) or larger

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Collection card (Whatman Protein Saver 903 Paper)

Acceptable: PerkinElmer 226 filter paper or blood spot collection card

Specimen Volume: 2 to 5 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.2. Let blood dry on the filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

3. Do not expose specimen to heat or direct sunlight.

4. Do not stack wet specimens.

5. Keep specimen dry

Specimen Stability Information: Ambient (preferred)/Refrigerated

Additional Information:

1. Blood spot specimens are acceptable but not recommended. Multiple extractions will

be required to obtain sufficient yield for supplemental analysis, and there is significant risk for test failure due to insufficient DNA.

2. Due to lower concentration of DNA yielded from blood spot, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

3. For collection instructions, see Blood Spot Collection Instructions

4. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

5. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Type: Extracted DNA

Container/Tube:

Preferred: Screw Cap Micro Tube, 2 mL with skirted conical base

Acceptable: Matrix tube, 1 mL

Collection Instructions:

1. The preferred volume is at least 100 mcL at a concentration of 75 ng/mcL.

2. Include concentration and volume on tube.

Specimen Stability Information: Frozen (preferred) 1 year/Ambient/Refrigerated

Additional Information: DNA must be extracted in a CLIA-certified laboratory or equivalent and must be extracted from a specimen type listed as acceptable for this test (including applicable anticoagulants). Our laboratory has experience with Chemagic, Puregene, Autopure, MagnaPure, and EZ1 extraction platforms and cannot guarantee that all extraction methods are compatible with this test. If testing fails, one repeat will be attempted, and if unsuccessful, the test will be reported as failed and a charge will be applied. If applicable, specific gene regions that were unable to be interrogated due to DNA quality will be noted in the report.

PRENATAL SPECIMENS

Due to its complexity, consultation with the laboratory is required for all prenatal testing; call 800-533-1710 to speak to a genetic counselor.

Specimen Type: Amniotic fluid

Container/Tube: Amniotic fluid container

Specimen Volume: 20 mL

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULAF / Culture for Genetic Testing, Amniotic Fluid. An additional 2 to 3 weeks are required to culture amniotic fluid before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Prenatal cultured fibroblasts (eg, products of conception), amniocytes, or other confluent cultured cells. This does not include cultured chorionic villi.

Container/Tube: T-25 flask

Specimen Volume: 2 Flasks

Collection Instructions: Submit confluent cultured cells from another laboratory

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Chorionic villi

Container/Tube: 15-mL tube containing 15 mL of transport media

Specimen Volume: 20 mg

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information: Specimen will only be tested after culture.

1. Specimens are preferred to be received within 24 hours of collection. Culture and extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing. An additional 3 to 4 weeks are required to culture fibroblasts before genetic testing can occur.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Type: Cultured chorionic villi

Container/Tube: T-25 flasks

Specimen Volume: 2 Full flasks

Collection Instructions: Submit confluent cultured cells from another laboratory.

Specimen Stability Information: Ambient (preferred) <24 hours/Refrigerated <24 hours

Additional Information:

1. Specimens are preferred to be received within 24 hours of collection. Culture and/or extraction will be attempted for specimens received after 24 hours and will be evaluated to determine if testing may proceed.

2. A separate culture charge will be assessed under CULFB / Fibroblast Culture for Biochemical or Molecular Testing.

3. **All prenatal specimens must be accompanied by a maternal blood specimen;** order MATCC / Maternal Cell Contamination, Molecular Analysis, Varies on the maternal specimen.

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81405-*IDS* (iduronate 2-sulfatase) (eg, mucopolysaccharidosis, type II), full gene sequence

88233-Tissue culture, skin, or solid tissue biopsy (if appropriate)

88240-Cryopreservation (if appropriate)

MPSWB Mucopolysaccharidosis, Blood
113435

Specimen Requirements:

Patient Preparation: For 6 hours before specimen collection, patient **should not** receive heparin.

Container/Tube:

Preferred: Lavender top (EDTA)

Acceptable: Yellow top (ACD)

Specimen Volume: 2 mL

Collection Instructions: Do not collect specimen from a line that may have been used to infuse heparin or has been flushed with heparin.

Specimen Minimum Volume: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
Whole blood	Ambient (preferred)	7 days
	Refrigerated	7 days

CPT Code Information:

83864

MPSBS Mucopolysaccharidosis, Blood Spot
65095

Specimen Requirements:

Patient Preparation: For 6 hours before specimen collection, patient **should not** receive heparin.

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Collection Container:

Preferred: Blood Spot Collection Card

Acceptable: Whatman Protein Saver 903 paper, PerkinElmer 226 filter paper, Munktell

filter paper, local newborn screening card, postmortem screening card, or blood collected in tubes containing ACD or EDTA and dried on filter paper

Specimen Volume: 2 Blood spots

Collection Instructions

1. Do not collect specimen from a line that may have been used to infuse heparin or has been flushed with heparin.

2. An alternative blood collection option for a patient older than 1 year is a fingerstick. For detailed instructions, see How to Collect Dried Blood Spot Samples.

3. Completely fill at least 2 circles on the filter paper card (approximately 100 microliters blood per circle).

4. Let blood dry on filter paper at ambient temperature in a horizontal position for a minimum of 3 hours.

5. Do not expose specimen to heat or direct sunlight.

6. Do not stack wet specimens.

7. Keep specimen dry.

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)

Specimen Minimum Volume: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time	Special Container
Whole blood	Ambient (preferred)	100 days	FILTER PAPER
	Frozen	100 days	FILTER PAPER
	Refrigerated	100 days	FILTER PAPER

CPT Code Information:

83864

MUC 82675 Mucor, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

MUG
82683

Mugwort, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time

Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

MULB
82864

Mulberry, IgE, Serum

Specimen Requirements:

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 0.5 mL for every 5 allergens requested

Collection Instructions: Centrifuge and aliquot serum into a plastic vial.

Specimen Minimum Volume: For 1 allergen: 0.3 mL
For more than 1 allergen: (0.05 mL x number of allergens) + 0.25 mL deadspace

Transport Temperature:

Specimen Type	Temperature	Time
Serum	Refrigerated (preferred)	14 days
	Frozen	90 days

CPT Code Information:

86003

RETZZ
614587

Multiple Endocrine Neoplasia Type 2 Syndrome, RET, Full Gene Analysis, Varies

Specimen Requirements:

Patient Preparation: A previous bone marrow transplant from an allogenic donor will interfere with testing. For instructions for testing patients who have received a bone marrow transplant, call 800-533-1710.

Specimen Type: Whole blood

Container/Tube:

Preferred: Lavender top (EDTA) or yellow top (ACD)

Acceptable: Green top (Sodium heparin)

Specimen Volume: 3 mL

Collection Instructions:

1. Invert several times to mix blood.

2. Send whole blood specimen in original tube. **Do not aliquot**.

Specimen Stability Information: Ambient 4 days/Refrigerated 4 days/Frozen 4 days

Additional Information:

1. Specimens are preferred to be received within 4 days of collection. Extraction will be attempted for samples received after 4 days and DNA yield will be evaluated to determine if testing may proceed.

2. To ensure minimum volume and concentration of DNA is met, the preferred volume of blood must be submitted. Testing may be canceled if DNA requirements are inadequate.

Specimen Type: Saliva

Patient Preparation: Patient should not eat, drink, smoke, or chew gum 30 minutes prior to collection.

Supplies:

DNA Saliva Kit High Yield (T1007)

Saliva Swab Collection Kit (T786)

Container/Tube:

Preferred: High-yield DNA saliva kit

Acceptable: Saliva swab

Specimen Volume: 1 Tube if using T1007 or 2 swabs if using T786

Collection Instructions: Collect and send specimen per kit instructions.

Specimen Stability Information: Ambient (preferred) 30 days/Refrigerated 30 days

Additional Information: Saliva specimens are acceptable but not recommended. Due to lower quantity/quality of DNA yielded from saliva, some aspects of the test may not perform as well as DNA extracted from a whole blood sample. When applicable, specific

gene regions that were unable to be interrogated will be noted in the report. Alternatively, additional specimen may be required to complete testing.

Specimen Minimum Volume: Whole blood: 1 mL; Saliva: See Specimen Required

Transport Temperature:

Specimen Type	Temperature	Time
Varies	Varies	

CPT Code Information:

81406

MRDMR 616031 Multiple Myeloma Measurable Residual Disease (MRD) by Flow Cytometry, Bone Marrow (Non-Orderable)

Specimen Requirements:

Only orderable as a reflex. For more information see MSMRD / Myeloma Stratification and Risk-Adapted Therapy with Reflex to Minimal Residual Disease, Bone Marrow

Specimen Type: Redirected bone marrow

Container/Tube:

Preferred: Yellow top (ACD solution A or B)

Acceptable: Lavender top (EDTA)

Specimen Volume: 4 mL

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Bone Marrow	Ambient (preferred)	72 hours
	Refrigerated	72 hours

CPT Code Information:

88184-Flow Cytometry; first cell surface, cytoplasmic or nuclear marker

88185 x 9-Flow Cytometry; additional cell surface, cytoplasmic or nuclear marker

88188-Flow Cytometry Interpretation, 9 to 15 Markers

MRDMM 65218 Multiple Myeloma Measurable Residual Disease (MRD), Flow Cytometry, Bone Marrow

Specimen Requirements:

Specimen Type: Redirected bone marrow

Container/Tube:

Preferred: Yellow top (ACD solution A or B)

Acceptable: Lavender top (EDTA)

Specimen Volume: 4 mL

Specimen Minimum Volume: 2 mL

Transport Temperature:

Specimen Type	Temperature	Time
Bone Marrow	Ambient (preferred)	72 hours
	Refrigerated	72 hours

CPT Code Information:

88184-Flow Cytometry; first cell surface, cytoplasmic or nuclear marker

88185 x 9-Flow Cytometry; additional cell surface, cytoplasmic or nuclear marker

88188-Flow Cytometry Interpretation, 9 to 15 Markers

MSP3 607837 Multiple Sclerosis (MS) Cascade, Serum and Spinal Fluid

Specimen Requirements:

Both serum and spinal fluid are required. Spinal fluid must be obtained within 1 week of serum collection.

Specimen Type: Spinal fluid (CSF)

Container/Tube: Sterile vial

Specimen Volume: 1 mL

Collection Instructions:

1. Submit CSF from collection vial number 4 (preferred); vial number 1, 3, 2 are also acceptable (in this order).

2. Label specimen as spinal fluid.

Specimen Type: Serum

Supplies: Sarstedt Aliquot Tube 5 mL (T914)

Collection Container/Tube:

Preferred: Serum gel

Acceptable: Red top

Submission Container/Tube: Plastic vial

Specimen Volume: 1 mL Serum

Collection Instructions:

1. Within 2 hours of collection, centrifuge and aliquot serum into a plastic vial.

2. Label specimen as serum.

Specimen Minimum Volume: Serum, Spinal fluid: 0.5 mL

Transport Temperature:

Specimen Type	Temperature	Time
CSF	Frozen (preferred)	14 days
	Refrigerated	72 hours
	Ambient	24 hours
Serum	Frozen (preferred)	14 days
	Ambient	14 days
	Refrigerated	14 days

CPT Code Information:

83521

83916 x2 (if appropriate)

MSDBS Multiple Sulfatase Deficiency, Blood Spot
618296

Specimen Requirements:

Submit only 1 of the following specimen types:

Preferred:

Specimen Type: Blood spot

Supplies: Card-Blood Spot Collection (Filter Paper) (T493)

Container/Tube:

Preferred: Blood Spot Collection Card

Acceptable: Whatman Protein Saver 903 Paper, PerkinElmer 226 filter paper, Munktell filter paper, or blood collected in tubes containing ACD or EDTA and dried on filter paper.

Specimen Volume: 2 Blood spots

Collection Instructions:

1. An alternative blood collection option for a patient 1 year of age or older is a fingerstick. For detailed instructions, see How to Collect a Dried Blood Spot Sample.

2. At least 2 spots should be complete (ie, unpunched).

3. Let blood dry on filter paper at room temperature in a horizontal position for a minimum of 3 hours.

4. Do not expose specimen to heat or direct sunlight.

5. Do not stack wet specimens.

6. Keep specimen dry.

Specimen Stability Information: Refrigerated (preferred) 60 days/Ambient 7 days/Frozen 60 days

Additional Information:

1. For collection instructions, see Blood Spot Collection Instructions

2. For collection instructions in Spanish, see Blood Spot Collection Card-Spanish Instructions (T777)

3. For collection instructions in Chinese, see Blood Spot Collection Card-Chinese Instructions (T800)